

THE FLAVOR MATRIX
The Science of Which Ingredients Taste Best
Together—and Why

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with
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Part I

Introduction

THE ART & SCIENCE OF FLAVOR PAIRING

The idea for this book came to me while I was cooking with a computer.

The year was 2012, and my kitchen companion was Watson, IBM's famous supercomputer. If you're like most people, you first met Watson when it competed—and dominated—on the TV show *Jeopardy!* You might not have known that the computer's talents extend beyond trivia. I certainly didn't when I first met Watson—but I was in for a surprise.

When IBM had approached The Institute of Culinary Education, where I work, to collaborate on a project that would mix computing and human creativity in a way that had never been attempted, I was skeptical. How could a trivia-playing computer teach me anything about cooking? After all, I had devoted my life to being at the cutting edge of the culinary world. What's more, cooking is a craft—an art, something that engages the senses and requires a human touch. What could a computer know about how a fresh tomato feels, or what a steak sounds like as it sizzles to perfect doneness?

It wasn't long before I realized that Watson was capable of much greater things than playing trivia. Since proving its revolutionary computing abilities on *Jeopardy!*, Watson had turned its prodigious talents to managing care for cancer patients, understanding financial markets, helping law enforcement solve cold cases, even providing music recommendations. And now, IBM wanted to see if Watson could help people to be more creative in the kitchen. They knew that Watson could answer any question posed to it, but what about the questions we didn't even know to ask?

The collaboration would work like this: Watson would create a random-seeming list of ingredients, which myself or another one of the chef-instructors at ICE would then use to create a dish that no one had ever seen before. The foods that Watson would select for us would be anything but random; rather, the machine would carefully select the most unexpected and delicious pairings it could come up with. Through a sweeping analysis of data pulled from academic articles, cookbooks and other sources, Watson would find combinations of ingredients

that it predicted would taste good when combined—but which rarely, if ever, appeared together in a recipe.

I enjoyed cooking with Watson. We produced some interesting dishes: roast duck with tomato, sage, olive, and cherries; lobster with pork, saffron, basil, carrot and balsamic; chicken with strawberries, mushroom and apple. These were combinations that I never would have come up with on my own, and our work together inspired me to reconsider what creativity in the kitchen actually looks like. But I knew I had only scratched the surface.

Watson didn't look at ingredients the same way that I did as a chef. Rather, it thought about combining them based on nothing but their inherent flavors. The computer seemed to be able to see the invisible filaments that bound different ingredients together. For the moment, the ability to see these connections was the exclusive domain of scientists and super computers. But I wanted to learn to see them for myself, and help others to do the same. None of my students at ICE, let alone my family or friends, had access to a supercomputer that could help them think more creatively about flavor-pairing, as Watson had helped me to do. How could I share this amazing experience with them?

Luckily, there was a vast body of data out there about flavor pairing, just waiting to be explored. But this information isn't part of the language of chefs—which is a shame, because the art and (yes) science of food pairing is in some ways the final frontier of modern cooking.

By most measures, the world of food has been revolutionized by science in the past few decades. Since the concept of “molecular gastronomy” was invented in the late 1980s, sophisticated new ingredients, devices, and techniques, from sodium citrate to sous-vide machines to spherification, have moved from elite restaurants to fast-casual service counters and home kitchens. Many people now cook with an understanding of how their efforts are transforming ingredients at a molecular level. The impact that science has had on the world of food cannot be overstated.

Except in one regard. In the midst of this epic disruption, the way we think about combining ingredients themselves has remained largely unchanged.

The old model of combining ingredients, and of pairing their unique profiles of flavors in new and creative ways, effectively shackles chefs to ingredients and flavors they already know. Historically, many chefs and ambitious foodies have the ability to flip through a mental Rolodex of ingredients, instantly recalling which ingredients and flavors work well together. This ability, the result of an intimate knowledge of food built up over years of cooking and eating, is called “taste memory.” But even the most extensive taste memory has its limitations. It’s great for cataloging what combinations have been enjoyed together in the past, but when it comes to deciding what *could* go together, taste memory is limited by the cook’s personal experience.

You could spend a lifetime familiarizing yourself with the tens of thousands of ingredients available around the world. The challenge is especially daunting given how many there are to choose from right in our own neighborhoods. Between 1980 and 2014, for instance, the number of ingredients in the average American grocery store tripled, from 15,000 items to around 44,000, according to the Food Marketing Institute. Farmer’s markets, meanwhile, are increasingly stocked with unique fruits and vegetables that many of us have never heard of before. USDA stats show that in 1994 there were 1,755 farmer’s markets across the United States alone. Twenty years later, the country had over 8,250 active markets.

All of this choice poses an acute challenge for the modern cook. Whether we’re overwhelmed by the sheer volume of our options, or simply confronting a mishmash of strange ingredients in an otherwise empty fridge, how are we to decide which of these ingredients we can put together in a way that will be interesting, and also taste good?

There are other books that list ingredients commonly use in the same dishes, but not one has ventured beyond convention to explore the untapped potential of how ingredients *could* be used together. This kind of mapping and exploration is exactly what I aim to do in the pages ahead. And rather than rely on taste memory, I’m going to rely on chemistry—specifically, on an analysis of ingredients’ shared molecular structures, one of the most exciting areas of food science today.

Over the past thirty years, researchers have changed the way we think about flavor. Food scientists have taught chefs that each ingredient has a complex network of chemical structures called volatile compounds that give that food its own unique flavor. These compounds are responsible for 80% of what we perceive when taking a bite or a sip. A simple ingredient like lettuce has about 20 compounds that combine to contribute to its flavor, while coffee has nearly 1,000.

Think of each of these volatile compounds as a single pixel in a photo. Alone, the pixel is just a single colored square—pink, say. But when it’s put together with hundreds of other colored squares, an image emerges. Create the image without the pink squares and your picture will be incomplete, no matter whether it shows a sunset, or a forest, or the inky depths of outer space.

In the same way, no single volatile compound can give us the “whole picture” of the product in which it’s found. For example, *Mesifurane*—a volatile compound prominent in strawberries—does not taste like fruit at all. It has the aroma of baked bread, butter and toasted almond. Yet if mesifurane was removed from a strawberry, the fruit’s taste would change dramatically. Almond croissants with strawberry jam wouldn’t be the same, either.

[TK image: Strawberry with bread, almond, butter, and Mesifurane compound.]

Volatile compounds such as mesifurane are the essential building blocks of flavor. So you may be surprised to learn that they have little to do with *taste*.

The words “taste” and “flavor” are often used interchangeably. But in fact they refer to fundamentally different things. Understanding the distinction is a crucial first step to mastering the art and science of pairing ingredients.

There are five—or six, depending on who you ask—basic tastes: salty, sweet, sour, bitter, umami, and fat. We perceive each of these tastes because of chemical reactions that take place on the tongue when we consume food. Chemical receptors, more commonly known as taste buds, are distributed over the tongue as well as on the soft palate, upper esophagus, cheeks, and even

the epiglottis (the piece of cartilage that covers your windpipe when you swallow). Each taste buds contains sensors for each taste and will fire signals back to the brain when ever it comes in contact with any of the five (or six) tastes. Contrary to popular wisdom, while certain parts of the tongue may be more sensitive to specific tastes, any taste can be perceived on any part of the tongue.

These tastes are only 20% of what we perceive as flavor. That's right: just one-fifth of what we experience when taking a bite of food comes from taste buds. The other 80% is reported by the nose. And whereas the taste component of flavor is created by tangible elements in our food- acids, sugars, fat and proteins; the aroma component comes from the atomic level in the form of volatile compounds.

Every bite of food contains hundreds if not thousands of volatile compounds, which I'll also refer to as "aromatic" compounds. And as their name suggests, it is the *smell* of these compounds that dictates flavor.

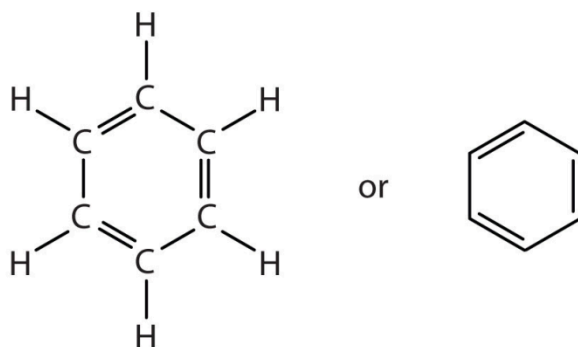
[Add sidebar: Volatile vs. Aromatic compounds]

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**Under the Microscope:
Volatile vs. Aromatic compounds**

A volatile compound is any chemical structure that has an odor. The term “aromatic compound” has a slightly different, more specific meaning: it refers to any hydrocarbon (see p. TK) that contains six carbon molecules arranged in a cyclic structure. This is also called a benzene ring or, on occasion, an “aromatic ring,” because it is the simplest aromatic hydrocarbon.



Benzene- found in ginger, mussels, eggplant and much more

Both volatile and aromatic structures play an important role in creating flavor in food. For our purposes, it is of little consequence whether a compound is technically “volatile” or “aromatic.” What matters is that they have a smell (and therefore a flavor). For the sake of simplicity, in the rest of this book I will refer to all such compounds as “aromatic.”

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Elements of Flavor: The 6 Basic Tastes

Bitter: Bitterness is the most sensitive of all tastes. This is because bitterness is the taste most likely to save your life. Nearly all toxic substances have a pronounced bitter taste. Your sensitivity to this taste functions as a safeguard against ingesting poison. With that said, some bitterness is desirable, as in coffee, tea, plants in the brassica family, olives, some nuts and cocoa.

Sour: Sourness can be an indication of spoilage or under-ripeness (and thus nutritional inferiority). Either way, sourness can warn that a food should be avoided. But, like bitterness, certain sour tastes in food can also be desirable, especially in dishes where sour can counteract richer tastes like sweet or fat. ‘Tart’ is the term commonly used to describe fruits that have a pleasantly sour taste.

Sweet: Our bodies like sweetness. When we detect it, it means that a food contains simple carbohydrates: sugars that can quickly be converted to fuel for our brain and muscle activity. The sensation of sweetness also sets off a series of chemical reactions that releases dopamine in the brain, which is why sweet treats literally make us happy.

Salt: Salt receptors play a role that reaches well beyond what is happening in the mouth. Without salt, cells can’t function properly. Yet too much or too little salt can have negative effects on our bodies, ranging from mild muscle cramps to kidney failure. Thus, as your body senses salt intake, it sets off signals to regulate fluids accordingly. Salt also “wakes up” the taste buds in your mouth, enhancing your perception of other tastes.

Umami: Our bodies’ fifth taste receptor, for the taste known as umami, was discovered around 100 years ago by a Japanese chemist, and has gained widespread recognition in America over the past 8 years. Umami (commonly translated as “pleasant savory taste”) is detected by dedicated glutamate receptors on the tongue, and while it is generally associated with monosodium glutamate, or MSG, which may be artificially added to foods, the taste is also naturally present in foods like mushrooms, tomatoes, parmesan cheese, soy sauce, and meats. .

Fat: The idea of fat as a taste, rather than simply a texture, remains a somewhat controversial topic. Fat is easily recognized in the mouth because it coats the tongue and provides a feeling of unctuousness. If this were the only way we perceived fat, counting it as a taste would qualify all textures as unique tastes, which would blur the definition of that term to the point of meaninglessness. However, recent research has found that the tongue can perceive free fatty acids, the compounds that comprise dietary fats. There has been a documented receptor cell response to fatty acids—and historically, receptor cell detection in the mouth has been the criteria for defining tastes. So while research is still ongoing, a growing number of signs point to fat as the sixth taste.

Other “Tastes”

The following “tastes” are commonly recognized as such because they are sensed in the mouth. But these “tastes” do not have dedicated chemical receptors; instead, they are sensed by the ordinary nerves in the mucous membranes inside our mouth. They are “tastes” without taste buds..

Spicy: Capsaicin, an odorless (and therefore flavorless) chemical compound found in all forms of chili peppers, creates the sensation of spiciness—a burning sensation—when it comes in contact with any mucous membranes. Though the burning sensation is first perceived on the lips and tongue, it is not a function of taste receptors; rather, it is carried to the brain by the same nerves responsible for movement and sensation in the face. The only way to relieve the burning is to wash the capsaicin off the tongue—and because capsaicin is fat soluble, whole milk always provides much more relief than ice water.

Astringency: Sometimes perceived as sourness or tartness, astringency creates a dry feeling in the mouth. The dry sensation is created in part by tannic acids, which bind to salivary proteins called mucins in the mouth, disrupting their natural lubricating properties and resulting in a rough, dry sensation on the tongue. Astringency is often associated with tea and wine, but drinks

with higher alcohol content can be astringent, as well. Tannic acids are also found in cranberry, pomegranate, walnuts, and under-ripe fruits.

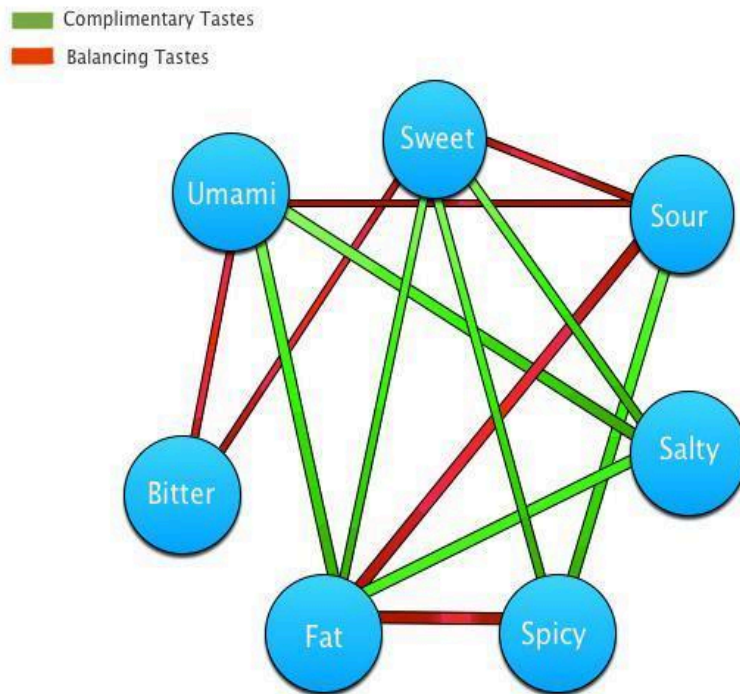
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Quick Guide: How to Pair Tastes

Any seasoned chef will tell you that the most delicious and dynamic dishes result from a combination of tastes. Controlling tastes in food is simple: If a pot of chili tastes flat, add a pinch of salt to draw out the flavors, and a dash of hot sauce to kick up the intensity. If the sauce for sautéed salmon is too buttery, a squeeze of lemon juice will cut through the fat.

There are two categories of tastes that pair well together: complimentary tastes, and balancing tastes. Complimentary tastes are the kind that accentuate one another; saltiness and sweetness are the best example. Balancing tastes oppose one another to help create balance; for instance, fattiness mellows out spiciness, and sourness plays down sweetness. The accompanying diagram maps out all of the different complimentary and balancing tastes.



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When it comes to perceiving flavor, our sense of taste would be nothing without our sense of smell. Test that statement for yourself: next time you sip your coffee, pinch your nose closed and see if it doesn't just taste like warm, slightly bitter water. Without your nose to sense the hundreds of volatile compounds, coffee, or any substance, becomes nearly flavorless.

All aromas contribute to flavor in the same way. Before a single bite enters your mouth, volatile compounds make their way to olfactory chemoreceptors in your nose and throat, bonding with these special cells, which in turn translate this chemical information into a biological signal; that new information is sent to your brain, where it is translated into the compounds' distinctive aromas. This takes place almost instantaneously, and can happen at a great distance; think about how much the aromas of burgers on a grill tell you about the food's flavor as the scent wafts through the summer air, even if you're sitting on the other side of a baseball field from where the food is being cooked.

Once the food enters the mouth, chewing releases yet more volatile compounds; over a thousand individual compounds may be found in a single bite. These signals, too, register in your nose and throat. Your brain combines this data with information from taste buds about what you are tasting.

Signals from the vibrations in your jaw and eardrums report data on a food's texture to your brain, rounding out your perception of flavor. In controlled experiments, researchers have shown that texture specifically effects your perception of salt and sweetness, but can also affect your perception of flavor more generally.

All three factors—taste, aroma and texture—help form a complete picture of flavor. But this one sense, aroma, is far more influential than the others.

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Elements of Flavor: Textures in Food

Texture has been proven to alter the way the brain perceives taste and flavor, and thus plays an overall role in the enjoyment of a meal. Just as a variety of smells, tastes, and colors can make a meal more pleasurable, so too can a variety of textures on a plate enhance a dish. What follows are the types of textures most commonly found in foods.

Creamy: Smooth or creamy textures may come in the form of vegetables, purees, gels and thick sauces. Creamy mashed potatoes, a silky butternut squash puree, even yogurt or mayonnaise lend a creaminess to a dish. Creamy textures provide richness and a pleasant mouthfeel, while the viscosity of creamy textured foods also can allow the taste and flavor of foods to linger on the palate.

Chewy (Meaty): Chewiness is often thought of as an undesirable quality, yet the chewiness of meats, grains and other vegetable are essential to plate composition. The act of chewing also helps to release aromatic compounds, which are so essential to creating flavor.

Crispy: Crispness or crunchiness is the off-speed pitch of the texture world. It changes things up and get the mouth and brain re-engaged with what they're perceiving. Crispness is a sign that something has been cooked well and the brain associates it with desirability, as demonstrated by experiments in the Cross Modal Research Lab at Oxford.

Fresh: Freshness is really a specific form of crispness. It can be the light, tender crunch of a leaf of arugula, or the hearty snap of a shaved radish. Fresh textures come from fresh vegetables, which signal wholesomeness to the brain (also demonstrated by the Cross Modal Lab) while also adding appealing colors to the plate. When raw, fresh vegetables also contain aromatic compounds that are destroyed by heat, allowing us to perceive flavors that we cannot get from cooked food.

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Despite the crucial role that aroma plays in creating food's flavor, we tend to save descriptions of it for talking about wine. A sommelier could rattle off over 100 different descriptors for the aromas found in wine. Although the same aromas, plus many more, are found in food, and although they translate almost directly into flavors, our vocabulary for *flavors* in food rarely ventures beyond the six basic *tastes*.

If we put the same thoughtfulness into categorizing the flavors we find in food as we do for the aromas in wine, our culinary experiences would be heightened immeasurably. If we gave the same care and attention to pairing ingredients that we give to pairing wine with meals, the resulting dishes could be truly revolutionary.

Of course, categorizing flavors is far more complicated than categorizing tastes. There are just so many more of one than the other. The tastes we experience are the product of a limited number of chemical receptors in our mouths, whereas the flavors we find in food are comprised of thousands of aromatic compounds. Just as you could spend a lifetime building your taste memory, you could spend a lifetime (or several) studying all the different flavors and aromas out there. Indeed, there are brilliant scientists who have devoted their entire careers to exactly that.

I am not one of these scientists. I'm just a chef who wants to understand how flavor works so I can become a better cook.

In the long term, my desire to master the science of flavor pairing, combined with my inspiring experience cooking with Watson, led me to write this book. But in the short term, that same curiosity led me down a scientific rabbit hole, as I immersed myself in articles from academic journals, entries in online chemistry databases, and the theories of the most brilliant minds in the world of molecular gastronomy. Follow me briefly while I explain the basics of what I found, and how I put this information together for your benefit in this book.

Because I am a chef, not a scientist, I allowed myself to make some curatorial choices about how to organize the vast number of aromatic compounds into manageable categories, and measure these categories' presence in a range of different ingredients. But I set about doing this as rigorously as anyone could. I first delineated the different categories of aromas we find in food, and linked them to the aromatic compounds that create these smells. I did this by using the taste memory I had gleaned as a professional chef and culinary educator working with a wide range of ingredients, while also drawing extensively on the latest research findings about which aromatic compounds are found in specific foods.

In order to connect ingredients' aromas with the molecular structures that cause them, I first had to broadly categorize the different types of aromatic compounds that are most commonly found in food. (You can see these different categories of aromatic compounds in the spread on page [TK page].) This is harder than it sounds; each group of aromatic compounds I identified actually contains thousands of different compounds with a wide range of scents. Assigning a specific aroma profile to any group of volatile compounds with complete accuracy is impossible. Some sort of editorial judgment is needed—but that seems right and proper for a book written by a human chef instead of a computer.

An additional complication for this phase of my research was the fact that the structure of a compound is not always predictive of its scent. For example, the compound carvone exists in two isomers- R and S. The structures are exact mirror images of one another. The R isomer of carvone has an intense spearmint aroma and is the main flavorant in *Wrigley's* famous chewing gum. The S isomer of carvone has the aroma of caraway seeds. Despite their identical structures, these two compounds have distinctly unique aromas.

[TK image of R and S isomers of Carvone]

Although the chemical structure of a given compound won't necessarily tell us what it smells like, the properties of specific *groups* of aromatic compounds can give us some clues about their aromas and their function. This is because 'functional groups' are the main distinguishing factor for most groups of aromatic compounds. Functional groups are the molecular structures attached

to carbon and hydrogen atoms make a compound aromatic. When these functional groups are detected by special proteins in the nose and throat, those sensors signal specific scents to the brain. For example, thiols (sulphur-containing functional groups) are stinky, at high concentrations, they smell of rotten eggs. Functional groups are the basic division of aromatic compounds as found in any chemistry textbook.

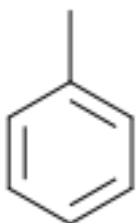
After understanding functional groups and their basic aromatic characteristics, I next classified the smells of food that are determined by these aromatic compounds as specific **primary aromas** and, when applicable, any notable subcategories of these aromas. (The final list of aroma groups I used is on page [TK page].)

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Elements of Flavor: Aromatic Compounds

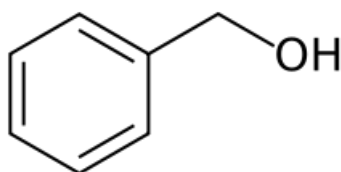
Aromatic Hydrocarbons (or Arenes)



Methylbenzene (Toulene), an aromatic hydrocarbon, is found in beer, annatto seed, and ginger

Aromatic hydrocarbons are molecules made of hydrogen and carbon atoms linked together in a circular formation, known as an aromatic, or benzene, ring. Aromatic hydrocarbons tend to have solvent (paint thinner) or sweet aromas, depending on the individual. They include benzene (solvent), Toulene (paint, solvent), Styrene (balsamic, rubber) are aromatic hydrocarbons that may be found in everything from garlic to whisky.

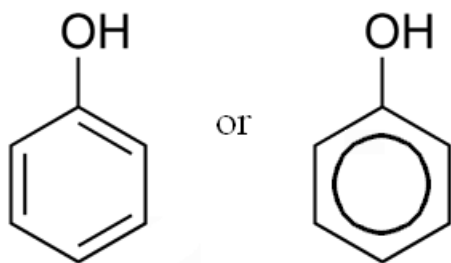
Alcohols



Benzyl Alcohol= found in apricots, clove, chicken, and mustard

In the context of aromatic compounds, alcohols refer to aromatic (benzene) rings with an attached hydroxyl (OH) group. This is a broad category that contains some well-known scents; examples of specific compounds include menthol (peppermint), linalool (floral and spicy), and ethanol (rubbing alcohol).

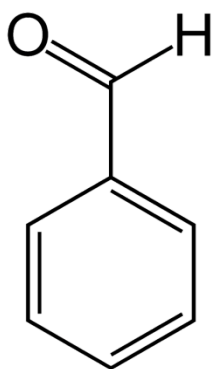
Phenols



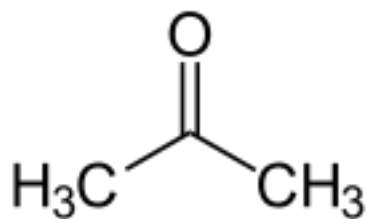
Phenol (*hydroxybenzene*)- found in cinnamon, coffee, and sake

Like alcohols, phenols are also hydrocarbons with an attached hydroxyl (OH) group. Because of the way the oxygen is bonded, phenols tend to have a higher acidity than alcohols. Many phenols have an astringent, bitter taste. You might know phenols as the bitterness in raw walnuts, cranberries, or kale. The phenols that most people know best are the tannins found in wine and tea. Other phenolic compounds range from Eugenol (the main odorant in cloves) and Thymol (thyme) to Raspberry ketone (raspberries) and Guaiacol (smoky flavor found in coffee, whiskey and smoke).

Carbonyls



Benzaldehyde (*aldehyde*)- imitation almond flavor

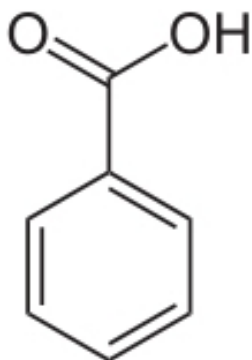


Acetone (*ketone*)- found in citrus, milk,

tomato

Carbonyls are hydrocarbons attached to oxygen molecules. These compounds are divided into important aromatic groups called **Aldehydes** and **Ketones**. Aldehydes and ketones create some the most recognizable scents in food: vanillin (vanilla), benzaldehyde (imitation almond flavoring), carvone (spearmint extract), and cinnamaldehyde (cinnamon extract).

Acids

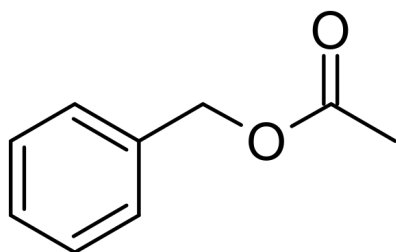


Benzoic Acid- found in honey, nutmeg, and thyme

Aromatic acids are acids with an attached aromatic or benzene ring. Their aromas vary from sour to grassy to buttery. Most aromatic acids are derived from phenols, aldehydes, or amino acids.

Acids play a role in a variety of scents, like butyric acid (vomit and rancid butter), acetic acid (vinegar), and cinnamic acid (cinnamon).

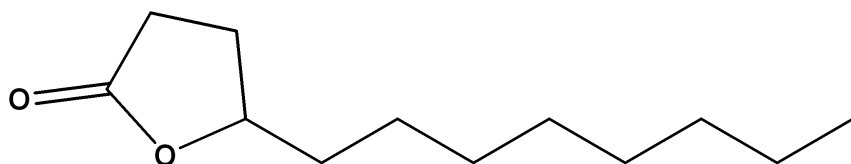
Esters



Benzyl Acetate- found in pear, blueberry, jasmine

Esters, organic compounds derived from acids, are found in the essential oils of plants. The vast majority of esters have strong fruity aromas, though some smell of pine or spearmint. Isoamyl acetate (banana), methyl butyrate (apple), octyl acetate (orange) and allyl hexanoate (pineapple) are common esters.

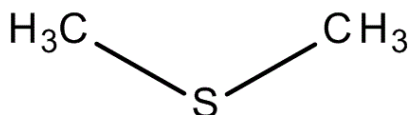
Lactones



γ-dodecalactone- found in strawberry, mushroom, chicken, and pork

Lactones are esters that contain a cyclic structure. They are used commonly in the flavor and fragrance industry. Most lactones have the intense fruity aromas like coconut, peach, and apricot. Lactones are largely responsible for the tropical scent of your favorite suntan lotion. In food they are found in tropical fruits, strawberries, mushrooms, chicken, pork, and milk products. Lactones found in food include δ-decalactone (apricot, coconut), (Z)-dairy lactone (peach, sweet) and γ-dodecalactone (apricot, peach, flower).

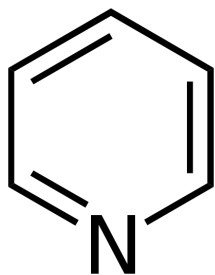
Sulphur Compounds



Dimethyl Sulfide- found in egg, asparagus, and field pea

For most, sulphur immediately brings to mind the not-so-appealing scent of boiled eggs. However, sulphur compounds, like dimethyl sulfide (cabbage, sulphur) and allyl methyl sulfide (garlic), play an important role in food items from alliums (onions, garlic, and chives) and brassicas (cabbage, cauliflower, broccoli, and brussels sprouts) to citrus fruits and cheeses. Sulphur compounds are also part of the “roasted” aroma found in chocolate, coffee, baked bread, popcorn, and roasted meats.

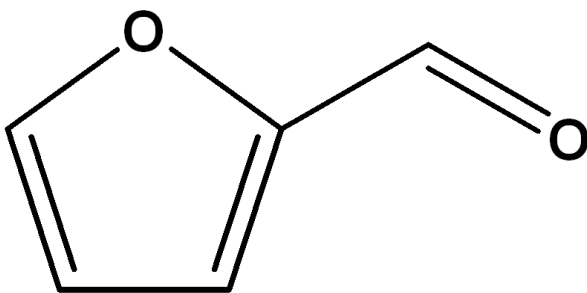
Bases



Pyridine- found in roasted meat, potato chips, bonito flakes

Aromatic bases are compounds that contain a nitrogen atom in an aromatic ring. The aroma of these bases can range from earthy and fresh cut grass aromas to cocoa. Pyridines and pyrazines, bases that are products of Maillard reactions (see p. [TK]), are the most common of the category. Compounds like 2-acetylpyridine (popcorn, roasted nut) and vinylpyrazine (roasted) contribute to the distinctive smell of roasted meats, toasted nuts and coffee.

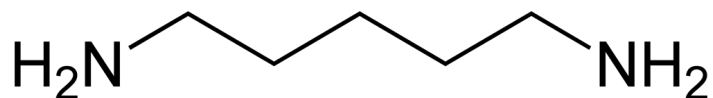
Furans



Furfural- found in beef, coffee, rum, tamarind, vanilla

Furans are molecules with an aromatic ring made up of 4 carbon molecules and one oxygen molecule. They tend to have a nutty aroma, and are a by-product of Maillard reactions. Furfural is one of the most common furans found in food and has the aroma of baked potato, almond, and baked bread; 2-ethylfuran has the aroma of butter and caramel, and is found in cooked meats and nuts.

Amines



Cadaverine- found in blue cheese, fish, and sauerkraut

Derived from ammonia, amines have non-cyclic structure and tend to have strong, often unpleasant scents. Putrescine (named for its putrid smell) and cadaverine (scent of rotting flesh) are both amines. As foul sounding as these compounds may seem, they are also found in fresh fish, beer, mushrooms, cheese, and wine. Remember, each compound is just a pixel in a larger picture, these compounds are not individually distinguishable in any of these products but they do contribute to the pleasant scent of each. For instance, the amine called indole has a fecal and mothball aroma, but it is an essential component in the smell of jasmine and contributes to the pairing of pork liver and jasmine.

[End dedicated section]

[Begin dedicated section]

Elements of Flavor: Primary Aromas

Floral- Floral aromas and fruit scents are very closely related because all fruit-bearing plants produce a flower at some point in their life cycle. Floral scents tend to be more “perfume-y” and lack the sweet notes of fruits. These scents range from orange blossom to rose to lavender. Floral aromas are most commonly found in fruits, spices, herbs and alcohol. For example, beta-damascenone, a key odorant in roses, can be found in apple, citrus, lemon balm and sage; cis-ocimene, found in Lilac, also plays a role in the flavor of caraway, basil, nutmeg and thyme.

Fruity: This is one of the broadest categories of scents. Fruity aromas include:

- **Citrus:** Citrus’ scent, in its purest form, is found in essential oils in the zest of lemon, lime, orange, grapefruit, and other citrus varieties. Compounds that create citrus aromas like limonene, myrcene and citronelle are also found in beer (via hops), olives, many herbs (basil, sage, mint, tarragon), ginger, fennel, and spices.
- **Tropical:** Lactones are the aromatic compounds that are found naturally in mango, pineapple, coconut, melon, and bananas. These same compounds are added to suntan lotions and mixes for blended rum drinks to create tropical aromas. These compounds can also be found in mushroom, pork, chicken, and coffee.
- **Melon:** Melons are some of the most highly aromatic fruits. Scents in melon can range from sweet and fruity flesh to rinds that smell of must, musk or rot. Among the many compounds responsible for the scent of melons, esters and aldehydes are prominent in offering aromas of fruit (melon and apple) and green (cucumber, cut grass). Similar compounds are found in cheese, citrus, rum, and tropical fruits.
- **Tree fruits:** This is the group of scents associated with flowering tree fruits like apple, pear, peach, cherry, apricot and fig. Most tree fruits are members of the Rosaceae family, meaning these fruits have similar DNA. Esters and lactones are the compounds most

common in these fruits. Tree fruits aromas are also found in cocoa, olive oil, vanilla, anise, and cumin.

- **Berry:** This is the sweet smell associated with strawberries, blueberries, raspberries, et al. Berry aromas which come from carbonyls and esters like raspberry ketone and ethyl formate are also found in jasmine, grapes, wine, tea, tree fruits, cooked rice, and bourbon vanilla.
- **Dried fruits or jams:** These processes offer a double layer of aromatic compounds, first from the fruit itself (esters, lactones), which is then layered with the byproducts of the drying or cooking process—like β -phellandrene, vanillin, and 3-ethylphenol. which create aromas of leather, smoke, and spice. Many of these compounds are also found in rosemary, cinnamon, mint, and buckwheat.

Vegetal: Aromas found in vegetables break down into three main categories:

- **Green:** Green is a common descriptor for wines made from under-ripe fruit. In food, green refers the crisp, somewhat tart aroma of green apple, bell pepper, or grass. As in wine, the scent is common in under-ripe fruit and green leaves, although the compounds responsible for ‘green’ (or grassy) aromas—(Z)-3-hexenal and (E)-2-hexenal—are among the most commonly found compounds in all foods. They can be found in most herbs and green vegetables, as well as in tea, tomatoes, bananas and cardamom.
- **Herbaceous:** This vegetal aroma occupies the blurry scent space between green and floral. Woody herbs like rosemary, thyme and sage belong in a different category of aromas (the terpene class); in the herbaceous category, by contrast, we find “softer” herbs like parsley, dill, chive, clover, or tarragon. Herbaceous aromas can also be found in hay and cereals like oat, bran, and dried corn. Herbaceous compounds like thymol (thyme), estragole (basil), and 1,8-cineol (bay leaf) are also found in tropical fruits, mustard, chicken, and cumin.

- **Earthy:** Earthy aromas are often described as smelling of dirt, the forest floor, mushrooms, or mold. On the more complex end of the spectrum, these aromas also have notes of tobacco or leather. When it comes to ingredients, such smells—and the aromatic compounds that create them—are typically found in mushrooms, beetroot, truffle, asparagus, and malt. Among the compounds that create earthy aromas are pyrazines and pyridines, the same compounds that form the basis for the aromas of caramel, roasted meats, and nuts.
- **Fruit-Like:** Fruit-like aromas come from mainly from vegetables that are technically classified as fruit. Many have a slightly sweet aroma, or simply lack the earthy and green aroma of most vegetables. The primary vegetables in this category are tomato, pepper, pumpkin, avocado, eggplant and corn. Major compounds found in this category include 3-methylbutanal (tomato), 2-isobutyl-3-methoxypyrazine (capsicum) and Hexanol (eggplant).

Terpene: The name of this large group of strong smelling aromatic hydrocarbons is derived from turpentine. Terpenes are described as smelling of resin, turpentine, or plastic. Found in the essential oils of a variety of plants and flowers, terpenes help create a wide range of flavors. There are over 400 naturally occurring terpenes.

- **Wood:** These scents immediately conjure thoughts of holidays—fresh cut pine, spruce, or fir. They are similar to resinous scents but have an added “fresh” or “green” aroma. They are typically found alongside pungent and/or spice scents. They are common in rosemary, sage, and ginger. They are also found in lavender, dill, fish, and berries.
- **Spice:** This category of terpenes describes the notes created by specific spices like black pepper, cinnamon, anise, nutmeg, allspice, and vanilla among others. There are thousands of aromatic compounds that create these aromas, and they can be found in most foods, although they are most prominent in spices, herbs and aromatic vegetables like black pepper, cardamom, turmeric and cinnamon.

- **Medicinal:** Medicinal aromas are sharp stinging scents, with notes of mint. Their aromas are most associated with camphor, solvent, or mothballs. On their own, they can be somewhat unpleasant, but like all terpenes, these are very common “background” scents and can be found in caraway, basil, and licorice
- **Petrol:** Petrol describes another set of terpenes associated with scents that are ostensibly undesirable in food, like gasoline or kerosene, but which can play an important role in the flavor of certain ingredients, such as herbs like oregano, marjoram, and epazote, as well as hyssop, camomile, coriander, and citrus.
- **Smoke:** These aromas are close to those of another terpene category, wood, and they often originate from wood itself. As aromatic compounds in wood break down, they created compounds called methoxyphenols, or guaiacols. These compounds make the smoky aromas that result from barrel aging, smoking, charring or roasting. They are found in significant concentration in vanilla whiskey, scotch, aged wines, and roasted meats or vegetables, and to a lesser extent in coffee, pork, mushrooms, corn, and honey.

Maillard: Louis-Camille Maillard was the French chemist who first identified the reactions that occur between amino acids and reducing sugars in browning processes. The compounds generated in these reactions are responsible for the desirable flavors in baked bread, roasted meats, and many other foods. While all aromas in this category are boosted by Maillard reactions, each has distinguishing characteristics.

- **Meaty, Caramel, Toasted/Roasted:** Maillard reactions create hundreds of aromatic compounds including pyrroles (nutty), pyrazines (roasted), pyridines (burnt), and furans (caramel, nutty). Furfural and hydroxymethylfurfural are often pointed to as the most important compounds created in Maillard reactions. These same compounds are found in plums, coconut, popcorn, rum, and shrimp.

- **Nutty:** A somewhat broad category of aromas, “nuttness” includes—predictably—the smell of all nuts. Specific scents can vary depending on variety. The distinctive scent of almonds comes from a single compound. Check the label on a bottle of almond extract and you will find benzaldehyde is the only ingredient listed. Benzaldehyde, an aldehyde, is found in most nuts, and also in apricot, cherry, lemon balm, hops, corn, and sage. Some toasted aromas fit into this category as well.
- **Chocolate:** Chocolate, one of the most complex aromas, is made up of nearly 700 different aromatic compounds. The exact aroma of chocolate is determined by variable factors including the variety of the cocoa bean and growing conditions as well as processing (particularly what the ground beans are mixed with). However, the recognizable “chocolaty” notes are the Maillard reactions associated with the roasting of the cocoa beans. The significant aromas come from aldehydes including 3-methylbutanal and vanillin as well as pyrazines. Many of these compounds can also be found in meats, as well as tamarind, tomatoes, cardamom, and eucalyptus.

Sulphur: The smell of sulphur is typically associated with hard boiled eggs. The strong smell of eggs is often a result of overcooking; as the egg cooks, amino acids break down and release sulphur. This smelly reaction is also responsible for the greenish-ring around the egg yolk. While the strong smell of sulphur is not desirable on its own, compounds like dimethyl disulfide, hydrogen sulfide, and benzothiazole play an important role in the flavor of brassicas, alliums, corn, cheeses, and truffle.

Pungent: This descriptor is often used to describe any strong, sharp scent, but it is used more accurately to describe the aromas associated with spiciness or heat in food. Mustard, horseradish, and wasabi are typical pungent ingredients. (Chili peppers do not fit in this category, as their aromas tend to be more fruity and vegetal. Capsaicin, the compound responsible for heat in chiles, is odorless—and hence flavorless.) Pungent aromas are partly based on sulphur compounds and can be found in turnip and rutabaga, brussels sprouts, arugula, watercress, radish, and broccoli.

Phenol: Phenol is a specific aromatic hydrocarbon, but it is also a commonly used descriptor for scents. In high concentration, phenolic scents are harsh and overwhelming, smelling of tar or gasoline. In lower concentrations, their smell can be identified in aged red wines, the peat smell of scotch, or the “tang” in pomegranate juice.

Savory: The one category of flavor that does not rely on aromatic compounds because Glutamate (umami) is odorless and sensed only on the tongue. However savory flavors are essential to the enjoyment of food and have been shown as effective salt replacers. These flavors are associated with cooked tomato, roasted mushroom, Parmesan cheese, seaweed, fish sauce and soy sauce.

Sour: Sour smelling compounds can “sting” the nose in high concentrations, like taking a big whiff of vinegar. Sour aromas come from acids and are common in dairy products, especially as they begin to age—in both good and bad ways. Soft-ripened cheeses, sour cream or spoiled milk are all associated with sour. They are also found in cocoa, coffee, alcoholic beverages, and bread.

Dairy: All dairy products have a similar aromatic profile at their base. After all, yogurt, crème fraîche, cheese, and even butter all begin as milk or cream. Aromatic acids play a major role in the flavor of dairy products, giving even the freshest dairy products a pleasant slightly sour smell. The distinctive aromas of dairy are also prominent in corn, pistachio, coconut, and whisky.

Marine: The typical aromas of seafood are best described as marine. ‘Fishy’ has a negative connotation as an aroma descriptor, and in reality seafood should never smell ‘fishy.’ Rather, it should smell simply of the sea. Sulphur compounds and amines (nitrogen-containing compounds) are most common in this group. They are found in lean and fatty fish, mollusks and crustaceans.

[End dedicated section]

[Main text continues]

In addition to categorizing the main groups of flavors and their underlying aromatic compounds, I also pinpointed the different levels of aromatic compounds in roughly 120 of the most common ingredients in the modern kitchen: from groups of ingredients such as alliums (which includes onions, garlic, chives and ramps) to specific, unique ingredients such as truffle and vanilla. Research journals contain detailed analyses of the hundreds of different aromatic compounds found in each of the ingredients, but I found the bulk of this data using the Volatile Compounds in Food (VCF) database, managed by Triskelion B.V., an independent research facility located in the Netherlands. The VCF database is the most comprehensive listing of aromatic compounds in foods available in the world today. I supplemented the information I found there with scientific papers on aroma research and volatile compounds.

[Add sidebar: Flavor Matrix Vocab 101]

Once I completed this process, I was left with spreadsheet upon spreadsheet cataloging the significant aromatic compounds in each of the ingredients that I wanted to study. Finally, I could compare them to each other.

That comparison boiled down to a single number: the percentage of aromatic compounds two different ingredients (or groups of ingredients) have in common. According to a relatively recent culinary innovation called “flavor pairing theory,” this simple metric can allow us to judge the relative compatibility of ostensibly dissimilar ingredients. Specifically, flavor pairing theory (which, like all great theories, is elegantly simple) tells us that if two ingredients share significant aromatic compounds—either in number or concentration—they will likely taste good together when combined in a dish. For you sports fans, consider this the analytics of flavor (and flavor pairing theorists like me are your Billy Beans).

Flavor pairing theory originated in 1999 with the pioneering chef Heston Blumenthal and his research team at The Fat Duck, a restaurant in Bray, England with its own dedicated research kitchen (which more like a lab than most kitchens you’ve seen). Blumenthal and his colleagues

stumbled upon their theory while experimenting with salty and sweet foods. They combined caviar and white chocolate, discovering to their surprise that the pairing tastes delicious.

Flavor pairing theory can be used to discover wild new combinations like white chocolate and caviar, or help explain old favorites, like pizza. Tomatoes, mozzarella, Parmesan, and baked wheat all have over 100 compounds in common—among them *4-methylpentanoic acid*, is one of the most significant, with its pronounced floral aroma. All of those shared compounds are the flavor equivalent of an orchestra playing together in perfect time. Each compound has a unique aroma that it contributes to the harmonious flavor of pizza, in much the same way that a tuba and flute make different sounds but both contribute to the full, rich sound of a Beethoven symphony.

[TK image: Pizza and methylpentanoic acid compound]

According to flavor pairing theory, finding the percentages of compounds shared among ingredients would allow me to judge how well certain ingredients pair together relative to others—but generating this percentage was only the next-to-final step in my research process. The last one involved organizing this information in a way that would be quickly and easily apprehensible—that you could use, at a glance, in your own kitchen.

I have consolidated all of my findings about flavor pairings into what I call a “flavor matrix,” making one for every ingredient or group of ingredients that I studied. The flavor map is a multi-layered pie chart that looks a bit like an artist’s color wheel. Instead of showing how primary colors can be blended together, however, flavor maps show how primary aromas combine to give each ingredient its unique flavor. Each flavor map essentially is the aromatic fingerprint of the ingredient it depicts; it reveals the true identity of that ingredient, and no two are exactly the same. Crucially, it also reveals which other ingredients pair best with the featured ingredient, based on its aromatic identity.

As I’ve compiled this information and have written up the results, I have been fortunate to have an excellent collaborator and partner in crime: my wife and creative partner, Brooke Parkhurst. Brooke is the person responsible for keeping everything in order. That responsibility extends far

beyond this book: From my wandering thoughts and technical jargon within these pages, to the lives of the three people (myself and our two children) who depend on her for everything, Brooke keeps us on track. As I spent over a year buried in chemistry tables and scientific papers, Brooke was always there to keep me tethered to the real world. She has been my translator throughout this process, from fixing my (excessive) typos to rearranging my thoughts so that everyone—experienced chefs and cooks alike—can understand and learn from them. This book would not have been possible without her.

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[Begin sidebar]

Quick Guide: Ingredient groups

You may come across some unfamiliar words in this book. That is because in certain cases it was advantageous to view a group of ingredients by their higher classifications. Many ingredients have nearly identical flavor compositions, a quirk that allows us to categorize them as a group rather than as individual ingredients. For instance, if you can see past the bitterness of one and the sweetness of the other, you will find that cranberries and blueberries have essentially the same flavor. Therefore, throughout this book I'll refer to cranberry and blueberry by their genus: *Vaccinium*. I've done this sort of condensation with the following classifications:

Allium: A group of aromatic plants that derive their flavor from sulphur compounds. The allium group includes: garlic, onion, shallot, leek, scallion, chive and ramp. For more information on Allium see page [TK].

Capsicum: The entire category of peppers, from mild bell peppers to spicy chilies. For more information on Capsicum, see page [TK].

Laurel: The leaf of the laurel tree, commonly known as bay leaf.

Southern Pea: Also known as field peas, this group of ingredients includes black-eyed peas, lima beans, crowder peas, zipper peas and more.

Vaccinium: The name for a genus of shrubs that grow a variety of culinary berries. Berries classified under *Vaccinium* are cranberry, blueberry, bilberry (or whortleberry), lingonberry (or cowberry), and huckleberry. For more information on *Vaccinium* see page [TK].

[End sidebar]

[Main text continues]

[Section break]

HOW TO USE THIS BOOK

Our goal in this book is twofold. We want to teach you about the science of flavor, as we hope we've started to do already. We also want to help you master it, and put it to use in your own kitchen.

In our home, Brooke and I have put the flavor pairing theory into practice with the help of hard data about aromatic compounds in food. It has revolutionized the way we cook. We want you to experience this revolution for yourself.

In the pages ahead, we will present you with a series of tools that will allow you to determine which ingredients pair best with one another, based on the science and theory of flavor pairing. Every four pages will focus on a new ingredient or group of ingredients, covering over 120 common ingredients in all.

The centerpiece of each section is the flavor map, an infographic containing all of the data about a given ingredient's aromatic compounds and compatible ingredients. At the center of each map—really, a pie chart—is the featured ingredient itself. Surrounding that ingredient are all of its primary aromas, each of which is represented by a “slice” of the pie chart. The more significant an aroma is to the overall flavor of the ingredient, the wider and larger that section will appear in the map. For example, in the olive flavor map, citrus is one of the biggest “slices” of the pie, stretching wider than almost any other flavor group. It also extends further from the center of the map than any other section. This is a reflection of the fact that olives share 60% of their aromatic compounds with citrus fruits. Indeed, of all ingredients, olives have the strongest affinity for foods with citrus flavors—a characteristic that is visualized on the flavor map.

[TK image: Flavor map “key,” with sections labeled]

Around the perimeter of the flavor map you will find all the ingredients that pair well with the featured ingredient. Ingredients are listed by their corresponding aroma (which, for the purposes

of the flavor matrix, I will often refer to simply as their “flavor”). For instance, on the Olive matrix, you’ll see that basil, lovage, parsley and tarragon extend further from the center of the matrix than other herbs, indicating that these secondary ingredients are the best choice to pair the primary ingredient, olive.

A quick glance at the olive flavor map will give you some ideas about specific ingredients to pair with olives. If you want to mix and match, however, it’s worth keeping in mind that some flavor groups go better together than others. To help guide your choices about combining multiple secondary ingredients with the primary one, I’ve created a **flavor pairing guide** (p. [TK page]). This tool will help you create more complex combinations of ingredients by matching three or more ingredients together.

The true value of the flavor maps is not to simply show which ingredients pair well. Flavor maps can also teach you how to *think* about flavors in food. They will help you to understand ingredients and flavors that complement one another, allowing you to pair foods and create dishes in new and innovative ways. For example, the fact that olives are great with citrus like lemons, orange and grapefruit also tells us that olives pair well with the *flavor of citrus*—meaning that coriander, passion fruit, Szechuan peppercorns, lemongrass, and sorrel could be incorporated to dishes with olives if you’re all out of lemons (or even if you aren’t).

As this example demonstrates, strong pairings in the flavor maps signal more than the mere fact that two ingredients will taste good when combined in a dish. It means there is a connection between the ingredients, one that is not always apparent. It shows that the ingredients have something in common. Maybe they are native to the same area, or have a botanical relationship or a similar flavor profile. These shared characteristics help to explain flavor compatibility, but they can also help you to be more flexible and creative in your use of different ingredients.

To take another example, there is a lot of aromatic overlap between chicken and pork. They make a strong pairing, but that doesn’t mean that these ingredients necessarily need to be combined in a dish. It is also a signal that they are similar, and easily interchanged in recipes.

Look at these pairings with a critical eye, and you will begin to understand the way that flavors and tastes are constructed—an understanding that will make you a more creative cook.

Alongside the Flavor Map in each section, I'll also list some quick facts about the featured ingredient: where it comes from, what common varieties you're likely to encounter, and what prominent aromas it has. I'll also give you a "best of" list of the other ingredients that pair best with the featured ingredient.

These pairing charts not only list recommended matches between ingredients; they also note the specific aromatic compounds that explain these matches, identify the aromas associated with each compound, and highlight additional ingredients that also contain those compounds. You can also find these basic pairing suggestions in the flavor map for olive, but by highlighting the molecular commonalities that explain certain pairings we hope to add another layer of discovery to your use of this book. (To show how we do this, I'll use another example from the olive section: The compound *b-caryophyllene*, found in the essential oils of most plants, has the aroma of fried spice and wood. *Myrcene*, which is found in thyme and parsley, has the aromas of balsamic, flower, fruit, herb, and must. These two compounds can be found together in olive and star anise. You'll also find both compounds in cinnamon, cilantro, fennel, ginger and turmeric, among others.)

Lastly, in each section you'll find an original recipe that we have created using the information embedded in each flavor map. For instance, in the olive section, you'll find instructions for using this classic hors d'oeuvre ingredient to make a unique bookend for any meal: a dessert of lemon curd with olive oil and crunchy olives. Sweetened with honey and dehydrated in the oven for an extra-intense flavor and a delightfully crispy texture, the olives in this dish will blow your mind. But even if you only use the recipes for inspiration about how you can incorporate the data in each flavor map into your own cooking, our creations will have done their job.

Our hope is that, once you've gotten comfortable with these flavor maps, you'll be able to use them at a glance to find new, striking ways to use common ingredients in your own cooking. And who knows: when you discover that shared flavor compounds make (for instance)

blueberries and horseradish an unexpectedly delicious pairing, and that these same molecules connect horseradish to pork, a roast pork sandwich with blueberry-horseradish jam may become your new favorite meal. See how many different dishes you can create with knowledge like this, and most importantly, enjoy the ride—wherever it takes you.

[Begin dedicated section]

Quick Guide: How to Pair Flavors

This page will be a handy reference tool as you utilize the flavor maps in this book. Flavor maps show you flavors in specific ingredients and highlight other ingredients they pair with; the flavor pairing chart that you see here will help you to create more complex combinations of ingredients by finding which *flavors* pair together. Just as the taste pairing guide on page [TK page] indicates that sweet and salty tastes make a great match, this flavor pairing guide shows that (for instance) citrus and sulphur are complimentary flavors.

[TK image: Flavor pairing guide]

Here's an example of how you could put the flavor pairing guide to use: In the olive section on p. [TK page], you might look at the flavor map and notice that olives and peaches or nectarines are an especially strong pairing. If you want to build a dish using more than these two ingredients, you'll want to find additional ingredients that pair best with these two. The flavor pairing guide tells us that Tree Fruit flavors (the flavor group where artichoke is located) pair nicely with Floral flavors. If you have another look at the olive flavor map, you'll see some ingredients in the Floral section that you could add to your olive-peach dish. The results could be any of the following combinations:

Olive + Peach + Szechuan Peppercorn + Lavender + Corn
Olive + Nectarine + Fennel Seed + Orange Blossom + Yogurt
Olive + Peach + Juniper + Camomile + Lamb

[End dedicated section]

Part II

Ingredients

Allium

Allium, a genus of flowering plants, includes a range of sulphur-containing aromatic plants. The main distinguishing factor between all of the plants is the concentration of sulphur compounds they contain. These plants in order of most pungent (highest concentration of sulphur to least are: Garlic, Onion Shallot, Leek, Scallion, Chive. In a direct comparison onions and garlic showed very similar pairings with garlic presenting slightly more Green and Woody aromas than onions. When selecting which Allium to use in a dish, the desired intensity of flavor can be a good guide (using the scale above).

There are approximately 126 different cultivars of of Onion. Common market forms of onion include yellow onion (most common), white onion (milder than yellow), and red onion (strongest flavor when raw, sweeter when cooked). Sweet onions, named after the areas they are grown (Vidalia, Maui or Walla Walla), have a milder onion flavor and slightly sweet taste.

With long exposure to heat—as in caramelizing onions or roasting garlic—Alliums lose their strong, pungent flavor for a more mellow roasted, sweet taste. Through the cooking process, heat takes the starches that make up the structure of the cell walls of Alliums and tears them apart, breaking the starches down into simple sugar molecules. These sugar contribute to the Maillard process, causing browning and creating a sweeter taste. These roasted, Maillard aromas overtake the sulphur compounds to become the dominant flavor in caramelized onions or roasted garlic.

Scallions and spring onions are regular onion varieties that are harvested before a bulb forms.

Shallots look like mini onions, though they are a different species in the onion family. Shallots are fall between red and yellow onions on the intensity scale.

Leeks, like shallot, are a separate cultivar in the Allium group. Leeks have the mildest flavor in the Allium group (apart from Chives)

Ramps, or Wild Leeks, are also a member of Allium, but only come into for a short period in the Spring in North America. Ramp flavor is often described as a cross between garlic and scallion.

Growing region Northern Hemisphere

Common varieties (in order of flavor strength)

Garlic, Onion, Shallot, Ramps, Leek, Scallion, Chive

Taste bitter to astringent

Main Aromas

Dimethyl sulfoxide- garlic

dipropyl disulfide- cooked meat, garlic, onion, pungent, sulphur

methyl propenyl disulfide- leek

pentanal- almond, bitter, malt, oil, pungent

Best Pairing Brandy, Vinegar, Wine, Citrus, Toasted Nuts and Cheese

Substitute all varieties of Allium may be substitute for one another

Pairing	Common Compounds	Aroma	Also Shared
Garlic + Apple	acetone	chemical, ether, nauseating, pungent	avocado, carrot, ce ginger, truff
	aniline	pungent	beet, buckwheat, f radish
Leek + Cocoa	dimethyl trisulfide	cabbage, fish, onion, sulphur, sweat	beef, beer, mustard sesame
	4-methyl-5-vinylthiazole	cocoa, fat, nut, popcorn, roast	hazelnut, passion f tequila
Garlic + Honey	dimethyl disulfide	cabbage, garlic, onion, putrid	cauliflower, cheese, parsley, rad
	allyl alcohol	pungent, mustard	brandy, brussels, tomato

Garlic Honey

Serves 4

Garlic honey is something that we discovered from a fantastic honey producer at the Union Square Farmers Market many years ago. It sounded so odd, we just had to grab a sample and were blown away by how delicious that unexpected combination turned out to be. In working on this book we began to see the prominent sulphur aromas that we knew in garlic were also prominent in honey and it reminded us of this delicious honey we enjoyed so many years ago.

4 oz raw honey

1 tablespoon minced garlic

½ teaspoon red pepper flakes

Combine the honey, garlic and pepper flakes in a small jar and mix well. Warming the honey slightly will make this process easier. Place the honey in the jar first, tighten the lid and drop the jar into a bowl of hot tap water for 10 minutes. Stir the garlic and red pepper flakes into warm and replace the lid store at room temperature 3-5 days before using.

Use the honey to drizzle over cheese and bread, or in marinades for vegetables or meats before cooking.

Artichoke

The flavor of cooked artichokes is dominated by earthy aromas. Sulphur, herb and citrus aromas contribute to the flavor as well, though they are less detectable to the nose.

Sulphur, citrus and maillard flavors make for some of the best pairings with artichoke.

Artichokes have an interesting ability to make other foods taste sweeter. Cynarin, a compound naturally present in artichokes, inhibits sweet-perceiving taste buds. This means that when a bite of food washes away the cynarin, your brain interprets the re-activation of taste buds as a mild increase of sweetness.

Growing Regions

Artichokes are native to the Mediterranean and the majority of their production is still based there. Buds of the artichoke begin to mature in late summer and continue through fall, though young artichokes are also harvested in Spring. This means that there are two opportunities a year to enjoy what many cooks and consumers consider to be an aphrodisiac and a delicacy- March through May and September through November.

Common Varietals Green Globe, Purple, Spiny

Taste Bitter, lightly sweet

Main Aromas

1-octen-3-one- earth, metal, mushroom

1-hexen-3-one- cooked vegetable, green, metal

decanal- floral, fried, orange peel, penetrating, tallow

Best Pairings

citrus, mint, basil, mushroom, sesame, wine, cheese, cocoa

Substitute Jerusalem artichoke, hearts of palm, chayote, asparagus

Pairing	Common Compounds	Aroma	Also Shared With
Artichoke + Plum	eugenol	burnt, clove, smoke, spice	pork, savory, tarragon, anise, ginger
	benzyl alcohol	boiled cherries, moss, roasted bread, rose, sweet	chile, date, mint, radish, walnut
Artichoke + Milk Product	furfural	almond, baked potatoes, bread, burnt, spice	beef, cherry, mustard, potato,
	diacetyl	butter, pastry, rancid, yeast	olive, peanut, pea, tomato, vinegar
Artichoke + Sesame	2-acetylthiazole	nut, popcorn, roast, sulphur	beer, clam, corn, shrimp
	heptanal	citrus, dry fish, fat, green, nut, rancid	allium, coriander, fish, lamb

Tahini Vinaigrette

Makes approximately 1 cup

This dressing, inspired by the unique pairings of artichoke, is a balance of toasted, savory and tart flavors. It's the perfect complement to artichokes but is equally delicious on asparagus or seafood, especially grilled fish, shrimp, or clams.

½ teaspoon fennel seed

1 teaspoon coriander seeds

½ teaspoon black peppercorn

4 fl ounces extra virgin olive oil

2 branches fresh thyme

1 garlic clove smashed

1 lemon, zest removed with a vegetable peeler and juiced

2 tablespoons tahini (sesame seed paste)

1 teaspoon minced jalapeno or ½ teaspoon chili paste

Garnish- toasted sesame seeds and/or grated parmesan cheese

Combine the fennel seeds, coriander seeds and black pepper in mortar and pestle and crush. Or, place the spices between two sheet of parchment crush with the bottom of a small pot. Transfer the cracked spices to a small sauté pan or pot.

Toast the spices in the dry pan over medium-low heat until very aromatic. Before the spices begin to smoke, remove the pan from the heat and add the olive oil, thyme, garlic and lemon peel (reserve the juice). Return the pot to the heat cook until the garlic begins to bubble. Remove again from the heat, cover the pot and set aside 15 minutes for the flavors to infuse.

In a small bowl, stir together the tahini, lemon juice and jalapeno. Strain the cooled olive oil and whisk into the tahini.

Asparagus

Asparagus, commonly thought of as a 'green' vegetable, is actually related to the *Allium* species (onion, garlic). Meaning that its appearance may be that of green vegetable, but asparagus flavor comes mostly from earthy and sulphur aromas. The sulphur aromas in asparagus become stringier as it is cooked so it is very important to avoid over cooking.

Green and White asparagus are grown from the same varietal, the white simply grown in the absence of light to prevent the development of chlorophyll. Often, hay or dirt is mounded over the rows of asparagus to block sunlight and keep the asparagus white. These tougher growing conditions are also responsible for the thicker stalks that are a typical characteristic of white asparagus.

Growing Regions- Native to the Mediterranean: Europe, Western Asia and North Africa. Also grown in California, Coastal South America, South Africa and Australia.

Common Varietals- Green, White

Taste Sweet to slightly bitter

Main Aromas

dimethyl sulfide- cabbage, gasoline, organic, sulphur, wet earth

Hexanol- cooked vegetable, flower, green, herb, rotten

hydroxy-2-butanone- butter, cream, green pepper, rancid, sweat

Best Pairing Beer, Bread, Seafood, Butter, Cheese, Egg, Ham, Vinegar

Substitute Artichoke, Broccoli or Cauliflower (especially thinly sliced stems)

Pairing	Common Compounds	Aroma	Also Shared With
Asparagus + Beer	1-octen-3-ol	cucumber, earth, fat, floral, mushroom	apple, cheese, chicken, basil, truffle
	4-vinylguaiacol	clove, curry, smoke, spice	citrus, coffee, fish, lovage, rice
Asparagus + Coconut	methylpyrazine	cocoa, green, hazelnut, popcorn, roasted, roasted nut	barley, beans, cabbage, shrimp, bread
	phenylacetaldehyde	berry, geranium, honey, nut, pungent	apricot, elderberry, melon, parsley, tamarind
Asparagus + Potato Chip	2,6-dimethylpyrazine	cocoa, coffee, green, roast beef, roasted nut	cheese, clam, nuts, pork
	2-acetylpyrrole	bread, cocoa, hazelnut, licorice, walnut	coconut, mushroom, , soy
Asparagus + Mustard	2-phenylethanol	fruit, honey, lilac, rose, wine	chicken, cocoa, lettuce, olive, tomato
	1-pentanol	balsamic, fruit, green, pungent, yeast	apple, lentil, mushroom, sherry

Beer-Pickled Mustard Seeds

Makes 1 1/2 cups

Asparagus is notoriously difficult to pair with wine, but its pairing with beer is off the charts. Here, we combine two of asparagus' favorite things: mustard and beer.

½ cup yellow mustard seeds

¼ cup brown mustard seeds

1 cup apple cider vinegar

2 tablespoons light brown sugar

1 teaspoon kosher salt

1/2 cup pale ale

optional: ground hot chili like cayenne, piment d'espette or pepperoncino

Place the mustard seeds in a small sauce pot and cover completely with cold water. Bring to a boil on the stove top. Remove from the stove immediately and drain in a fine mesh strainer. Rinse the Mustard seeds thoroughly with cold water.

Return the mustard seeds to the pot, add the vinegar, sugar and salt. Bring the mixture to a simmer and remove from the heat. Stir to prevent the seeds from sticking to the bottom of the pot. When cooled slightly, stir in the beer.

Transfer the mixture to a glass jar or plastic container and refrigerate 24 hours before using. For a spicy mustard, stir in ground chili to taste.

Avocado

Avocado trees are botanically related to Laurel (Bay) and Cinnamon trees, causing these ingredients to show some unexpected affinities for one another. The flavor of avocados is created by the high concentration of monounsaturated fats. These lipids provide aromas of green, dairy, citrus and floral. As the fats oxidize, Maillard aromas are created.

Growing Regions

Native to Mexico, Avocados are grown in tropical and Mediterranean climates throughout the world. Avocados can be consistently harvested year-round, thanks in part to the fact that they ripen after being picked.

Common Varietals (In US) Hass, Fuerte, Choquette, Tonnage, Hall

Taste fat

Main Aromas

Hexanal fresh, fruit, grass, green, oil

(E)-2-hexenal fat, floral, fruit, green grass, pungent

2,4-hexadienal cut grass, earth, fish, green

Best Pairing

cocoa, chilies, fruits (esp. citrus), butter, cream, roasted meats, seafood

Substitute

chayote, artichoke, mashed peas or beans

Pairing	Common Compounds	Aroma	Also Shared With
Avocado + Tequila	2-methyl-1-propanol	apple, bitter, cocoa, plastic, solvent	allium, brassica, corn, mango
	Methyl heptenone	citrus, mushroom, pepper, rubber, strawberry	annato, ginger, lemon balm, tomato
Avocado + Cocoa	acetoin	butter, cream, green pepper, rancid	litchi, passion fruit, vinegar
	furfuryl alcohol	burnt, caramel, cooked	almond, coconut, malt, shrimp
Avocado + Apple	1-hexanol	banana, flower, grass, green, herb, nut, resin	lettuce, turnip, celery, dill, peas
	2-ethyl-1-hexanol	citrus, green, oil, rose	cherry, dried bonito, endive, pineapple

Benne Seed Avocado Salad with Fig Vinegar

Makes 4 servings

Classic pairings for avocado—chile and lime—meet some unexpected new friends here. Cocoa maximizes Maillard flavors in the beef while fig plays up avocado's natural fruitiness. This is equally delicious served as an avocado salad without the meat. Don't overlook the fig vinegar- it's a hidden gem in this recipe and a flavorful secret weapon to keep in your pantry. Serve this salad on it's own, with the Cocoa Rubbed Beef on page XX or with grilled shrimp.

Cocoa and Chile Beef Rub (p. xx)

1 tablespoon cocoa powder

1 tablespoon kosher salt

1 teaspoon ancho chile powder

1 teaspoon ground black pepper

Benne Seed Salad

2 avocado, sliced

1 lime

¼ cup toasted white sesame seeds

4 radishes, thinly sliced

2 cups baby arugula

2 tablespoons extra virgin olive oil

Fig vinegar (recipe follows) *or balsamic vinegar*

Preheat oven to 375° F

Combine cocoa, salt, ancho, and pepper in a small bowl and mix well.

Cut the avocado into wedges (4 pieces per half). Squeeze a bit of lime over the avocado.

Then season wedges with the cocoa-chile mixture. Let the pieces stand for 1 minute.

Pour the toasted sesame seeds into a shallow plate. Dip one side of the avocado wedges into the sesame seeds.

Toss the arugula and radishes with olive oil and salt. Divide among plates and top with the avocado wedges. Drizzle each plate with fig vinegar.

Fig Vinegar

Makes 2 cups

¼ cup dried fig, chopped

1 cup water

2 cups white wine vinegar

Combine the figs and water in a small sauce pot. Bring the mixture to a simmer and cook until nearly dry. Add the vinegar and return to a simmer.

Remove the pot from the heat and cover tightly. Set aside 30 minutes to infuse. Mash the dried figs with a fork, then press the mixture through strainer. Reserve the liquid in glass jar, up to 6 months. Store in a cool, dark place.

Beef

Beef flavor is affected by the specific breed of animal and, to a greater extent, the animal's diet. The fatty acids derived from diet- linolenic, oleic, caproic- are the main contributors to taste perception.

- Grass-fed beef have higher concentrations of linolenic acid (clean fatty flavor)
- Grain-fed beef develop more oleic and linoleic acid (fatty, fried flavor)
- Corn-finished cattle develop less caproic acid (cheesy-fatty flavor)

Aging also plays a significant role in beef flavor. Beef has very little flavor before aging. Retail cuts of meat have undergone a minimum of 7 days aging, where distinctive “meaty” flavor develops. After 14 days, *1-octen-3-ol* a key aromatic in beef flavor, shows an increase of more than 10x.

Growing Regions

Globally in temperate to cold climates. Top beef-producing countries: US, Brazil, EU, China, Argentina

Common Breeds

Angus (US)/Aberdeen (UK), Charolais (France), Chianina (Italy), Hereford (Global), Kobe (Japan), Limousin (France),

Taste meaty, sweet

Main Aromas

Trimethylpyrazine- cocoa, earth, must, potato, roast

1-octen-3-ol- cucumber, earth, fat, floral, mushroom

3-methylbutanoic acid- cheese, fecal, putrid fruit, rancid, sweat

heptanal- citrus, dry fish, fat, green, nut

Best Pairing

nuts, dried fruits, butter, cream, mustard, allium, cocoa

Substitute

venison, lamb, bison, tuna

Pairing	Common Compounds	Aroma	Also Shared With
Beef + Cocoa	acetophenone isopentanal	almond, flower, meat, must, plastic cocoa, cooked vegetable, fresh, green, malt, spice	buckwheat, cauliflower, corn, tamarind, wine beer, hazelnut, ginger, whisky
Beef + Beer	naphthalene limonene	mothball, tar balsamic, citrus, fragrant, fruit, greenery, herb	beans, capsicum, peanut, sage, shrimp apple, caraway, chervil, fig, okra, sweet potato
Beef + Cabbage	hexane 2-pentylfuran	alkane butter, floral, fruit, green bean	honey, milk products, pistachio, thyme artichoke, clam, parsley, tea, tomato
Beef + Grape	trimethyloxazole pseudocumene	nut, sweet pesticide, plastic	chicken, cocoa, oats, potato, soy sauce coffee, crab, vanilla, walnut, watercress

Cocoa and Chile-Rubbed Beef

Serves 4

Like most meats, beef derive a lot of flavor from the maillaird reactions that take place during the cooking process. In the recipe, cocoa and dried chilies augment those roasted aromas, packing even more flavor into the beef.

1 tablespoon cocoa powder

2 teaspoons ancho chili powder

1 teaspoon sesame seeds

1 teaspoon ground black pepper

2 pounds trimmed flank steak

1 tablespoon canola oil

serve with Benne Seed Avocado Salad (page XX)

Combine cocoa, salt, ancho, and pepper in a small bowl and mix well. Season the beef on both sides with the mixture (you may not use all of it).

Heat canola oil in a large sauté and sear beef on all sides until golden brown, 3-4 minutes per side. Remove from heat and rest on a rack to cool.

Beetroot

Beetroot is an important crop globally as sugar beets are grown and processed to extract sucrose (table sugar). Culinary uses of the beetroot focus on other varieties as noted below. Like most root vegetables, beet may be eaten either raw or cooked. Roasting elicits the natural sugars from beets giving them a very pleasant earthy-sweet taste. Geosmin, the aroma compound most responsible for the earthy flavor in beets, is also responsible for the earthy aroma that fills the air when rain falls on dry soil.

Though less commonly eaten, beet leaves have excellent flavor. Young leaves may be added to salads or eaten raw. Larger leaves may be sautéed like spinach or other greens or chopped and added to soups and stews. Swiss chard is also a member of the beet family.

Growing Regions- Beets are native to Europe, and Western Asia and today are grown worldwide.

Common Varietals- Chioggia, Detroit Dark Red, Formanova, Golden, Lutz Green Leaf

Taste sweet

Main Aromas

4-methylpyridine- ash, sweat

isovaleraldehyde- almond, cocoa, cooked vegetable, malt, spice

geosmin- beet, earth

Best Pairing

citrus, toasted nuts, cheese, yogurt, seafood, wine, vinegar

Substitute

carrot/parsnip, cabbage

Pairing	Common Compounds	Aroma	Also Shared With
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Beet + Lychee	isoamyl alcohol	bitter, burnt, cocoa, gasoline, malt	cherry, cheese, coffee, mustard, bread
	ethanol	alcohol, floral, ripe apple, sweet	allium, carrot, crab, olive, raspberry
Beet + Oyster	dimethyl sulfoxide	garlic	blackberry, cucumber, mint, onion
	capronaldehyde	fresh, fruit, grass, green, oil	camomile, ginger, parsley, radish
Beet + Lemon Balm	phenylacetaldehyde	berry, geranium, honey, nut, pungent	apricot, elderberry, corn, sage
	carvone	basil, bitter, caraway, fennel, mint	celery, fennel, savory, orange

Beet and Carrot Salad with Orange-Hazelnut Dressing

Serves 4

Beets show a particular affinity for the earthy flavor of fellow roots, like carrots and jicama, as well as fruit-like aromas. Toasted nuts both in the salad and the dressing enhance the roasted flavor of cooked beets.

2 cups, raw red beets, shredded

2 cups carrot, shredded

2 cups jicama *or* green cabbage, shredded

4 pieces scallion, thinly sliced

½ cup toasted cashews, chopped

1 tablespoon chopped parsley

¼ cup Orange-Hazelnut Dressing, recipe follows

Shred the vegetables with the fine julienne attachment on a mandolin or by grating on a box grater.

Combine beets, carrots, jicama or cabbage, and scallions in a large bowl. Add the orange-hazelnut dressing as needed (about $\frac{1}{4}$ cup). Toss well and season to taste with salt and pepper. Just before serving, mix in the cashews and parsley.

Orange-Hazelnut Vinaigrette

Makes 1 cup

$\frac{1}{4}$ cup white wine vinegar
1 tablespoon shallot, minced
2 tablespoons hazelnuts, finely chopped
 $\frac{1}{2}$ cup canola or vegetable oil
2 tablespoons orange juice
1 teaspoon orange zest
1 teaspoon Dijon mustard
 $\frac{1}{4}$ cup extra virgin olive oil

Combine the shallots and vinegar in a small bowl with a pinch of salt and pepper, mix well. Set aside to macerate while preparing the hazelnut oil.

In a small saute pan, combine the chopped hazelnuts and oil and place over medium. When the nuts begin to bubble in the oil, remove from the heat and set aside to cool.

Whisk the orange juice, zest and mustard into the vinegar and shallots. Stream in the hazelnuts in oil and the olive oil while continuing to whisk. The mixture should thicken and emulsify.

Berries

Berries represent a wide category of edible fruits. Botanically, berries are defined as a fruit produced by the ovary of a single flower. By this definition, bananas, tomatoes, grapes, eggplant, capers, watermelons and pumpkins are all berries. This chapter, however, will focus on classically defined berries in three main genera rather than the individual berries. This big picture view will help to focus on the similarities among all berries and identify the particular characteristics in each genus.

Fragaria- a genus of the rose family which produces varieties of Strawberry such as: Honeoye, Earliglow, Allstar, Frais de Bois

Rubus- a separate genus of the rose family that produces raspberries, blackberries, dewberries, loganberry and boysenberry

Vaccinium- a genus heath family that includes blueberry, cranberry, and lingonberry

Growing Regions

(Strawberry) China, United States, Mexico, Turkey, Spain

(Blueberries) US, Chile, Peru, Argentina, Mexico

(Blackberry) Chile, Mexico, Ecuador, US, Serbia, Hungary,

(Raspberry) Russia, Poland, Serbia, US, Ukraine, Mexico, UK

Common Varietals-

(Strawberries)- Honeoye, Earliglow, Allstar, Frais de Bois

(Blueberries) Earlyblue, Bluetta, Blueray, Jersey

(Blackberry) Thornfree, Lochness, Chester Thornless,

(Raspberry) red, yellow-gold, purple, black

Taste- sweet to sour

Main Aromas

Strawberry

Furaneol- burnt, caramel, cotton candy, honey, sweet (mainly responsible for toasted)

2,5-dimethyl-4-methoxy-3(2H)-furanone (mesifurane)

ethyl butanoate- apple, butter, pineapple, red fruit, strawberry

Raspberry

raspberry ketone- citrus, raspberry

β -ionone- cedarwood, floral, raspberry, seaweed, violet

geraniol- geranium, lemon peel, passion fruit, peach, rose

Black Berry

2-Heptanol- citrus, coconut, earth, fried, mushroom, oil

p-cymen-8-ol- citrus, must, sweet

2-heptanone- bell pepper, blue cheese, green, nut, spice

α -terpine- berry, lemon, wood

Blueberry

(Z)-3-hexenal- apple, bell pepper, cut grass, green, lettuce

(E,E)-2,4-hexadienal- citrus, fat, floral, green

(E,Z)-2,6-nonadienal- cucumber, green, lettuce, wax

methyl 2-methylbutanoate- apple, fruit, green apple, strawberry, sweet

butyl acetate- apple, banana, glue, pungent, sweet

2-methylbutyl acetate- apple, banana, pear

Cranberry

α -terpineol- anise, fresh, mint, oil, sweet

Benzaldehyde- bitter almond, burnt sugar, cherry, malt, roasted pepper

methyl 2-methylbutanoate- apple, fruit, green apple, strawberry, sweet

Best Pairing citrus, melon, apricot, peach, nectarine, chocolate, arugula, wine, vinegar, cream, yogurt

Substitute other berries, currants, cranberry, grape, kiwi

Pairing	Common Compounds	Aroma	Also Shared With
Vaccinium + Thai Basil	limonene	balsamic, citrus, fragrant, fruit, greenery, herb	celery, fennel, orange, turmeric, sesame
	3-octanone	butter, mold, herb, resin	coffee, peanut, turkey
Strawberry + Mushroom	1-nonanol	fat, floral, green, oil	chicken, cauliflower, basil, cheese
	butanoate	apple, banana, butter, cheese, ester, floral	apple, gooseberry, olive, wine
Rubus + Cumin	β -caryophyllene	fried, spice, wood	carrot, star anise, pistachio, mint
	p-cymene	citrus, fresh, gasoline, solvent	cardamom, eucalyptus, ginger, black pepper, walnut

Chicken Mushroom Burger

Makes 5 burgers

This recipe has traveled around the world, literally. The pairing of chicken, mushroom and strawberry is something that was discovered by Michael Laiskonis and Florian Pinel as we worked on Chef Watson with ICE and IBM. It was a pairing that they took particular interest in and that I used as the inspiration for this recipe. Eventually, others

took notice and felt like it was an “idea worth sharing,” and this recipe became the subject of a talk on the science of flavor that I gave at TEDxWarsaw. Now, you can taste it for yourself.

2 tablespoons olive oil
2 cloves garlic, chopped
1 shallot, minced
8 ounces fresh mushrooms, roughly chopped
1 pound ground chicken
6 tablespoons dry breadcrumbs
2 large eggs, beaten
2 tablespoons chopped fresh basil
1 teaspoon salt
freshly ground pepper to taste
4 burger rolls softened butter to spread on the rolls

Heat the olive oil in a large sauté pan. Add the garlic and shallot and cook over medium heat until softened. Add the mushrooms and sauté until lightly browned. Set aside to cool.

Combine the cooled mushroom mixture, ground chicken, breadcrumbs, eggs, basil, salt and pepper in a bowl. Blend well and form into 5 patties. Cook the patties in a sauté pan with a small amount of oil in the pan approximately 5 minutes per side over medium to medium-high heat. Or cook on the grill, or roast in a 425-degree oven until cooked through, about 12 minutes.

Split the 4 rolls, spread with softened butter, and arrange, cut side up, on the baking sheet with the tomatoes. Broil until the buns are toasted and the cheese is slightly melted.

Strawberry ‘Ketchup’

1 tablespoon butter
2 tablespoons shallot, minced
2 tablespoons fresh ginger, minced
2 cloves garlic, minced
2 cups mushrooms, chopped
4 cups strawberries, chopped
1 teaspoon crushed coriander seeds
½ teaspoon ground black pepper
kosher salt, to taste

Melt the butter in a sauce pot. Add the shallots, ginger and garlic. Sauté until tender and fragrant. Add the mushrooms, season lightly with salt and cook until tender and dry, about 6 minutes.

Stir in the chopped strawberries, coriander and black pepper. Cook until the strawberries become very soft, then blend with an immersion blender or transfer into a food processor and process until smooth. Return to the pot and simmer until thickened. Season to taste with salt, pepper and a pinch of sugar if necessary.

Brassica Oleracea

(Cabbage, Brussels Sprouts, Kohlrabi, Kale, Collard Greens)

Brassica Oleracea is a separate species of the Brassica genus that includes what are generally referred to in culinary terms as "cruciferous vegetables." Broccoli and Cauliflower are members of the Oleracea species but will be examined in a separate chapter. The leafy members of the Oleracea species, like most Brassica, derive the majority of their flavor from sulphur compounds. The release of sulphur compounds increases with cooking hence, the distinctive smell of boiled cabbage or brussels sprouts. Green aromas are also found throughout the species while the dark green leafy varieties (kale, collard greens) contain phenolic compounds as well.

Growing region

Native to Central Europe, Brassica Oleracea is grown throughout the world with significant production in the US and Southeast Asia. In temperate climates, Brassica plants can be harvested year-round. Meanwhile, in cooler climates, plants mature in late Fall or Winter.

Common varietal Cabbage, Brussels Sprouts, Kohlrabi, Kale, Collard Greens

Taste bitter

Main Aromas Earthy, Sulphur

allyl isothiocyanate- garlic, pungent, sulphur

butyl isothiocyanate- green, pungent, sulphur

(E)-4-hydroxycinnamic acid- astringent, balsamic, phenol

Best Pairings

toasted nuts, cheese, mushrooms, grains, anchovy

Substitute

Any variety of Brassica (Oleracea or Rapa) may be substituted for another, including broccoli and cauliflower

Pairing	Common Compounds	Aroma	Also Shared With
Kohlrabi + Pork	indole	burnt, fecal, medicine, mothball	cauliflower, lemon, shrimp
	2,6-dimethylpyrazine	cocoa, coffee, green, roast beef, roasted nut	asparagus, barley, coffee, hazelnut
Brussels Sprouts + Apple	2-methyl-1-butanol	fish oil, fusel oil, green, malt, wine	beer, camomile, rye bread, vinegar
	methyl octanoate	fruit, orange, sweet, wax, wine	cabbage, chile, mussels
Kale + Coconut	1-tetradecanol	coconut	citrus, crab, gooseberry, scallion
	2-acetylfuran	balsamic, cocoa, coffee, smoke, tobacco	beef, cocoa, oyster, pumpkin

Umami Vinaigrette

Makes 2 cups

I love using this as a dressing for salads that include kale, cabbage, shaved brussels or other brassica. It's also perfect for coating for brussels sprouts, broccoli or cabbage before roasting.

4 fl oz red wine vinegar
1 tablespoon shallot, chopped
2 garlic cloves
4 anchovy filets
1 tablespoon Dijon mustard
½ teaspoon mushroom powder (optional)
¼ cup grated parmesan cheese
8 fl oz canola or vegetable oil
4 fl oz extra virgin olive oil

Combine vinegar, shallot, garlic, anchovy, mustard, mushroom powder, parmesan cheese in a blender. Blend on low.

When the vinegar mixture is smooth, slowly add the oils to the blender with the machine running. When all the oil is added, the mixture should be emulsified. Store in a glass jar, refrigerated up to 4 weeks.

Brassica Rapa

Brassica Rapa is a group of plants with a number of subspecies (outlined below) that are grown for both oil production and culinary uses. The seeds of Brassica Rapa plants are pressed to create Canola Oil. Those with culinary uses have flavor dominated by sulphur-containing compounds that create aromas that range from cabbage to mustard. These compounds include *allyl isothiocyanate* (also found in radish, horseradish, mustard) that can mimic the spicy effects of capsaicin.

Growing Regions

Southeast Asia, Korea, China, Japan, United States are top producers.

Common Varietals Napa cabbage, bok choy, rapini, mizuna, komatsuna, tatsoi, turnip, field mustard, choy sum

Taste bitter, peppery

Main Aromas

Phenylethyl isothiocyanate- horseradish, mustard

Eugenol- burnt, clove, smoke, spice

Longifolene- flower, vegetable

Best Pairing Berries, Butter, Cream, Cheeses, Allium

Substitute spinach, kale, broccoli, cauliflower

Turnip, Apple, Chicken, Hazelnut and Radish Pan Roast

Serves 4

This is one of those lovely meals that comes together in just about 30 minutes and leaves you with only one pan to clean. As if that weren't enough to convince you, it packs an outrageous combination of flavor and textures and is just as delicious with the chicken omitted and served as a vegetable side.

8 boneless, skinless chicken thighs
2 teaspoons kosher salt
1 teaspoon fresh thyme leaves
1 teaspoon mustard powder
½ teaspoon ground black pepper
2 tablespoon canola or vegetable oil
2 tablespoons butter
8 baby turnips, cut in half
1 golden delicious apple, peeled, cored and cut into 8 wedges
8 radishes, cut in half
4 garlic cloves, peeled and smashed
6 branches fresh thyme
½ cup apple cider vinegar
Garnish:
¼ cup toasted hazelnuts
thinly sliced fresh radishes

Place the chicken in a bowl. Add the salt, thyme, mustard and pepper; toss well to coat.

Place a large sauté pan over high heat. Place a resting rack set over a baking pan next to the stove.

When the sauté pan is hot, add the oil. Carefully place the chicken into the hot oil and cook in batches until well browned on both sides. As the chicken browns, remove it to the rack to rest.

When all the chicken is browned, turn the heat to medium and pour the excess oil out of the pan and wipe out any loose brown bits.

Place the pan back over the heat and add the butter. Immediately, add the turnips, apples, radishes, garlic, and thyme. Toss well and sauté on the stove top for approximately 3 minutes. Return the chicken to the pan along with any juices that may have collected on the resting pan. Toss well in the pan and transfer the pan to the oven.

Cook in the oven 20 minutes or until the chicken registers a temperature of 165°F internally. When the chicken is cooked add the apple cider vinegar and butter to the pan and stir well to deglaze.

Divide among plates and garnish with hazelnuts and radishes.

Cane Syrup, Molasses; related: Sorghum, Maple Syrup

Cane syrups are produced from fresh-pressed sugar cane juice that has been boiled so that water evaporates and sugar caramelizes. Throughout this cooking process, the syrups develop deep, roasted notes. In consecutive boiling, the syrups become darker, eventually developing a slightly sulphurous and bitter flavor, as in blackstrap molasses. In maple syrup production, the sap of maple trees is carefully boiled to expel water but is stopped before any caramelization begins. Cane syrups can (and should be) used as a more flavorful alternative to plain sugar or corn syrup.

Varieties of these syrups may be made from juiced Sorghum (sorghum molasses); date palm sap (palm syrup/sugar); maple sap (maple syrup); or coconut sap (coconut syrup/sugar).

Growing regions

Brazil, India, China, Thailand, Pakistan are top producers.

Common varietals Sweet sorghum, Cane syrup, Blackstrap molasses, Maple Syrup

Taste sweet, tannic

Main Aromas

ethyl n-hexanoate- anise, caramel, fruit, spice, yeast

ethyl phenylacetate- floral, fruit, honey, rose, sweet

isomaltol- burnt, caramel, fruity

Flavors specific to Maple Syrup

2-methyloxolan-3-one- sweet, rum, baked bread

vanillin- vanilla, sweet

Flavors specific to Molasses

Dimethyl sulfide- cabbage, gasoline, organic, sulphur, wet earth

Best Pairings Port Wine, Tamarind, Sherry Vinegar, Apple, Grains, Orange,

Substitute Any syrups or unrefined sugars may be substituted for one another.

Pairing	Common Compounds	Aroma	Also Shared With
Molasses + Mango	furaneol	burnt, caramel, cotton candy, honey, sweet	grapefruit, oat, peanut, sherry
	2-methylbutanoic acid	butter, cheese, fermented, rancid, sour	chicken, coffee, mint, pistachio, rice
Molasses + Garlic	2-propanol	must, solvent	cider, lentils, pear, tofu, whisky
	coumaran	sweet, balsamic	chestnut, lemon balm, licorice, plum
Molasses + Fish	2,6-dimethoxypheno l	medicine, phenol, smoke	bonito flake, pork, soy sauce
	propanoic acid	fat, pungent, rancid, silage, soy	breadcrumb, calvados, mushroom, vinegar

Molasses-Stained Swordfish Kebabs

Serves 4

This glaze is particularly delicious on seafood and swordfish stands up best to the grilling, though it may be brushed onto salmon before broiling or used to marinate chicken or beef for the grill.

3 cloves garlic, Minced garlic
2 tablespoons Sherry or Red Wine vinegar
2 tablespoons molasses or cane syrup
1 teaspoon fresh thyme leaves
finely grated zest of ½ orange
½ teaspoon ground cumin
2 lbs swordfish, cut into 1½ inch cubes
1 red onion
1 bell pepper
6 slices thick cut bacon (optional)
8 (6-inch) bamboo skewers, soaked in cold water
Kosher salt
freshly ground black pepper

In a small bowl, whisk together the garlic, vinegar, thyme, orange zest, cumin and molasses. Divide the mixture into containers.

Cut the onion, pepper and bacon into squares approximately the same size as the swordfish. Arrange all the fish, onion, peppers and bacon in alternating fashion on the skewers. Fill the skewers as much as possible, leaving just ½ inch of the wood showing on each end.

When all of the skewers have been assembled, lay them out on a plate or baking and well with kosher salt and very light seasoning of pepper. Brush all sides of the skewers with one container of the molasses mixture.

Cook the skewers on a well-oiled, hot grill 3 minutes per side or until cooked through-swordfish is best served medium well. Remove the skewers from the grill and place on a rack to rest immediately brush each skewers with some of the fresh molasses mixture. Alternatively, skewers may be cooked under a broiler.

Capsicum (Peppers)

Capsicum, or Peppers, are members of the same family of Nightshades that also includes tomato, eggplant and potato thus giving these three ingredients a particular affinity for one another. Spiciness in peppers is derived from Capsaicin, a flavorless compound that produces a burning sensation wherever it comes in contact with the body. Capsaicin is hydrophobic, meaning it cannot be rinsed away by water, but will bond with fat which is why drinking milk or downing a shot of blue cheese dressing is the best way to eliminate the burn of spicy foods.

The flavor of Capsicum is mainly derived from fruity and floral aromas. Flavor in particular varieties can range from highly vegetal and green, as in green bell peppers, to the sweet and berry-like Aji Panca from Peru (though its flavor is typically overshadowed by intense heat).

Growing Regions

Native to North, Central and South America. China, Mexico, Turkey, Indonesia, Spain are now top producers.

Common Varietals (*Species*)

Capsicum annuum- bell peppers, sweet Italian peppers, serrano, cayenne, paprika and jalapenos

Capsicum baccatum- South American varieties like aji Amarillo, aji limon, criolla sella

Capsicum chinense- Habaneros, Scotch bonnets, Trinidad Scorpions, Bhut Jolokia, Carolina Reaper

Capsicum pubescens- Rocoto and Manzano pepper

Capsicum frutescens- Brazilian Malagueta, Tabasco

Taste sweet, spicy

Main Aromas

2-isobutyl-3-methoxypyrazine- bell pepper, earth, floral, green pepper, peasy

2,3-butanedione- butter, caramel, fruit, sweet, yogurt

3-carene- bell pepper, lemon, pungent, resin, rubber

Best Pairings

stone fruits, aromatic spices, citrus, mint, wine, cheese, sour cream, yogurt

Substitute

Most peppers may be substituted for one another.

Pairing	Common Compounds	Aroma	Also Shared With
Ancho + Pumpkin	2-(sec-butyl)-3-met hoxypyrazine	bell pepper, carrot, earth, green	beetroot, cucumber, peanut
	3-pentanone	ether, sweet	allium, cream, mushroom, pear, sage
Jalapeno + Peach	pentadecane	alkane	coconut, leek, mustard, pork
	limonene	balsamic, citrus, fragrant, fruit, greenery, herb	basil, camomile, coffee, star anise
Bell Pepper + Dill	camphor	camphor, earth, pine, spice	fennel, lemon balm, tomato
	sabinene	herb, pepper, turpentine, wood	celery, cumin, ginger, pistachio

Banana and Pepper Hot Sauce

Makes about 1 cup

This recipe makes the most of the fruity aromas found in capsicum in a most unexpected way. Here, flavors line up nicely with bananas lending pleasant sweetness to your new favorite hot sauce.

1 teaspoon vegetable oil

4 red jalapeno peppers or habanero, minced (for a milder sauce remove the seeds)

2 tablespoons cilantro stems, minced

4 garlic cloves, minced

½ cup rice wine vinegar

1 green (under-ripe) banana, chopped

0.35 gram xanthan gum (optional)

Heat the oil in small sauté pan over medium-high heat.

Add the peppers, cilantro, and garlic to the hot oil and cook until fragrant and tender, but not browned.

Degalze with vinegar and mix in the chopped banana. Mix well can continue cooking until the vinegar is reduced by half, about 1 minute.

Transfer the mixture to a bender or food processor. Puree until smooth. With the blender running add the xanthan gum and mix 10 seconds more. Transfer to a jar and reserve refrigerated up to 4 weeks.

Caramel

Caramel flavor is derived exclusively from Maillard reactions. Toasted, roasted and nutty aromas are created as they decompose under the stress of heat. The longer that sugar is left exposed to these temperatures, the darker and more flavorful it will become. However, in relative quick succession, the sugar will burn and bitter acrid flavors develop.

Taste sweet, slightly bitter

Main Aromas roasted, smoke, phenol

5-(hydroxymethyl)furfural- caramel, cardboard, fat, must, spice

maltol- caramel, cotton candy, malt, roasted bread, roasted nut

diacetylformoin- caramel

Best Pairing

citrus, vinegar, wine, roasted meats/bacon, chocolate, nuts

Substitute

molasses, sorghum, maple syrup, coffee, bourbon

Pairing	Common Compounds	Aroma	Also Shared With
Caramel + Fish Sauce	guaiacol	burnt, medicine, phenol, smoke, wood	beer, cabbage, citrus tomato
	diacetyl	butter, caramel, fruit, pastry, rancid, sweet, yeast	apricot, brandy, cream rye bread

Caramel + Tamarind	maltol	caramel, cotton candy, malt, roasted bread, roasted nut	chestnut, cocoa, pecan, pork
	5-methylfurfural	almond, caramel, cooked, roasted garlic, spice	banana, beef, orange, potato
Caramel + Peanut	furaneol	burnt, caramel, cotton candy, honey, sweet	mango, popcorn, soy sauce, strawberry
	4-hydroxydecanoic acid lactone	fat, fruit, lactone, peach, sweet	coconut, parmesan, plum, sherry
Caramel + Apple	vanillin	sweet, vanilla	fennel, lemon, oats, pineapple
	furaneol	burnt, caramel, cotton candy, honey, sweet	chestnut, coffee, passion fruit, pistachio

Fish Sauce Peanut Brittle

Makes about 2 cups

Try grinding this up as a savory-sweet crunchy finish for meats or vegetables. Brooke and I are really in love with the brittle as a topping for steamed buns filled with tender pork or beef.

2 cups sugar

6 tablespoons fish sauce

4 dried arbol chiles or 2 teaspoons red pepper flakes

1 cup chopped peanut

In a heavy saucepan, stir together the sugar, fish sauce and chiles. Place the pot over medium heat, swirling the mixture occasionally (do not stir). If you notice crystals forming around the edge of the pan, wipe the inside of the pot with a brush moistened with water to wash the crystals back into the mixture.

Continue cooking the sugar mixture at a simmer until it has a deep brown color (12-15 minutes). Carefully judge the color as the fish sauce will make the caramel look darker than it really is. The temperature should be between 330°F and 350°F. When fully cooked, remove the pot from the heat and stir in the peanuts.

Immediately pour the mixture out onto a greased sheet of wax paper. Cool completely to harden, then break into smaller portions.

Carrots

The sweet, tender root vegetable that is the modern carrot is only possible after centuries of selective breeding. Wild carrots produce a bitter, woody root, hinting at the reason why they were originally cultivated almost exclusively for their leaves and seeds. The orange carrot that we know today is also something of a newcomer; the original carrot varieties were red and yellow.

Woody and resinous aromas play an important role in carrot flavor along with earthy and citrus notes. Though less commonly eaten, carrot tops have mild herbaceous flavor that taste like parsley with hints of carrot flavor.

Growing Regions

Native to the Middle East, today carrot cultivation has spread to nearly every part of the world with China, Uzbekistan, Russia, US and Ukraine making up the top 5 producing nations.

Common Varietals Chantenay, Danvers, Imperator, Nantes *All species contain variations that may be purple, red, orange, yellow or white in color*

Taste sweet

Main Aromas

α -terpinolene- pine, plastic, sweet

α -pinene- cedarwood, pine, resin, sharp, turpentine

β -myrcene- balsamic, fruit, geranium, herb, must

Best Pairing

citrus, wine, vinegar, radish, olive, coriander, cardamom, star anise, coffee, cilantro, basil, mint, tarragon

Substitute

radish, parsnip, celery, sweet potato

Pairing	Common Compounds	Aroma	Also Shared With
Carrot + Balsamic	myrcene	balsamic, flower, fruit, herb, must	annatto, cilantro, turmeric, rosemary
	6-allyl-4-methoxy-1, 3-benzodioxol	balsamic, carrot, spice, warm	black pepper, blueberry, dill, parsnip
Carrot + Basil	terpinolene	pine, plastic, sweet	cardamom, lime, ginger, mango
	bornyl acetate	herb, pine, sweet	black currant, cinnamon, pistachio
Carrot + Coffee	4-vinylguaiacol	clove, curry, smoke, spice	cider, lovage, pork, sage
	5-methylfurfural	almond, caramel, cooked, roasted garlic, spice	coconut, corm, rum, tamarind

Coffee and 5 Spice-Roasted Rainbow Carrot Salad with Herbed Yogurt Dressing and Arugula

Serves 4

Carrots love spice. The aromas that come together to make the flavor of carrot are heavy on spice and citrus notes. In this recipe exotic spices like star anise and cinnamon get an extra kick by the use of coffee as a spice. Herbs and yogurt help to smooth the whole thing out and tie all these wildly unexpected flavors together.

1 pound rainbow carrots, peeled and trimmed
3 tablespoons extra virgin olive oil
1 tablespoon kosher salt
1 tablespoon coffee grinds
1 teaspoon Chinese 5 spice powder
4 cups loosely packed arugula, lightly dressed with extra virgin olive oil and lemon juice
2 ounces lemon juice
1 tablespoon minced ginger
6 ounces olive oil
Herbed Yogurt (recipe below)

Preheat the oven to 425 degrees.

Prepare a baking sheet, lined with aluminum foil. In a large bowl, combine the carrots, olive oil, salt, coffee and 5 Spice. Spread into a single layer on the preheated baking sheet, being sure not to overcrowd the sheet. Roast until the carrots are tender and browned around the edges, 25 to 30 minutes. Remove from the oven and cool slightly on the baking sheet.

For the dressing, whisk together the lemon juice, ginger and olive oil. Toss the roasted carrots with the dressing before serving.

Spoon the herbed yogurt (recipe below) onto a serving plate. Arrange the carrots on top of the yogurt and place a small pile of lightly dressed arugula on top of the carrots (so that the carrots are still visible under the greens). Top with toasted almonds, if using.

Herbed Yogurt Dressing

Makes 1 cup

2 teaspoons chives, thinly sliced

2 teaspoons dill, stemmed and finely chopped

Zest of 1 lemon

Pinch kosher salt

Freshly ground black pepper

1 cup Greek yogurt

Combine all ingredients in a bowl and mix well. Refrigerate until ready to serve.

Broccoli, Cauliflower, Romanesco

Broccoli, Cauliflower and Romanesco are members of the Brassica Oleracea species outlined in a separate chapter. However, these plants have their own unique characteristics that separate them from their leafy brothers and sisters.

These three varieties are distinguished mainly by color and the shape of the flowering head. The shape of each impacts its appearance and texture after cooking. The tightly formed heads of cauliflower and romanesco result in a firmer texture when cooked. However, the higher water content of cauliflower requires careful cooking to prevent it from becoming mushy.

All three varieties derive flavor from sulphur compounds similar to other brassica with lesser aromas of green, vegetable and citrus.

Growing regions- Cool weather crops that do not handle Summer heat well. They are native to the Northern Mediterranean but are now grown around the world with China, India, Spain, Mexico, Italy being the top producers.

Common Varietals broccoli, cauliflower, romanesco,

Colors- white, orange, green, purple

Taste- slightly bitter with nutty & slightly sweet undertones

Main Aromas earthy, sulphur, roasted

3-butenylisothiocyanate- green, pungent, sulphur

cis-3-Hexen-1-ol- bell pepper, grass, green leaf, herb, unripe banana

nonanal- citrus, fat, floral, green, paint

Best Pairings

citrus (esp. Lemon), egg, milk, cream, cheese, cocoa, curry, coriander, fish sauce

Substitute

broccoli, cauliflower, romanesco for each other *or* asparagus, brussels sprouts, broccoli
rabe, kale

Pairing	Common Compounds	Aroma	Also Shared With
Cauliflower + Cocoa	4-methylacetophenone	bitter almond, floral, fruit, spice, sweet	capsicum, celery, lemon, passion fruit
	2-butenal	pungent	caviar, cheese, daikon, peanut
Broccoli + Fig	(E,E)-2,4-heptadienal	fat, fish, nut, plastic	allium, beef, tomato, walnut
	indole	burnt, fecal, medicine, mothball	blue cheese, hazelnut, okra, rice, shrimp, paste
Broccoli/Cauliflower + Peanut	(E,E)-2,4-decadienal	coriander, deep fried, fat, oil, oxidized	chile, clam, lime, mushroom, yogurt
	dimethyl trisulfide	cabbage, fish, onion, sulphur, sweat	oyster, parsnip, pumpkin, radish

Broccoli, Peanut and Pumpkin Stir Fry

Serves 4

Fans of Thai cooking may already be comfortable with the pairing of broccoli and peanut, but pumpkin and fish sauce don't typically get invited to the party. To bring even more unique pairing to this dish, try serving it with grilled clams or oysters.

Vegetables

- 2 heads broccoli
- 1 small pumpkin or acorn squash
- 4 cloves garlic, peeled and smashed
- 2 tablespoons fish sauce
- 3 tablespoons vegetable oil

Sauce

- 1 cup chicken stock
- 2 tablespoons fish sauce
- 1 tablespoon lime juice
- 1 tablespoon honey
- 1 tablespoon corn starch

Stir Fry

- 2 tablespoons vegetable oil
- 3 cloves garlic, sliced
- 6 scallions, sliced into ½ inch pieces
- 1 piece fresh ginger (thumb-sized) peeled and thinly sliced
- 2 whole dried arbol chiles
- 2 tablespoon cilantro leaves
- ¼ cup toasted peanuts, chopped

Separate the heads and stems of the broccoli. Peel the stems and thinly slice, cut the head into bite-sized florets

Peel, seed and slice the pumpkin or squash.

Combine the broccoli stems and florets and pumpkin in a mixing bowl. Add the garlic, fish sauce and oil. Toss well to coat. Spread the vegetables into a single layer on a roasting pan. Use two pans if needed, it is important the vegetables not be too crowded.

Roast the vegetables in the oven 12 minutes, they should still be al dente. Set aside to cool.

Combine the sauce ingredients in a bowl and whisk well to combine. Set aside.

Heat a large sauté over high heat. Add the oil, garlic, scallion, ginger and chiles and cook, stirring quickly to prevent browning. When the mixture is very aromatic, add the roasted vegetables to the pan toss well. Continue cooking over highest heat. When the pan begins to sizzle rapidly add the sauce. Stir well until the mixture boils and thickens.

Divide among plates and garnish with cilantro and peanuts.

Cocoa Bean

The subject of the flavor of the cocoa bean could fill the pages of this book on its own. This chapter will take the wide view, highlighting general flavor characteristics found in beans and how that flavor is transformed in various products.

Cocoa beans grow in pods on the cocoa tree which is native to the Americas. At harvest, cocoa beans have a thin, fleshy and fruit-like covering. The flesh may be eaten fresh and has a mild tropical fruit flavor. Fresh beans are bitter and unpalatable. After harvest, beans are dried and fermented for 5 to 14 days. It is during this process that cocoa's distinctive flavor begins to develop. However, the flavor is not fully developed until Maillard reaction take place as the beans are roasted. Exact cocoa flavor can vary according to the origin of the beans (terroir) and other variables in the fermentation and roasting process.

Cocoa flavor is not derived from a single compound or even a few compounds for the matter. The flavor of cocoa is made up of nearly 600 different aromatic compound with just 25 or so playing the most significant role (the top three compounds are below).

Chocolate flavor depends on the composition of the chocolate with dark chocolate exhibiting the most cocoa aromas. Milk chocolate exhibits more lactones (fruity aromas) along with cocoa. White Chocolate contains only cocoa butter, which is odorless, meaning it's flavor is derived from milk and vanilla.

Growing regions

Native to the Americas, grown in tropical climates throughout the world. 70% of global production comes from western Africa. Prominent growing regions include: Cote d'Ivoire, Ghana, Indonesia, Nigeria, Cameroon, Brazil, Ecuador, Mexico, Peru, Dominican Republic

Common varieties- Dark, bittersweet, semi-sweet, milk, white

Taste bitter

Main Aromas

3-methylbutanoic acid- sweaty, rancid

furaneol- burnt, caramel, cotton candy, honey, sweet

2-ethyl-3,5-dimethylpyrazine- potato chips

Best Pairing

coffee, caramel, nuts, whisky, mushroom, cauliflower, red wine, dried fruit

Substitute

different forms of cocoa/chocolate, roasted nut purees, vegetable ash

Pairing	Common Compounds	Aroma	Also Shared With
Cocoa + Beetroot	phenethylamine	ammonia, fishy	blue cheese, citrus, rhubarb, sake
	benzaldehyde	bitter almond, burnt sugar, cherry, malt, roasted pepper	almond, avocado, cinnamon, lemon thyme, sesame seed
Chocolate + Passion fruit	linalool	bergamot, citrus, coriander, lavender, lemon, rose	coriander, fennel, mint, pistachio, tarragon
	2-acetylfuraethyl	balsamic, cocoa, coffee, smoke, tobacco	barley, jalapeno, oyster, pumpkin
Chocolate + Peanut Butter	3-hydroxy-2-methyl-4-pyrone	caramel, cotton candy, malt, roasted bread, roasted nut	chestnuts, citrus, pineapple
	pseudocumene	pesticide, plastic	banana, nectarine, vanilla, walnut

Chocolate Mousse with Crisp Beet Meringue

Yield: Serves 6

One of our favorite things to do is to fit unexpected flavors in familiar places. There is nothing more comforting than a creamy bowl of chocolate mousse. Beets, chamomile and orange zest punch up the flavor while beet meringues, add crunch and make a dramatic presentation.

8 ounce semi sweet chocolate

1 tablespoon butter

6 ounces heavy cream

1 orange, zested

1 tablespoon dried chamomile (optional)
¼ cup beet puree
6 egg whites
2 tablespoons sugar

Combine the chocolate and butter in a heat-proof bowl. Bring the heavy cream, orange zest and chamomile to a boil. Cover the pot and steep for 10 minutes.

Strain the boiled cream over the chocolate and let stand 3 minutes. Stir to combine and continue whisking until smooth. Whip the egg whites to soft peaks, then add the sugar and continue whisking until stiff peaks form. Gently fold the whipped egg whites into the chocolate mixture one-third at a time until fully incorporated. Refrigerate until ready to serve.

Beet Meringue

Yield: 1 13x9 baking pan

3 large egg whites
pinch cream of tartar (optional)
1/3 cup granulated sugar
¾ cup powdered sugar
½ cup roasted beet pureed until smooth

Whisk the egg whites and cream of tartar in a bowl until frothy. Gradually add the granulated sugar while whisking until stiff peaks form and the whites are smooth and glossy. Fold in the powdered sugar.

Preheat an oven to 150°F. Prepare a baking sheet with a silpat or parchment paper sprayed with nonstick spray. Place dollops of the meringue all over the tray. Drop

spoonfuls of the beet meringue in between meringue, then gently swirl with a spatula and smooth into a thin layer. Place in the oven 6 hours or until crisp.

Citrus

Most varieties of citrus known today are the result of centuries of careful breeding or hybridization. Mandarin orange, Pomelo and Citron are the only original species of citrus still commercially available today.

In citrus, color can not always be counted on to be a distinguishing factor. When grown in tropical regions that do not experience frost or cold, Winter temperatures, many varieties of citrus will remain green until fully mature. Further, color is not an indicator of ripeness in citrus fruit as oranges may turn color well before ripening.

Citrus flavor is based largely on woody or pine aromas. The word ‘Citrus’ is derived from the Greek word for ‘Cedar,’ as their aromas were found to be so similar. Citrus flavor is most prominent in the essential oils found in the outer, colored layer of the rind (known as ‘zest’). Citrus pulp is less aromatic but contains the sugars and acid that give the fruit its sweet or sour taste.

Growing regions

Citrus may be native to Southeastern Asia or Australia and New Zealand. Citrus trees arrived in Mediterranean Europe and North Africa by the time of Alexander the Great and were brought to Florida by Spanish conquistadors. Today Brazil, China, United States, Mexico, India and Spain are the top citrus producing countries. Though available year-round, all varieties of citrus are at their peak during winter months.

Common varieties

Grapefruit, pomelo, lemon, Meyer lemon, lime, orange, blood orange, clementine, mandarin/tangerine, Buddha’s Hand, Yuzu

Taste sour to sweet

Main Aromas

Citrus: *limonene*- balsamic, citrus, fragrant, fruit, herb

Orange: *valencene*- citrus, green, oil, wood

Lemon: *γ -terpinene*- bitter, citrus, gasoline, resin, turpentine

Lime: *β -pinene*- pine, polish, resin, turpentine, wood

Grapefruit: *myrcene*- balsamic, fruit, geranium, herb, must

Pairings

cilantro, ginger, bell pepper, cauliflower, broccoli, Brussel sprouts, dark chocolate

Substitute

coriander seed, cardamom, chamomile, apricot, lemongrass, lemon balm

Preserved Citrus with Sage and Chili

Preserving citrus captures, the intense flavor found in the rinds of citrus fruits. Over time, salt and sugar break down the pulp and leave the preserved rinds floating in a pleasantly pungent brine. Preserved citrus rinds may be used to a punch of flavor to meat and seafood dishes, salads or cocktails.

2 mandarin oranges

2 lemons

2 limes

2 teaspoons coriander seed

1 teaspoon fennel seed

4 branches fresh sage

3 whole dried chilies (Arbol or Thai) *or* 1 teaspoon red pepper flakes

5 tablespoons kosher salt

7 tablespoons sugar

Wash the fruit well in warm water. Cut the fruit into quarters and remove any visible seeds. Place the fruit in a bowl with your spices, salt and sugar. Mix thoroughly. Pack the mixture into clean jars or heavy-duty zip-close bags. If using jars, fill to top. If using bags, press out all of the excess air. Seal tightly.

Store at a cool temperature (below 70°F or in the refrigerator) and leave to cure approximately three weeks (refrigerated preserves will take longer), occasionally turning the jars or bags to shift the contents. Rinse citrus well before using.

Corn

Corn flavor is the result of a combination of over 200 different aromatic compounds. Sulphur compounds play the largest role in Corn flavor, though cooked corn does not present the strong, 'off' flavors often associated with sulphur. Sweetness and texture in taste perception of corn also contribute significantly to the overall flavor. Corn may be consumed fresh or dried and ground to make grits (coarsely ground dried corn), polenta (medium grind), or cornmeal (fine ground).

Nixtamalization is the process of soaking and cooking corn in an alkaline solution to improve the flavor and nutrition. The result of nixtamalized corn is known as hominy; it can also be ground into masa dough for making tortillas. In this process pyrazines (a group of aromatic compounds associated with roasted flavors) are created. The pyrazines contribute to the unique popcorn-like flavor of nixtamalized corn.

Growing regions

Native to Mexico, corn is grown abundantly throughout the world. With US, China, Brazil, EU, and Argentina being the top 5 producers. Corn typically matures in summer months, though varies known as 'winter corn' mature in cooler weather, making fresh corn available year round.

Common varieties Dint corn, Flint corn, Pod corn, Popcorn, Flour corn, Sweet corn

Taste sweet

Main Aromas

Dimethyl Sulfide- cabbage, gasoline, organic, sulphur, wet earth

Hydrogen Sulfide- rotten egg

Methanethiol- cabbage, garlic, gasoline, putrid, sulphur

Best Pairings

lime, avocado, beans, basil, tomato, tropical fruit, mushroom, caramel, nuts

Substitute

different forms of corn (grits, polenta, cornmeal), barley, peas

Pairing	Common Compounds	Aroma	Also Shared With
Corn + Coconut	2-phenylethanol	fruit, honey, lilac, rose, wine	allium, chicken, tam truffle
	ethyl acetate	aromatic, brandy, contact glue, grape, sweet	capsicum, fig, musse vinegar
Corn + Vanilla	octadecane	alkane	citrus, lamb, must pecan, rosemar
	4-hydroxybenzaldehyde	roast	apple, coffee, hor pineapple, shern
Corn + Apple	β -damascenone	boiled apple, floral, fruit, honey, tea	beer, elderberry, le balm, tea, whis
	hexyl acetate	apple, banana, confectionery, fruit, grass, herb, pear	chestnut, oregan mushroom, whis

Corn and Coconut Creme Brulee

Serves 6

This dessert takes advantage of two of corn's more unique pairings- coconut and vanilla- to make one unforgettable dessert. If you love the combination as much as I do, try the combination out in different forms like cake or cookies made with cornmeal and coconut.

2 ears corn, cleaned

2 tablespoon butter

1 teaspoon kosher salt

15 fl oz coconut milk

8 fl oz water

16 fl oz milk

3 eggs

3 egg yolks

1 teaspoon vanilla extract

½ cup granulated sugar

½ cup light brown sugar

Cut the kernels from the ear of corn. Cut each corn cob into 3 pieces and reserve.

Place a sauce pot over medium heat; add the butter and melt. Stir in the corn kernels and salt. Sauté until just tender, about 5 minutes, being careful not to let the corn or butter brown. Add the corn cobs, coconut milk and water to the pot. Bring the mixture to a boil, then reduce the heat to simmer. Cover the pot and cook at a simmer for 15 minutes.

Remove the corn cobs from the pot and transfer the mixture to a blender. Process on high until smooth. Strain the mixture through a fine mesh sieve into a bowl. Whisk in the milk, eggs and vanilla. Adjust seasoning to taste.

Pour the mixture in to brulee dishes or ramekins, filling to just below the rim. Place the cups inside a large baking dish. Fill the base of the baking dish with water, then cover

with plastic wrap. Bake in a 325°F oven until just set, about 35 minutes. Refrigerate until ready to serve.

To serve, combine the sugars in a blender or food processor and blend well. Spread a thin layer of the sugar mixture on the top of the baked custards and caramelize with a torch or under a preheated broiler. Serve immediately.

Cress

Cress refers to a variety of leafy green plants with a distinctive bitter, peppery taste. Included in this group are *Watercress* (Brassica), *Nasturtiums* (Tropaeolum) and Garden Cress (Brassica). Cress flavor comes mainly from sulphur compounds, many of which are also found in horseradish and mustard. The sulphur compounds are layered with aromas of wood and green/vegetal.

Growing regions Belgium, France, UK, Italy, United States

Common varieties upland cress, garden cress, watercress, nasturtium, nasturtium flowers

Taste bitter; sharp, mustardy

Main Aromas

Phenylethylisothiocyanate- horseradish, mustard

5-(methylthio)pentanenitrile- broccoli, cabbage

Benzenemethanethiol- burnt wood, smoke

Best Pairings

cheese, lemon, orange, berries, grains, bacon, mustard, seafood

Substitute

arugula, mustard greens, dandelion greens, radicchio

Pairing	Common Compounds	Aroma	Also Shared With
Watercress + Oyster	dimethyl sulfoxide	garlic	allium, beer, bee crème fraiche
	1-penten-3-ol	burnt, fish, green, meat, wet earth	dill, lemon, mustard
Cress (any) + Strawberry	vinylbenzene	balsamic, gasoline, plastic, rubber, solvent	lentil, olive, pea, s
	pseudocumene	pesticide, plastic	bread, crab, hazel nectarine
Nasturtium + Pineapple	allyl isothiocyanate	garlic, pungent, sulphur	beer, cucumber, blackberry
	acetic acid	acid, fruit, pungent, sour, vinegar	annatto, coffee, vanilla

Grilled Oysters with Watercress Butter

Makes approx. 8 ounces

Watercress butter is a perfect pairing for oysters, but make this in a big batch and keep it in the freezer (it keeps for months) and try it on grilled or roasted fish, chicken and beef. It's also delicious softened to room temperature and spread on bread.

2 cup watercress leaves

1 tablespoon fresh dill, minced

1 lemon, zest finely grated, juice reserved

1 shallot, minced

6 slices bacon cooked crisp and crumbled, fat reserved (optional)

¼ cup crème fraîche

2 teaspoons Dijon mustard

8 ounces butter, softened to room temperature

12 oysters, scrubbed clean and shucked

½ cup fresh breadcrumbs

Finely chop the water cress by hand and place in a small bowl. Add the dill, lemon zest and shallot to the watercress.

In a separate bowl, beat together the butter, lemon juice, mustard and reserved bacon fat (if using). When the butter is smooth, fold in the watercress mixture and crumbled bacon. Season to taste with salt and pepper.

Prepare a hot grill. Prepare a baking tray that will fit on the grill with a layer of rock or kosher salt, approximately ¼ inch thick. Press the oysters into the slat on the tray so they sit level. Top each oyster with approximately 1 tablespoon of the butter and sprinkle each with breadcrumbs. Cook on the grill with the lid closed until the butter is melted and bubbling and breadcrumbs lightly browned.

Crustacean (Crab, Lobster, Shrimp, Crawfish, Prawns, Langoustine)

Crab, Lobster, Shrimp, Crawfish, Prawns, and Langoustine are all members of the same class of Crustacean, a group of marine animals that include over 67,000 species. While the cooked meat of crustaceans can vary in texture from animal to animal, the flavor across the class is very similar with the greatest variations in flavor being created by the environment and diet of the specific animal.

Crustacean flavor is mainly created by Nitrogen-containing compounds. Specifically, amines, which are derivative of ammonia and Thiazines which also contain a sulphur atom.

Growing region

Crab- China, Venezuela, Philippines, Indonesia, United States

Lobster- US, Norway, Caribbean, Japan, Australia

Shrimp- China, Thailand, Brazil, Ecuador, Mexico

Crawfish- China, US, Vietnam, France, Northern EU

Common varieties (Taste)

Crab Blue (tender, sweet), Dungeness (meaty), Jonah (meaty, flaky), Snow Crab (sweet, delicate leg/claw meat), Stone Crab (sweet, large meaty claw) Peekytoe (sweet, mild)

Langoustine (sweet/slightly briny, delicate)

Lobster (sweet, meaty)

“Cold Water/Clawed” northern US and EU; claw meat, smaller tail

“Warm Water/Spiny” Caribbean, South America, Australia/New Zealand, large tail, no claws

Shrimp

Brown (mild flavor), White (tender, sweet), Pink (mild, sweet), Rock (sweet, lobster-like flavor)

Main Aroma

N,N-dimethyl-1Butanamine- fish

5,6-Dihydro-5-methyl-4H-1,3,5-dithiazine- fried onion, leek, meat, savory, seafood, sulphur

3-hydroxybutan-2-one- butter, cream, green pepper, rancid, sweat

Best Pairings

wine, cream, butter, garlic, nuts, breadcrumbs, lemon, apple

Substitute

mollusks or fish

Pairing	Common Compounds	Aroma	Also Shared With
Shrimp + Lamb	4-hydroxynonanoic acid lactone	apricot, cocoa, coconut, peach, sweet	beans, fenugreek, mushrooms, oregano
	2-methylpyridine	ash, sweat	allium, katsuobushi, oats, peanut, rice
Crayfish + Pumpkin	2,3-pentanedione	bitter, butter, caramel, cream, fruit, sweet	coffee, hazelnut, okra, rye bread
	2,5-dimethylpyrazine	burnt plastic, cocoa, medicine, roast beef, roasted nut	asparagus, barley, pork, rice
Crab + Cherry	2-ethyl-1-hexanol	citrus, green, oil, rose	grape, lettuce, milk, olive, tomato
	1,3-dimethylbenzene	fried, medicine, nut, plastic, rancid	avocado, dill, endive, walnut
Lobster + Beef	2,4-dithiapentane	sulphur, garlic	litchi, milk, mushroom, truffle
	octanal	citrus, fat, floral, green, pungent	artichoke, olive, pomegranate, shallot

Shrimp and Lamb Gumbo

Serves 4

This is one of the most surprising pairings that I have ever come across! With the Chef Watson team, I originally discovered the shrimp and lamb combination when creating a Creole-inspired, chickpea-dough dumpling. Yes, weird stuff! But when those same flavors are transformed into a deliciously rich and unique gumbo it all makes sense.

2 tablespoons vegetable oil
1 lb ground lamb
1 tablespoon kosher salt
2 teaspoons turmeric
2 teaspoon file powder
1½ teaspoons ground black pepper
1 cup yellow onion, diced
½ cup minced celery
½ cup green bell pepper, diced
½ jalapeno, seeds removed and diced
1 tablespoon minced garlic
1 cup canned chopped tomatoes

4 cups chicken stock
1 lb okra, sliced
1 lb shrimp, peeled, deveined and chopped
1 tablespoon chopped parsley
1 tablespoon lemon juice

Heat the oil in a large pot set over high heat. Add the lamb, salt, turmeric, file and pepper and cook, stirring well to break the meat into small pieces. Continue cooking until all the liquid has evaporated from the pot and meat begins to brown, 8-12 minutes.

Stir the onion, celery, pepper and jalapeno into meat and continue cooking until the vegetables are tender and aromatic, about 6 minutes more. Then add the garlic and cook 1 minute more.

Add the tomato and cook until dry.

Add the chicken stock and bring to a boil. Reduce the heat to simmer and season to taste with salt and pepper. When the seasoning is adjusted, stir in the okra and simmer 5

minutes. Then add the shrimp and cook 4 minutes more. Stir in the lemon and parsley just before serving.

Cucumber

Cucumbers are members of the gourd family, making them closely related to pumpkins, squash and melon. Cucumbers have the mildest flavor of this family due to high water content and low concentration of aroma compounds. The key aromas in their flavor are green vegetal and melon. While raw cucumbers are often sought out for their crisp texture, Cucumber flavor may be heightened by brief cooking via sauté or stir fry.

Growing region

China, Turkey, Iran, Russia, Ukraine are top producers.

Common varieties slicing, pickling (“Kirby,” “Liberty”), burpless

Taste slightly bitter

Main Aromas

trans-2-nonenal- cucumber, cut grass, fat, paper, watermelon

trans, cis-2,6-nonadienal- cucumber, green, lettuce, wax

cis-6-nonenal- melon

Best Pairings

citrus, coriander, mint, dill, mushroom, tomato, cheese, yogurt, seafood

Substitute

Crisp lettuce, zucchini, jicama, green beans

Pairing	Common Compounds	Aroma	Also Shared With
Cucumber + Blueberry	tetradecanal	flower, hay, honey, wax, wood	allium, beer, chicken, pistachio, scallop
	(E)-2-nonenal	cucumber, cut grass, fat, paper, watermelon	camomile, fig, mango, pecan, tamarind
Cucumber + Hazelnut	2-(sec-butyl)-3-methoxy pyrazine	bell pepper, carrot, earth, green	asparagus, beetroot, celery, parsnip, zucchini
	2-pentylfuran	butter, floral, fruit, green bean	avocado, chile, pork, shrimp
Cucumber + Persimmon	dodecanoic acid	fat, fruit, metal, wax	apricot, dill, lime, yogurt
	(Z)-2-penten-1-ol	banana, green, plastic, rubber	clam, lettuce, prickly pear, thyme

Cucumber, Berry and Hazelnut Salad with Tamarind Vinaigrette

Serves 4

Cucumbers have a particular affinity for sour flavors, making a tamarind vinaigrette the perfect pairing for cucumber.

2 English ('Hothouse') cucumbers, striped and thinly sliced

½ red jalapeno, thinly sliced

½ cup red onion, thinly sliced

1 tablespoon fresh dill or mint, chopped

¼ cup dried berries (blueberry, cranberry or mix), chopped

kosher salt and freshly ground pepper

¼ cup tamarind vinaigrette (recipe follows)

¼ cup chopped hazelnuts or sesame seeds, toasted

Combine the cucumbers, jalapeno, red onion, herbs, and dried berries in a large bowl. Season with salt and pepper. Toss well. Store covered and refrigerated until ready to serve.

Just before serving, add the vinaigrette and toss again, garnish with toasted nuts.

Tamarind Vinaigrette

Makes approx. 1½ cups

1 garlic clove, minced

1 tablespoon fresh ginger, minced

1 lime, juiced

½ teaspoon fine sea salt

¼ cup tamarind paste

½ cup extra virgin olive oil

½ cup canola oil

In a small bowl, whisk together the garlic, ginger, lime juice, and salt. Set aside 10 minutes for the flavors to combine.

After resting, whisk in the tamarind and oils. Season to taste with salt and pepper. Store refrigerated up to 3 weeks.

Egg

Eggs are comprised of two parts: the white and the yolk. Egg whites are composed of approximately 90% water, 10% protein and trace amounts of fat, and nutrients. The two main proteins in egg whites, ovalbumin and ovotranferrin, coagulate (become 'cooked') at 180°F and 140°F respectively, making the general temperature required to set an egg yolk approximately 165°F. Egg yolks contain a variety of unsaturated and saturated fatty acids and a range of vitamins and minerals. The color of the egg yolk is created by Lutein, a carotenoid whose concentration is determined by the animals diet. Egg yolks coagulate at 158°F.

When raw, eggs have little to no flavor. Through cooking sulphur compounds are released which create the distinctive flavor eggs. The longer eggs are cooked, the more sulphur compounds are released which accounts for the strong smell of over-cooked hard boiled eggs.

Growing region

China, United States, India, Japan, Mexico are top producers of chicken eggs.

Common varieties chicken, quail, duck, goose

Taste bland, though can be effected by animal's diet

Main Aromas

Hydrogen Sulfide- rotten egg

Methional- cooked potato, soy

phenyl acetaldehyde- berry, geranium, honey, nut, pungent

2-acetyl-1-pyrroline- nut, oil, popcorn, roast

Best Pairings

citrus, cream, cheese, mushroom/truffle, beef, chicken, roasted/smoked flavors, asparagus

Substitute

prepared mayonnaise, vegetable puree, cream, butter, in baking: applesauce

Pairing	Common Compounds	Aroma	Also Shared With
Egg + Truffle	dimethyl sulfide	cabbage, gasoline, organic, sulphur, wet earth	cauliflower, chile, oyster, parsley
	2-butanone	ether, fragrant, fruit, pleasant, sweet	chicken, potato, rice scallop, walnut
Egg + Caviar	ammonia	pungent	beetroot, cabbage, celery, radish, rhubarb
	2,4-decadienal	deep fried	date, hazelnut, pear sesame seed, tomato
Egg + Vanilla	thiophene	garlic	crab, coffee, pork, whisky
	1,3-dimethylbenzene	fried, medicine, nut, plastic, rancid	avocado, basil, beans, cream, watercress

Cured Egg Yolks

Makes 6 yolks

Cured egg yolks are an excellent way to add egg flavor to a variety of dishes where it would not otherwise be possible. Try these cured yolks grated like over pasta or salad. Or

use as a finishing season on cooked meat. Porcini cured yolks are particularly delicious on grilled beef.

2 cups kosher salt

1 cup granulated sugar

6 egg yolks

variations:

- Citrus: add 2 tablespoons finely grated citrus zest to cure mix
- Porcini: add 1 tablespoon porcini powder to cure mix
- Spice: add 1 tablespoon finely ground black pepper, plus 1 teaspoon each ground coriander, fennel seed and red pepper flake
- Umami: 1 teaspoon porcini powder, plus ¼ cup crushed bonito flakes

In a small bowl, whisk together the salt sugar, and any additional seasonings.

Spread half of the salt mixture in the bottom of a tupperware or glass baking dish. Use your thumb to make 6 indentions into the mixture.

Carefully place an egg yolk in each indentation. Use the remaining mixture to cover each yolk. Cover the container and refrigerate for 3 days.

After curing, remove the yolks from the container and transfer to bowl filled with room temperature water. Remove the yolks one-by-one, gently rinsing to remove any remaining cure mix. Place on paper towels and pat dry. Transfer to a cooling and leave 2 hours to air dry at room temperature.

After drying, move the yolks to a sealable plastic container and reserve, refrigerated up to two weeks. *To use*, grate the yolks over pastas, salads, or cooked meats.

Eggplant

Eggplant earned its name because early European cultivars of the plant were small oblong and white—like an egg. Eggplant is a member of the nightshade family *Solanum* which includes tomatoes and potatoes. They prefer similar growing conditions to a tomato, so these two ingredients are often found in recipes together.

Eggplants have very little aroma when raw and they're largely inedible. When cooked, Maillard reactions and Earthy aromas of cooked vegetable provide the majority of flavor.

Contrary to popular belief, salting eggplant will not draw out 'bitterness' from the flesh. Salting before cooking helps to collapse cell walls, reducing the 'spongy' structure of the eggplant and making it less likely to absorb excess liquid during cooking. This means that when roasting or grilling eggplant the salting process is really unnecessary.

Growing region

Eggplant was first cultivated in southern and eastern Asia. Today it is an important part of cuisines of from Asia, throughout the Mediterranean, and the Americas. China, India, Iran Egypt and Turkey are the top producers globally.

Common varieties Italian (oval), Japanese (slim), Fairy Tale (small)

Varieties may come in purple, white, or green

Taste bitter to slightly sweet when cooked

Main Aromas

Hexanol- cooked vegetable, flower, green, herb, rotten

Toluene- glue, paint, solvent

3-carene- bell pepper, lemon, pungent, resin, rubber

Best Pairings

tomato, mushroom, cheese, yogurt, olive, wine, lemon, orange, broccoli, basil

Substitute

zucchini, summer squash, portobello mushroom

Pairing	Common Compounds	Aroma	Also Shared With
Eggplant + Walnut	1,2-dimethylbenzene	geranium	artichoke, beer, lentils, southern peas, scallop
	acetophenone	almond, flower, meat, must, plastic	blue cheese, cocoa, corn, tea
Eggplant + Pomegranate	heptanal	citrus, dry fish, fat, green, nut, rancid	apricot, beef, dill, sesame seeds
	β -bergamotene	tea	basil, carrot, citrus, pepper
Eggplant + Elderberry	linalool	bergamot, citrus, coriander, floral, lavender, lemon, rose	cumin, hyssop, lemon balm, sake
	heptane	alkane, burnt matches, floral, plastic	chestnut, port, thyme, yogurt
Eggplant + Lamb	heptane	alkane, burnt matches, floral, plastic	chestnut, milk, thyme
	heptanal	citrus, dry fish, fat, green, nut, rancid	cilantro, ginger, corn, pistachio, orange

Fried Eggplant with Roasted Red Pepper, Walnut and Pomegranate Molasses Sauce

Serves 4

This recipe is based on the pairing of eggplant and pomegranate which we just found supremely enticing. The fried eggplant make a great hors d'oeuvre simply topped with a dollop of Muhammara, a Syrian red pepper sauce flavored with pomegranate syrup. Or go all out and assemble it like an eggplant parm, alternating layers of eggplant and muhamamara.

2 lbs eggplant (Italian or Japanese) sliced ½ inch thick
kosher salt
freshly ground black pepper
2 cups all purpose flour
2 eggs
3 cups breadcrumbs
canola or vegetable oil, as needed
½ cup muhammara, plus more as needed (recipe follows)
¼ cup torn basil leaves, for garnish
shaved pecorino cheese (or parmesan) for garnish

Lay the eggplant slices out on a baking pan and season with salt and pepper.

Set up 3 shallow bowls or baking dishes. Add the flour to the first and season with salt and pepper. Crack the eggs into the second; add ½ cup water and a pinch of salt to the eggs and beat thoroughly. Place the the breadcrumbs in the third pan and season with salt and pepper.

Dip the eggplant slices into the flour. Pat off any excess flour, then dip into the eggs, then press into the breadcrumbs to coat each side. Set aside on a clean pan while you batter the remaining eggplant.

Fill a large, wide sauté pan with ½ inch of oil and place over medium heat. Heat the oil to 350°F or until breadcrumbs sprinkled into the oil sizzle immediately. Add as many eggplant slices as will fit into the pan without touching. Cook until golden brown on one side then flip and brown on the second side. Remove to a rack to drain while you fry the remaining pieces.

When all the eggplant is cooked, arrange the slices on a platter and spoon over the muhammara. Garnish with the basil and pecorino.

Muhammara (Syrian Roasted Pepper, Walnut and Pomegranate Sauce)

makes 2 cups

1 lb roasted red bell peppers, skins and seeds removed

1 cup walnuts, toasted

1 tablespoon pomegranate molasses

2 tablespoons extra virgin olive oil

1 teaspoon ground cumin

1 teaspoon Aleppo pepper or sriracha, to taste

kosher salt, as needed

Combine the peppers, walnuts, pomegranate molasses, olive oil and cumin in a blender. Process until smooth. Season to taste with salt and Aleppo pepper.

Fennel

Fennel is a member of the botanical family that includes carrot, celery and parsley. Like carrot, fennel may be consumed raw or cooked and the entire plant has viable culinary uses, from the bulb to the woody tops and, finally, to the feathery leaves. Fennel seeds and their flowers or pollen also find their way into dishes around the world.

The compound *anethole*, gives fennel its distinctive anise, or licorice-like flavor. While limonene, the prominent compound of citrus, plays a secondary role in fennel flavor.

Growing region

Native to Mediterranean, India, Mexico, China, Iran, Bulgaria are now the top producers of fennel.

Common varieties Florence Fennel, Bronze Fennel

Taste sweet

Main Aromas

Anethole- anise, sweet

Limonene- balsamic, citrus, fragrant, fruit, herb

α -phellandrene- citrus, mint, pepper, turpentine, wood

Best Pairings

citrus, basil, dill, cilantro, ginger, lemongrass

Substitute

celery, lovage, caraway, hoja santa

Pairing	Common Compounds	Aroma	Also Shared With
Fennel + Hyssop	antheole	anise, sweet	dill, coriander, elderberry, rhubarb, shiso
	linalool	bergamot, coriander, lavender, lemon, rose	camomile, celery, mint, turmeric
Fennel + Plum	4-methoxybenzaldehyde	almond, anise, mint, sweet	anise, basil, coffee, hazelnut, peppercorn
	1,8-cineol	camphor, cool, eucalyptol, mint, sweet	gin, lemongrass, passion fruit, star anise
Fennel + Cranberry	limonene	balsamic, citrus, fragrant, fruit, greenery, herb	black currant, caraway, chervil, lovage, pecan, pineapple
	1-isopropyl-4-methylbenzene	citrus, fresh, gasoline, solvent	curry leaf, lime, yogurt, parsley

Bourbon Roasted Fennel and Plums with Coffee Ice Cream

Serves 4

Fennel doesn't often find its way into dessert. But when it's paired with the sweetness of plums- and bourbon and coffee lend their roasty and toasted aromas- a sweet finish to a meal is a welcome landing spot for this aromatic veggie.

2 tablespoons butter

2 tablespoons light brown sugar

2 plums cut in half, pit removed

1 cup fennel, thinly sliced

2 fl ounces bourbon

½ teaspoon kosher salt

Optional Garnish:

Micro basil

Toasted pecans

Scoop the ice cream into 4 separate bowls and place in the freezer until ready to serve.

Melt the butter in a medium cast iron skillet set over medium heat. When the butter melts it will begin to foam. Let it continue cooking until the foaming subsides and it takes a light brown color.

Add the sugar and stir until melted. Place the plum halves cut-side in the pan and continue cooking for 3 minutes. Scatter the sliced fennel over the plums and gently toss to coat with the brown sugar mixture.

Cook an additional 3 minutes, stirring occasionally. Remove the pan from the heat and stir in the bourbon and salt.

Transfer to a bowl and cool 5 minutes, then spoon over ice cream.

Fig

Fig was one of the first plants to be cultivated by humans, with evidence of fig trees being planted and cultivated one thousand years before humans planted wheat or rye. While the trees are mainly grown for their fruit, another valuable commodity are their highly aromatic leaves used to wrap around other foods before grilling or roasting to impart their flavor during the cooking process. Fig fruit is thin-skinned, with tender flesh and very small seeds that can sometimes be gritty. The fruit is typically consumed whole, either raw or cooked. Their flavor comes from a mix of fruit and woody aromas.

Growing region

Native to Middle East and Western Asia. Turkey, Egypt, Algeria, Morocco, Iran

Common varieties Black Mission, Brown Turkish, Kadota (green)

Taste sweet

Main Aromas

(E)-2-hexanal- green banana, fresh, herbal, spice

5-methylfurfural- almond, caramel, cooked, roasted garlic, spice

β-Caryophyllene- fried, spice, wood

Best Pairings

lemon, orange, berries, roasted/smoked meats, wine, tea,

Substitute

apricot, date, cherry, cantaloupe

Pairing	Common Compounds	Aroma	Also Shared With
Fig + Avocado	2-phenylethanol	fruit, honey, lilac, rose, wine	cocoa, corn, mezcal, watermelon
	3-hydroxy-2-butanone	butter, cream, green pepper, rancid, sweat	beef, blue cheese, cherry, olive, vinegar
Fig + Olive	ethanol 6-methyl-5-hepten-2-one	citrus, mushroom, pepper, rubber, strawberry	carrot, citrus, lemongrass, thyme, walnut
	(E,E)-2,4-decadienal	coriander, deep fried, fat, oil, oxidized	apricot, camomile, cauliflower, chicken, mushroom
Fig + Clam	ethanol	alcohol, floral, ripe apple, sweet	apple, grape, peach, pineapple, pistachio, vinegar
	geranylacetone	green, hay, magnolia	apricot, capsicum, passion fruit, sage

Fig, Olive and Walnut Relish

Makes approx. 2 cups

This mix makes a wonderful accompaniment to cheese and charcuterie. It is also delicious on grilled or roast chicken or fish, avocado toast or a cauliflower steak.

½ cup pitted nicoise olives, cut in half

1 lemon, juiced and zested

½ teaspoon fresh thyme leaves

pinch granulated sugar

2 tablespoons extra virgin olive oil

¼ cup toasted walnuts, chopped

1 cup Black Mission or Brown Turkish figs, diced

kosher salt

freshly ground pepper

Combine the olives, lemon juice and zest, sugar, thyme and olive oil in a bowl. Mix thoroughly.

Add the walnuts and diced figs and gently fold to incorporate. Season to taste with salt and freshly ground pepper.

Fish

Fish can be divided to a few simple categories to classify for culinary purposes.

Fresh or Saltwater environments. Fish that live in saltwater environment produce more glycine and glutamate (umami-creating amino acid) to counteract the salinity in their environment. Meaning that in general saltwater fish tend to have a richer, fuller taste.

Fatty or Lean Fish. Much of a fish's flavor is carried by the natural oils present in the flesh. Lean fish will have a more mild 'fish' flavor while fatty fish taste more 'fishy.' Freshness is essential in fatty fish as the natural oils (fatty acids) are quick to breakdown or go rancid and create off-flavors in the fish.

High Activity or Low Activity. Like most animals, the more a muscle is used, the more flavorful it tends to be. High activity fish like: mahi, salmon, mackerel, cobia and tuna tend to be more flavorful than low activity haddock, pollock and cod. Bass, snapper, grouper and tilefish are considered medium activity fish.

Like crustaceans, the flavor of fish is produced by Nitrogen-containing compounds. Specifically, amines, which are derivative of ammonia and Thiazines which also contain a sulphur atom.

Growing region Global

Common Species

Lean Fish: Flounder, Sole, Cod, Snapper, Bass, Perch, Halibut, Pike, Catfish, Grouper, Mahi, Monkfish, Shark, Skate, Trout, Tuna

Fatty Fish: Anchovy, Carp, Herring, Mackerel, Pompano, Sable, Salmon, Sardines

Taste bland

Main Aromas

2,4,6-tribromophenol- saltwater fish, brine-like

2,6-nonadienal- cucumber, green

1,5-octadien-3-ol- earth, geranium, green, herb, mushroom

Best Pairings

lemon, orange, coconut, melon, toasted nuts, breadcrumbs, olive/capers, cream

Substitute different varieties of fish may be substituted for one another

Coffee Cured	Pairing	Common Compounds	Aroma	Also Shared With
	Lean Fish + Melon	1-penten-3-ol 2,3-butanediol	burnt, butter, fish, green, meat, oxidized, wet earth fruit, herb, onion	apricot, clam, olive bell pepper, date, f vinegar
	Lean Fish + Peanut	2-decenal naphthalene	fat, fish, orange, paint mothball, tar	carrot, cilantro, lar walnut avocado, chili, kiw sage
	Fatty Fish + Coffee	phenol 2-ethyl-1-hexanol	medicine, phenol, sharp, smoke, spice citrus, green, oil, rose	cocoa, olive, por sesame seed annatto, lettuce pomegranate, swe
	Fatty Fish + Beer	pyrrolidine formic acid	alkaline pungent	barley, caviar, poler citrus, brandy, pea

Salmon

Serves 4

This home-cured fish makes for the ultimate weekend bagels and lox buffet. After all, who's not already having coffee with their favorite morning deli food?

3 tablespoons fresh ground coffee, plus more as needed

1 cup kosher salt

½ cup granulated sugar

Zest of 1 lemon

1 lb salmon filet, skin removed

Rub the coffee liberally onto both sides of the salmon. set the coffee-rubbed aside while you prepare the cure mix.

In a small bowl, combine the salt, sugar and zest. Mix well.

Place a piece of plastic wrap about 18 inches long on the table. Mound half of the salt mixture in the center of the plastic wrap and press into a rectangle, slightly bigger than the piece of fish. Place the fish on top of the salt and cover with the remaining mixture. Wrap tightly in the plastic and refrigerate for 6 to overnight.

Remove the fish from the cure and quickly rinse under cold water. Pat dry and refrigerate until ready to serve.

To serve, cut the fish into slices approximately $\frac{1}{4}$ inch thick. Lightly brush the slices with olive oil and place them between two sheets of plastic wrap. Pound very lightly until the slices are $\frac{1}{8}$ inch thick.

Serve on toasted everything bagels with cream cheese.

Or serve plated with mint, olives, pomegranate seeds and toasted sesame seeds.

Game Meats

Game Meat is the term commonly used to refer a category of animal that are hunted for both sport and food. Many animals classified as 'game' are now domesticated and raised as a food source, such as venison, elk and boar. This chapter will focus on those animals. An accurate depiction of the flavor of all game meats is difficult because environmental factors can create dramatic differences in the flavor of wild animals.

The compounds responsible the typical flavor of game meats like venison, elk and boar are not particularly appetizing on their own. In isolation, the compounds responsible for game meat flavors have aromas like 'pungent, solvent, or fecal.'

Venison refers to any portion of meat that comes from deer. The majority of venison sold in North America is raised in New Zealand and may be labeled under the appellation name Cervena. Venison meat is deep red color and extremely lean. It has a robust meaty flavor and relatively mild 'game' flavor.

Elk are a separate breed, mainly raised in North America, and have a flavor that is very similar to venison.

Boar or wild pigs grow more slowly in captivity than their relatives, the domesticated pig. Boar tend to yield less meat than domesticated pigs, though the meat they produce is thought to be of higher nutritional value. It is generally leaner and with a greater concentration of essential amino acids. Boar meat has a red coloring that is more similar to beef than pork and a bolder flavor than pork.

Growing region New Zealand, Ireland, Great Britain, Germany, North America

Common Species Venison, Elk, Boar

Taste umami

Main Aromas

Acetone- chemical, ether, nauseating, pungent

n-butanol- alcohol, fruit, medicine, solvent

isobutanol- apple, bitter, cocoa, fusel, solvent

skatole- fecal, mothball

Best Pairings Earthy: mushroom, parsnip, winter squash; bourbon, coffee, rosemary, juniper, red wine

Substitute Any game meat may be substituted for another; or lamb, beef

Pairing	Common Compounds	Aroma	Also Shared With
Game Meat + Eucalyptus	octadecane	alkane	cashew, dill, mustard, radish, scallion
	linalool	bergamot, coriander, lavender, lemon, rose	basil, cardamom, citrus, coriander, nutmeg, pomegranate
Game Meat + Cilantro	heptanal	citrus, dry fish, fat, green, nut	beer, fig, sour cream, potato
	a-pinene	cedarwood, pine, resin, sharp, turpentine	lemongrass, jalapeno, juniper, shiso, pomegranate
Game Meat + Pineapple	γ -dodecalactone	apricot, flower, fruit, peach, sweet	chervil, mushroom, nectarine, olive
	butyric acid	butter, cheese, rancid, sour, sweat	apple, butter, coffee, sour cherry

Game Meat Seasoning

Makes approximately ½ cup

The citrus and spice notes in the rub are the perfect compliment for game meats like venison, elk or boar. Simply rub generously onto meat before cooking. It is also delicious on meaty fish like tuna or swordfish.

2 tablespoons black peppercorn

2 tablespoon juniper

1 tablespoon coriander seed

1 tablespoon dried rosemary leaves

2 teaspoon dried orange peel

3 tablespoons kosher salt

1 tablespoon sugar

Combine the peppercorn, juniper, coriander seed, rosemary and orange peel in a mortar and pestle or spice grinder. Grind until thoroughly broken down, but not too fine.

Transfer to a small bowl and mix in the salt and sugar. Reserve in an air tight container up to 3 months.

Ginger Root

Ginger belongs to the family Zingiberaceae, which includes other pungent spices like turmeric, cardamom and galangal (remember- plants from the same family tend to pair together very well). The knobby root of the ginger plant has a pungent flavor with strong aromas of wood/pine and spice.

Gingerol, the compound that gives raw ginger its spicy taste, is a relative of capsaicin, the active component in chili peppers. Cooking transforms gingerol into zingerone, a compound with less spice and a strong pine aroma.

Growing region

India, China, Nepal, Indonesia, Nigeria are top producers.

Common varieties

Indian Ginger, Jamaican Ginger, Chinese Ginger

Taste sweet and spicy

Main Aromas

Zingiberene- fresh, sharp, spice

β-sesquiphellandrene- wood

humulene- balsamic, hop, spice, wood

Best Pairing

allium, citrus, pinenut, origanum, pepper, basil, coriander, parsley

Substitute

lemongrass, rosemary, galangal, szechuan peppercorn

Pairing	Common Compounds	Aroma	Also Shared With
Ginger + Blueberry	borneol	camphor, fragrant, green, polish	cardamom, eucalyptus, lime, sage
	2-undecanone	fresh, green, orange, rose	chicken, coconut, lamb, passion fruit, turmeric
Ginger + Cumin	β -phellandrene	mint, turpentine	black currant, dill, fennel, pink peppercorn
	β -bisabolene	balsamic, citrus, spicy	anise, carrot, oregano, star anise
Ginger + Pistachio	geraniol	geranium, lemon peel, passion fruit, peach, rose	basil, cinnamon, hyssop, lemongrass
	α -farnesene	boiled vegetable, floral, sweet, wood	calvados, lemon, quince, rosemary

Ginger Mule with Pistachio Vodka

Yield: makes 4 drinks

Ginger and pistachio make for a surprising pairing. The resinous flavor in pistachios works incredibly well with the pine flavor of ginger. The homemade pistachio vodka in this recipe is a simple task that is well worth the wait.

2 ounces ginger syrup, recipe below

2 ounces lime juice

6 mint leaves

6 ounces pistachio vodka, recipe below

4 ounces ginger beer

In the bottom of a cocktail shaker, muddle the mint leaves with ginger syrup and lime juice. Add vodka and fill with ice. Shake to chill. Strain into rocks glasses filled with 3 or 4 large ice cubes. Top each glass with approximately 1 ounce ginger beer.

Ginger Syrup:

1 piece fresh ginger, skinned and thinly sliced (about ½ cup)

4 cups water

2 cups granulated sugar

Combine all ingredients in a medium saucepot. Bring the mixture to boil. Reduce the heat to simmer and cook 5 minutes. Remove from the heat and allow to cool completely.

Strain and reserve refrigerated, up to 4 weeks.

Pistachio Vodka

¾ cup chopped pistachio, toasted

18 fl oz vodka

Combine the chopped pistachio and vodka in a glass jar or bottle. Shake vigorously for 30 seconds.

Store the jar in a cool, dark place for 7 days. Shaking the jar once each day. After a week, strain and reserve in a cool, dark place until ready to use.

Grain- Wheat, Barley, Oat, Farro, Quinoa

Grain or cereal grains, are the edible seeds of grass that have been cultivated by humans as a food source for nearly 9,000 years. All grains are made up of 3 parts:

- Bran, the edible outer seed covering
- Endosperm, stored starch that makes up the bulk of the grain
- Germ, like the yolk of an egg, it is the embryo of the seed and has a high oil and nutrient content

Flavor-creating aromatic compounds are found mostly in the bran and germ portions of grain, rendering refined grain products like white flour nearly flavorless until they undergo Maillard reactions in the cooking process. Compounds in the bran contribute roasted and toasted aromas (nut and cocoa) while smaller germ has aromas of fat or lightly fishy.

Growing regions of Common Varietals

Barley *Russia, Germany, France, Canada, Spain*

Oat *Russia, Canada, Poland, Finland, Australia*

Wheat *China, Russia, India, US, Canada*

Farro (Emmer Wheat) *Armenia, Morocco, Spain, Italy*

Quinoa *Bolivia, Peru, Ecuador*

Taste bitter to slightly sweet

Main Aromas

2-methylpropanal- caramel, cocoa, green, malt, nut

3-methylbutanal- almond, cocoa, fermented, hazelnut, malt

heptanal- citrus, dry fish, fat, green, nut

Best Pairing

honey, citrus, roasted meats, butter, cheese, seafood, toasted nuts

Substitute All varieties of grain may be substitute for one another or lentils, split peas, dried beans.

Pairing	Common Compounds	Aroma	Also Shared With
Barley + Swiss Cheese	2-methylbutanal	almond, cocoa, fermented, hazelnut, malt	apple, beer, olive, rye bread
	4-hydroxypentanoic acid lactone	herb, sweet	beef, mushroom, soybean, tomato
Oat + Coconut	methyl dodecanoate	coconut, fat	allium, cocoa, mustard, passion fruit
	methypyrazine	cocoa, green, hazelnut, popcorn, roasted	cashew, chicken, peanut, sesame seed, shrimp
Rice + Clam	geranylacetone	green, hay, magnolia	cream, macadamia, sage, tomato
	2-nonenal	paper	asparagus, ginger, olive, peas, pork

Creamy Coconut Oats with Shrimp and Jalapeno

Serves 4

Coconut and oats may sound like something well-suited for the breakfast table or maybe even the dessert course, but we thought this pairing needed to explore its savory side. Here, creamy oats get a spicy kick from jalapeno and are then finished off with grilled shrimp and toasted sesame. All of the flavors in this dish build on the aromas naturally present in grains.

2 tablespoons butter

4 pieces scallion, thinly sliced

2 cloves garlic, minced

1 teaspoon minced jalapeno

1 cup rolled oats

2 cups coconut milk

16 grilled or sautéed shrimp

Garnish: Cilantro, Toasted Sesame Seeds

Melt the butter in a saucepot. Add the scallion, garlic and jalapeno and sauté over medium heat until tender and very aromatic, about 3 minutes.

Add the oats and toast with the aromatics. Stir in the coconut milk and bring the mixture to simmer. Cook at a simmer until tender and creamy. If the mixture becomes too thick, add water and continue cooking to desired consistency.

Serve topped with grilled shrimp or fish and garnish with cilantro and sesame seeds.

Grape (Table Grapes)

Table grapes are grown to be consumed fresh or cooked. Different varieties are grown for wine making or for drying to then produce raisins. Varieties of table grapes can be grown for specific size, shape or color.

All varieties of table grapes share certain flavor characteristics. Hexanol, a major contributor to grape flavor, is also a key aroma in green vegetables (green beans, lettuce) and pomegranate. Other prominent aromas in grape flavor are berry and citrus. Grapes have an affinity for sour flavors, spices with citrus notes, and fruit-like vegetables (peppers, pumpkin and tomato).

Growing region

Native to Asia. China, EU, Turkey, US, and Chile are the top producers of table grapes.

Common varieties

Thompson Seedless, Flame Seedless, Concord, Delaware, Cotton Candy, Muscadine

Taste sweet to tart

Main Aromas

1-hexanol- cooked vegetable, flower, green, herb, rotten

(Z)-3-hexen-1-ol- bell pepper, grass, green leaf, herb, unripe banana

β -ionone- cedarwood, floral, raspberry, seaweed, violet

octanal- citrus, fat, floral, green, pungent

Best Pairings

arugula/cress, walnut, mustard, vinegar, honey, stone fruit

Substitute

melons, fig, gooseberry, strawberry

Spice Roasted Grapes <i>Makes</i>	Pairing	Common Compounds	Aroma	Also Shared With
	Grape + Pumpkin	ethyl tetradecanoate	ether, soap, wax	anise, apple beer, coconut passion fruit
		ethyl acetate	aromatic, brandy, contact glue, grape, sweet	allium, beetroot, chestnut pumpkin, radish
	Grape + Pork	3-methylphenol	fecal, medicine, phenol, sharp, urine	cheese, cocoa, hazelnut mushroom, rum
		acetophenone	almond, flower, meat, must, plastic	cauliflower, clam, pomegranate vanilla
	Grape + Elderberry	4-hydroxynonanoic acid lactone	apricot, cocoa, coconut, peach, sweet	asparagus, beef, chile, corn watercress
		linalool oxide D	citrus, floral, fragrant, green, sweet	fig, lemon balm, sage, strawberry

approximately 2 cups

Herbs and spices here accentuate some of the lesser flavors in grapes (citrus and wood) while vinegar takes down sweetness to give them a savory edge. Serve with sautéed fish, on a cheese plate, in a salad or with roasted vegetables (like pumpkin, beets or cauliflower) or roasted meats.

- 2 tablespoons extra virgin olive oil
- 1 shallot, thinly sliced
- 1 teaspoon fennel seeds
- 3 pods whole cardamom
- 1 dried red chile (arbol or thai)
- 6-8 branches fresh rosemary
- 3 cups red or green grapes, cut in half

kosher salt

freshly ground pepper

¼ cup red or white wine vinegar

Preheat oven to 425° F

Heat the olive oil in a saute pan over medium heat. When hot, add the shallots and saute 1 minute, until just softened. Add the fennel seeds, cardamom and pepper flakes and thyme and cook 30 seconds more.

Add the grapes to the pan, toss well and season with salt and pepper. Transfer the pan to the oven and roast for 5 minutes.

Remove the pan and immediately deglaze with the vinegar. Toss well and serve warm.

Green Beans

There are over 130 varieties of green beans; they may range in color from green to yellow, purple, red or streaked. After cooking, red and purple varieties tend to revert back to a dark shade of green.

Green beans are classified by how they grow. Bush Beans grow on small plants, low to the ground while Pole Beans grow tall or ‘climb’ and require a trellis or other support. A single variety of green bean may have both Bush and Pole variants. Green beans exhibit classic green vegetable aromas produced by hexanol with much lighter fruity aromas coming from hexanal.

Growing region

China, Indonesia, India, Turkey, Thailand are top producers.

Common varieties

Bush bean (Blue Lake, Bush Kentucky Wonder, Derby, Kenyan Bean, Purple Teepee)

Pole bean (Algarve, Blue Lake, Golden Gate, Kentucky Blue, Kentucky Wonder, Scarlett Wonder)

Haricot Vert French for ‘green bean’ refers to tender, young Bush Beans.

Taste bitter

Main Aromas

n-hexanol- cooked vegetable, flower, green, herb, rotten

n-hexanal- fresh, fruit, grass, green, oil

l-octen-3-ol- cucumber, earth, fat, floral, mushroom

Best Pairings

fennel, mushroom, toasted nuts, allium, wine, cheese, butter

Substitute

green peas, edamame, fava beans, asparagus, cactus

Pairing	Common Compounds	Aroma	Also Shared With
Green Bean + Vanilla	vanillin	sweet, vanilla	cinnamon, fish, fennel, pineapple, pork,
	1-phenylethanone	almond, flower, meat, must, plastic	allium, corn, lentil, walnut
Green Bean + Hazelnut	2-pentylfuran	butter, floral, fruit, green bean	artichoke, jalapeno, parsley, shrimp
	2-(sec-butyl)-3-methoxy- pyrazine	bell pepper, carrot, earth, green	beetroot, cucumber, parsnip, pumpkin
Green Bean + Dill	β -damascenone	boiled apple, floral, fruit, honey, tea	crème fraiche, lemon, savory, vinegar
	(Z)-3-hexen-1-ol	bell pepper, grass, green leaf, herb, unripe banana	chervil, endive, peas, radish

Crunchy Green Bean “Granola”

Makes 2 cups

Vanilla is a particularly tasty pairing for green beans. There are historical recipes from Venice that combine green beans with tomato and vanilla. In this recipe, green beans lean to the sweet side and lend crunch to a granola which makes a fantastic topping for dishes savory or sweet. For a particularly strong pairing, try this mixture as a garnish on pureed pumpkin or squash soup.

2 quarts water
1 tablespoon sugar
2 tablespoons kosher salt
1 lb thin green beans (haricot vert)
vegetable oil, as needed
2 tablespoons butter
1 cup hazelnuts, chopped
¼ cup unsweetened, shredded coconut
2 tablespoons sesame seeds
¼ cup honey
1 teaspoon vanilla extract

Preheat oven to 150°F or a food dehydrator to 130°F

Bring the water, sugar and salt to a boil in a medium saucepot.

Trim the green beans, then add to the boiling water and cook for 60 seconds. Remove the green beans and place in a colander to drain for approximately 30 seconds, then spread the beans on cooling rack to cool to room temperature.

When cooled brush or spray the green beans with a very light coating of oil. Place in the oven or dehydrator to dry. At 150°F the beans will take about 6 hours, at 130°F they will take approximately 10 hours.

When the beans are completely dried, remove from the oven and coarsely chop. Reserve in an air tight container.

Prepare a baking pan with lightly oiled parchment paper.

Melt the butter in a small sauce pot. Add the hazelnuts, coconut, and sesame seeds and stir well to coat. When all of the nuts are well toasted stir in the honey. Continue cooking over medium heat until the honey begins to simmer. Cook 1 minute at a simmer. Remove from the heat and stir in the vanilla. Immediately transfer the mixture to the prepared baking sheet and press out to single layer. Bake in a 250°F oven 30 minutes. Remove and cool completely. Coarsely chop the nuts and mix with the green beans.

Honey

Specific honey flavor is determined by where bees source their nectar to then produce honey. The flavors can range from the spiced flavor of clover honey to citrusy orange blossom honey. Many commercial honey are blended, or have a mixture of honey from multiple floral sources. In this chapter we will look at the flavor characteristics shared by all varieties of honey.

Honey contains Maillard-like flavors created by furanones, which also have strong antimicrobial properties. Floral aromas are persistent throughout all varieties of honey as are vegetal aromas. Honey generally pairs well with all ingredients and should be a considered a higher flavor option for sweetening.

Growing region

China, Turkey, Argentina, Ukraine, United States are top producers.

Common varietal

Depends on floral source; (in the US) Alfalfa, Avocado, Blueberry, Buckwheat, Clover, Eucalyptus, Fireweed, Orange Blossom, Sage, Tupelo, Wildflower

Taste sweet

Main Aromas

1-(2-furanyl)-ethanone- balsamic, cocoa, coffee, smoke, tobacco

2, 3-butanediol- cream, floral, fruit, herb, onion, rubber

3-hydroxy-2-butanone- butter, cream, green pepper, rancid, sweat

Best Pairings

citrus, dried fruit, wine, mustard, tea, lettuce/greens, vinegar, alcohol

Substitute

maple syrup, molasses, cane syrup, treacle

Pairing	Common Compounds	Aroma	Also Shared With
Honey + Sage	2-methyl-1-propanol	apple, bitter, cocoa, fusel, plastic, solvent	apple, mezcal, tamarind, vinegar
	carvacrol	caraway, spice, thyme	lemon balm, nutmeg, orange, pistachio
Honey + Jalapeno	methyl 2-methylbutanoate	apple, fruit, green apple, strawberry, sweet	cherry, lime, persimmon, prickly pear, thyme
	linalool	bergamot, coriander, lavender, lemon, rose	apricot, cardamom, coriander, juniper, mint, star anise
Honey + Mushroom	2-phenylethanol	fruit, honey, lilac, rose, wine	asparagus, barley, chervil, endive, mustard, watercress
	4-methoxybenzaldehyde	almond, anise, mint, sweet	anise, basil, date, fennel

Spicy Honey Syrup

Makes 2 cups

With its pairing affinity for alcohol, honey is a particularly good option for mixing cocktails. This spicy honey syrup is particularly great in tequila, gin and vodka drinks. Or for a non-alcoholic option use this syrup to sweeten lemonade.

1 cup honey
1 cup water
1 Jalapeno, thinly sliced
1 tablespoon serrano
4 branches fresh thyme
1 teaspoon black or Szechuan peppercorn
1 pod star anise

Combine all ingredients in a small sauce and bring the mixture to a simmer. Cook at a low boil for 5 minutes. Allow the mixture to cool completely in the pot. Strain into a clean bottle or jar and store refrigerated up to 1 month.

Jerusalem Artichoke

Jerusalem Artichokes or Sunchokes are the starchy root of a plant in the sunflower family. Their flavor is often described as being similar to that of cooked artichoke, hence their name. However, modern Jerusalem Artichokes have a richer, nuttier flavor, with main aroma compounds contributing woody, pine and nutty, scents. They are also extremely versatile and may serve as a more flavorful substitute in recipes that call for potato. While they do require peeling, a good scrub will do the trick to remove any grit from the skin. Jerusalem artichokes can add a delicious, nutty crunch to salads if thinly sliced and added raw. Peak season is November through February.

Growing region

Native to North America, US and Canada remain the top producers and they are grown to a less extent in Europe.

Common varieties

Mammoth, Oregon, Columbia, Stampede, Fuseau

Taste sweet to slightly bitter

Main Aromas

β -bisabolene- balsamic, woody

α -pinene- cedarwood, pine, resin, sharp, turpentine

terpinolene- sweet, pine, citrus

Best Pairings

lemon, orange, mushroom, roasted meat, allium, cream, butter

Substitute

crosnes, jicama, water chestnut, potato

Pairing	Common Compounds	Aroma	Also Shared With
Jerusalem Artichoke + Cherry	diacetyl	butter, caramel, fruit, pastry, sweet	beetroot, cider, tu vinegar
	hexadecane	alkane, root	eggplant, citrus, m thyme
Jerusalem Artichoke + Gin	β -farnesene	citrus, oil, wood	carrot, cumin, lemo pomegranate
	pyridine	burnt, pungent, rancid	cabbage, mushroom soy sauce
Jerusalem Artichoke + Crab	propylbenzene	mothball	garlic, honey, pe watercress
	anisole	anise	beef, cream, poblan

Gin and Brown Butter Emulsion

Makes about 10 ounces

The juniper and botanicals of gin coupled with the maillard powerhouse of brown butter come together to make the perfect flavor match for roasted Jerusalem artichokes. But this sauce should become a regular in your repertoire, try it on roasted meats, potatoes, fish or other root vegetables as well.

8 ounces butter

4 garlic cloves, smashed

4-6 branches woody herbs (sage or thyme or rosemary or oregano)

2 tablespoons shallot, minced

4 fl ounces gin

2 tablespoons heavy cream

juice of 1 lemon, as needed

kosher salt

freshly ground black pepper

Melt the butter in a small sauce pot over medium heat. Simmer until all the water has evaporated and the butter look clear. Continue cooking until the butter takes on a deep brown color and has a nutty, roasted aroma.

Remove the pot from the heat and carefully add the garlic and herbs to the butter. Set aside to infuse and cool.

Transfer a spoonful of the browned butter to a separate pan. Add the shallots to the pan and sauté over medium heat until tender and sweet smelling. Remove the pan from the heat and deglaze with the gin. Carefully place the pan back on the heat and reduce by half. Stir in the heavy cream and reduce 1 minute more.

Strain the butter into a heat proof container.

Transfer the gin mixture to a blender and blend on low. Pour the brown butter into the blender in slow steady stream while the machine is running. When all the butter has been added the mixture should be emulsified and thickened. Season to taste with salt, pepper and lemon juice. Reserve warm.

Kiwifruit

Kiwifruit, once labeled as Chinese Gooseberry in Europe, is technically classified as a berry. The tender green or golden-colored flesh has a flavor that is unique among all fruits. Kiwi flavor comes from a mix of green vegetal and fruity (apple, pineapple and berry) aromas.

Kiwi also contain *actinidin* an enzyme that works as a natural tenderizer for meat. Rub a cut kiwi directly onto raw meat and then apply the marinade of your choice. Actinidin will immediately go to work breaking down connective tissue. It is best to use this trick for short marinade times as the meat can become mushy after just 24 hours.

Growing region

Kiwi are native to north and eastern China., China, Italy, New Zealand, Chile, Greece are now top producers globally,

Common varieties

Hayward- fuzzy skin, green flesh

SunGold- smooth skin, yellow flesh

Taste sour and sweet

Main Aromas

ethyl butanoate- apple, butter, pineapple, red fruit, strawberry

trans hex-2-enal- fat, floral, fruit, green grass, pungent

Hexanal- fresh, fruit, grass, green, oil

Best Pairings

orange, lime, cranberry, strawberry, wine, brandy

Substitute

honeydew, fig, grape, strawberry

Pairing	Common Compounds	Aroma	Also Shared With
Kiwi + Whisky	ethyl butanoate	apple, butter, cheese, pineapple, strawberry	melon, orange, passion fruit, vinegar
	vinylbenzene	balsamic, gasoline, plastic, rubber, solvent	blueberry, cinnamon, peach, pecan
Kiwi + Cilantro	2-undecenal	sweet	caviar, chestnut, lamb, yogurt
	octanal	citrus, fat, floral, green, pungent	lime, ginger, corn, sesame seed
Kiwi + Jalapeno	3-carene	bell pepper, lemon, resin, rubber	basil, cardamom, fennel, parsley, peppercorn
	(E)-2-hexen-1-ol	bell pepper, fat, geranium, green, spice	celery, dill, radish, tamarind

Sun Gold Kiwi and Cranberry Relish

Makes 2 cups

Cranberry and kiwi growing seasons just barely overlap; but thank goodness they do. The flavor blend together perfectly making a unique combination of fruit, spice and tannins. Cilantro and jalapeno add extra flavor intrigue and round out the taste profile perfectly.

1 cup granulated sugar

2 cups water

zest of 1 orange
1 pc ginger thinly sliced
12 oz fresh cranberry
1 cup diced SunGold Kiwi (about 4 kiwi)
1/2 fresh jalapeño minced
1 tablespoon minced cilantro

Combine the sugar, water, orange zest and ginger in a small saucepan and bring to a boil. Reduce the heat to simmer and cook 2 minutes.

Add the cranberries to the simmering liquid and cook until they all burst, but have not completely broken down, 4-5 minutes.

Strain the liquid and reserve for another use (like cocktails!). Discard the ginger and orange.

Transfer the cranberries to a bowl and cool to room temperature. Fold in the diced kiwi, jalapeño and cilantro. Season to taste with salt and pepper. Serve with any grilled or roasted meat.

Lamb & Mutton

Lamb and Mutton are meat derived from domesticated sheep. To be classified as lamb, the animal must be one year-old at the time of harvest. Mutton are one year in age or more, though typically over two years at the time of harvest.

Lamb and mutton are both red meat with an appearance very similar to beef. The meat of lamb is somewhat lighter in color, has a more delicate flavor and is more tender than that of mutton.

The flavor of lamb and mutton is unique from beef due to branched-chain fatty acids that develop naturally in the animal. The fatty acids tend to have strong aromas that some find unappealing. They're often described as sweaty, barnyard fatty or sour. These compounds increase with the age of the animal, so most consumers prefer the taste of lamb to mutton. As the majority of lamb's flavor is found in fat, lean cuts from the loin have the mildest flavor.

Growing region

New Zealand, Australia, UK, India, Ireland are top producers.

Common Species

Cheviot, Dorset, Hampshire, Oxford, Shropshire

Taste umami, slightly sweet

Main Aromas

(E,E)-2,4-decadienal- coriander, deep fried, fat, oil, oxidized

(E)-2-nonenal- cucumber, cut grass, fat, paper, watermelon

4-methyl-3-thiazoline- burnt, earth, garlic

Best Pairings

toasted breadcrumbs, lemon, orange, mustard, soft herbs, butter, tea

Substitute

beef, game meats

Pairing	Common Compounds	Aroma	Also Shared With
Lamb + Shrimp	trimethylpyrazine	cocoa, earth, must, potato, roast	almond, asparagus, okra, sesame
	2-methylpyridine	ash, sweat	beans, oats, wild rice
Lamb + Strawberry	butanoic acid	butter, cheese, rancid, sour, sweat	bread, mushroom, tarragon, wine
	heptanal	citrus, dry fish, fat, green, nut, rancid	artichoke, chervil, eggplant, garlic, olive
Lamb + Mint	5-hydroxydodecanoic acid lactone	fruit, sweet	blue cheese, macadamia, sherry
	heptanoic acid	apricot, floral, rancid, sour, sweat	coffee, lemon, licorice, sauerkraut

Lamb Seasoning Rub

Lamb has its strongest pairings with Maillard and Earthy flavors. In this seasoning mix, coffee and mushroom powder amplify those specific flavors. Fennel and mint are more classic flavors for lamb that round out the mix. Rub this on meat before grilling, roasting or cooking sous vide.

2 tablespoons freshly ground coffee beans

1 teaspoon fennel seed

1 teaspoon dried onion

finely grated zest of 1 lemon
1 tablespoon dried mint
1 teaspoon mushroom powder
4 tablespoons kosher salt
2 teaspoons freshly ground black pepper

Combine the coffee, fennel seed, dried onion and lemon zest in a mortar and pestle or spice grinder and grind until well mixed but not a fine powder. Add the mint, mushroom powder, salt and pepper and mix well. Store in an airtight container up to 1 month.

Lemongrass

Lemongrass provides a distinctive flavor to the cuisines of Southeast Asia and India. Fresh lemongrass has bright lemon and fruity aroma. It is often found in recipes alongside chiles, coconut, and lime—all great pairings for the dominant citrus aromas. Beyond the citrus, aromas of wood and spice are prominent in the flavor lemongrass.

Lemongrass only contributes aromas to a dish, as its stalks are woody and tasteless. To use lemongrass, it should be sliced extremely thinly to make it more tender or it should be “bruised” with the back of knife and added to a dish to impart flavor, then removed before serving.

Growing region

Tropical climates throughout the world.

Common varieties

over 50 species, ornamental lemongrass (*Cymbopogon citratus*) is most common

Taste bland (flavor is fully dependent on aroma)

Main Aromas

β-citral- citrus, lemon

Geranial- flower, fruit, lemon, mint, pungent

β-myrcene- balsamic, fruit, geranium, herb, must

Best Pairings

cardamom, cinnamon, pepper, rosemary, citrus, soft herbs, alcohol

Substitute

citrus zest, coriander, ginger, lemon balm, lemon verbena

Pairing	Common Compounds	Aroma	Also Shared With
Lemongrass + Cauliflower	limonene	balsamic, citrus, fragrant, fruit, greenery, herb	cherry, clam, kiwi, peach, pomegranate
	nonanal	citrus, fat, floral, green, paint	chicken, coriander, mustard, strawberry
Lemongrass + Chervil	geraniol	geranium, lemon peel, passion fruit, peach, rose	allium, basil, cheese, tomato
	neral	citrus, lemon	cardamom, curry, ginger, sage, thyme
Lemongrass + Rosemary	β -myrcene	balsamic, fruit, geranium, herb, must	anise, carrot, dill, pistachio, star anise
	piperitone	fresh, mint	cinnamon, mint, pepper, raspberry

Lemongrass Bitters

These lemongrass bitters are delicious in cocktails. But try introducing them into your baking game, adding the bitter into any recipe that calls for vanilla extract. You can use it in addition to, or in place of, the vanilla. It is especially good in chocolate desserts and cookies.

1 stalk lemongrass

8 pods cardamom

1 cinnamon stick, crushed

1 tablespoon coriander seeds

4 branches fresh rosemary
8 fl oz grain alcohol or vodka

Trim the root end from lemongrass and discard. Cut from the trimmed end, slice 3 inches of the stalk as thin as possible. Reserve remaining lemongrass for another use.

Combine the sliced lemongrass, cardamom, cinnamon stick, coriander, and rosemary in a glass jar. Pour over the alcohol and secure the lid. Shake vigorously for 30 seconds. Store the jar in a cool, dark place.

Leave the mixture in a cool dark place to infuse for 3 weeks. Take the jar out every couple days and shake vigorously to promote infusion. After 3 weeks, strain the mixture through cheesecloth into a clean glass jar or bottle. Store, tightly sealed in a cool, dark place.

Lettuce

Lettuces were first cultivated as a food source for humans by the ancient Egyptians. While lettuce can add color, texture and bulk to a meal, it has very little flavor. Most lettuces contain a mere 20 aromatic compounds (strawberries have 400+). Lettuce flavor can be increased by cooking, like grilling romaine or sautéing leaf or butter lettuce. Heat will also cause lettuce to wilt, so a loss of texture is the cost of increased flavor.

Growing region

China, United States, India, Spain, Italy are top producers.

Common varieties

Romaine, Iceberg, Leaf (or Loose leaf), Butterhead, Summercrisp

Taste lightly bitter to sweet

Main Aromas

α -copaene- spice, wood

2-ethyl-1-hexanol- green, citrus, oil, rose

(E)-2-hexen-1-ol- bell pepper, fat, geranium, green, spice

Best Pairings

blueberry, apple, mushroom, mint, buttermilk, yogurt, fish

Substitute

cucumber, other greens- cress, brassicas

Pairing	Common Compounds	Aroma	Also Shared With
Lettuce + Mustard	tridecane	alkane	dill, eggplant, lamb, rice, walnut
	phenethyl isothiocyanate	pungent	broccoli, horseradish, mushroom, wasabi
Lettuce + Tamarind	2-methyl-1-propano	apple, bitter, cocoa, fusel, plastic, solvent	allium, apple, honey, mango, sake
	2-phenylethanol	fruit, honey, lilac, rose, wine	blue cheese, cashew, elderberry, peanut, truffle
Lettuce + Scallop	3-pentanol	fruit, green	chicken, oats, olive, peas, tomato
	1,2-dimethylbenzene	geranium	artichoke, beans, dill, persimmon, walnut

Tamarind Glazed Lettuce with Creamy Dressing

Serves 4

As noted above, lettuce flavor is greatly enhanced by cooking. This recipe amps up flavor with a little cooking, but really kicks the taste in overdrive by glazing a spicy, sour and sweet tamarind dressing directly onto the lettuce then finishing the salad with a cooling herb dressing.

1 head iceberg lettuce cut into quarters

OR 2 heads Romaine Hearts, cut in half

2 fl oz tamarind concentrate

2 fl oz rice wine vinegar

2 tablespoons brown sugar
½ jalapeno, minced
2 cloves garlic, chopped
1 tablespoon ginger, chopped

1 recipe creamy herb dressing (recipe follows)

Garnish, cooked bacon crumbles, diced tomatoes and crumbled blue cheese or feta

Preheat oven to boil

Cut the lettuce in half and arrange on a baking sheet cut side. Lightly season each piece with salt and set aside.

To prepare the glaze, combine the tamarind, rice vinegar, jalapeno, garlic and ginger in a blender and puree until smooth.

Pat the lettuce wedges dry with a paper towel. Brush each with a generous coating of the tamarind glaze. Place the lettuce on the center rack of the oven below the broiler. Cook 1-2 minutes. Remove the lettuce and apply a second coating of the glaze and place under the broiler for 1 minute more.

Divide the wedges between plates and top with dressing and garnishes.

Creamy Herb Dressing

½ cup sour cream
1 cup buttermilk*
1 tablespoon minced chives
1 tablespoon minced basil
1 tablespoon minced dill

1 clove garlic, minced

1 teaspoon kosher salt

pinch ground cumin

juice of 1 lemon

Worcestershire Sauce, to taste

Tabasco, to taste

**if you do not have buttermilk, mix 1 cup milk with 1 teaspoon distilled vinegar*

Combine all ingredients in a small bowl and whisk well to combine. Season to taste.

Melon (Watermelon, Cantaloupe, Honeydew)

Melons belong to the Cucurbitaceae family, which also includes squash, pumpkin, zucchini and cucumber. Melons are native to Africa and Southwest Asia, but reached Europe before the end of the Roman Empire, possibly earlier. Melons were also among the first plants brought to the Americas by European settlers.

The most commonly eaten melons- watermelon, cantaloupe (European and American species), honeydew- grow in a variety of species that can vary slightly in size, shape, and color. Aroma plays a major role in melon flavor, so much so that a ripe cantaloupe can be smelled through its rind. A slight give to the flesh, when the whole melon is squeezed can also indicate ripeness in cantaloupe or honeydew. Ripe watermelons have a hollow sound and a pronounced pale yellow spot on their bottom. Melons reach ripeness in late Spring through Summer.

While specific flavor varies by melon, the fruity aromas outlined below are largely responsible for melon flavor across the category.

Growing regions of Common Varietals

Watermelon- China, Turkey, Iran, Brazil, Egypt

Cantaloupe- Guatemala, Brazil, Honduras, US, Canada; Spain,

Honeydew- Mexico, Guatemala, Costa Rica, US

Taste sweet

Main Aromas

ethyl 2-methyl propyl acetate- apple, banana, floral, herb, plastic

(Z)-6-nonenal- melon

cis,cis-3,6- nonadienal- fat, soap, watermelon

isobutyl acetate- apple, banana, floral, herb, plastic

Best Pairings

mushroom, vanilla, toasted nuts, mustard, ham/bacon, wine

Substitute

apricot, grape, berries

Pairing	Common Compounds	Aroma	Also Shared With
Melon + Clam	1-penten-3-one	green, herb, metal, mustard, pungent	date, endive, tea
	1-hexanol	cooked vegetable, flower, grass, herb, rotten	coriander, hazelnut, mustard, toasted bread
Melon + Olive	isopentyl acetate	apple, banana, glue, pear	brandy, cocoa, sake
	(E)-2-nonenal	cucumber, cut grass, fat, paper, watermelon	camomile, cucumber, fish, pecan
Melon + Corn	nonanal	citrus, fat, floral, green, paint	broccoli, caraway, pistachio, pork
	(E)- β -ionone	cedarwood, floral, raspberry, seaweed, violet	citrus, fenugreek, seaweed, watercress

Bacon Tapenade and Melon Bruschetta

Serves 6

*Bacon and bruschetta are **guaranteed** to be good. This meaty, smoky tapenade makes the ultimate pairing for honeydew melon and a thin, crisped slice of sourdough. But don't be afraid to go "All-American" and make this your go-to sandwich spread. A sophisticated*

cheese plate would love this too! If gluten-free is your thing, top cubed watermelon with the tapenade.

2 cups Niçoise olives, pitted

1 anchovy filet, rinsed

1 clove garlic

3 tablespoons capers, chopped

2 eggs, hard-boiled, grated

1 tablespoon parsley, chopped

Extra virgin Olive oil, as needed

1 lemon, juiced

8 strips thick cut bacon, cooked crisp and crumbled

12 slices sourdough bread

Melon Honeydew or Cantaloupe, thinly sliced or shaved into strips with a vegetable peeler

In the bowl of a food processor, combine the olives, anchovy and garlic and pulse until no large pieces remain, but do not make a smooth paste. Transfer to a bowl and stir in the capers, egg and parsley. Stir in olive oil until the mixture is spreadable but still thick. Season to taste with lemon juice and freshly ground black pepper. Gently mix in the bacon crumbles.

Brush sourdough with olive oil or reserved bacon fat and toast until charred on the grill or in the oven at 450° F.

Spread some of the tapenade on each piece of bread and top with ribbons of melon.

Milk

This chapter will focus on the flavor profile of milk and milk products including cream, yogurt, sour cream, creme fraiche and butter. The taste of these products will vary due to a number of factors including animal diet, production method, fat content, fermentation or others. Flavor however, is largely consistent across the category. There are sweet and floral aromas, along with sour scents that comprise the majority of milk flavor.

Soured milk (yogurt, sour cream, crème fraiche, buttermilk) gets its distinctive ‘tang’ from higher concentrations of lactic acid which is mostly odorless (flavorless). Lactic acid esters, however, create aromas of dairy and are lightly floral. The aromatic compound *Diacetyl*, a byproduct of fermentation, is a secondary factor in the flavor of these products. Diacetyl has a rich, buttery aroma and is the main component of imitation butter flavoring.

Growing Region

(dairy cow) United States, India, China, Brazil, Germany

Common Forms

Milk (skim, 2%, whole, cream, heavy creamy), sour cream/crème fraiche, yogurt, buttermilk, butter

Taste lightly sweet to tangy

Main Aromas

2-Propanone- chemical, ether, nauseating, pungent

2-Butanone- ether, fragrant, fruit, pleasant, sweet

2-Heptanone- bell pepper, blue cheese, green, nut, spice

Acetaldehyde- ether, floral, green apple, pungent, sweet

Best Pairings

tropical fruits, roasted meats, dried fruit, honey, toasted nuts, seafood

Substitute

coconut milk, almond milk, kefir, soy milk

Pairing	Common Compounds	Aroma	Also Shared With
Cream + Oyster	2-heptanone	bell pepper, blue cheese, fruit, green, nut, spice	blue cheese, caviar, hazelnut, pea
	1-octanol	bitter almond, burnt matches, fat, floral, metal	cauliflower, coco mustard, sage
Yogurt + Walnut	Hexyl methyl ketone	fat, fragrant, gasoline, mold, soap	curry, miso, truffle, wine
	2-pentanone	burnt plastic, ether, fruit, kerosine, pungent	carrot, coffee, musk pumpkin seed, sh
Milk + Pork	1-heptanol	chemical, green, putrid, wood	apple, chestnut, dill sherry
	vanillin	sweet, vanilla	asparagus, fennel, peanut
Sour Cream + Clam	2-undecanone	fresh, green, orange, rose	allium, date, lemon turmeric
	hexanoic acid	cheese, goat, oil, pungent, sweat	cashew, coffee, musk polenta

Walnut Spiced Yogurt Hummus

Makes 2 cups

Welcome to a new world of hummus. Chickpeas are replaced by walnuts and thick Greek yogurt to make one of the creamiest and most umami-filled spreads we have ever tasted. Try this as an amazing base for roasted shrimp, as part of a crudité platter or for a really sexy pita sandwich.

2 cups toasted walnut pieces

6 cloves garlic, chopped

¼ cup extra virgin olive oil

1 lemon

1 teaspoon turmeric

pinch cumin

1 tablespoon miso (optional)

½ cup greek yogurt

Toast the walnut pieces and set aside to cool to room temperature.

Combine the the olive oil and garlic in a small sauté pan over medium heat. Cook until the garlic begins to sizzle lightly in the oil and smell sweet, 3-5 minutes. Remove the garlic and oil from the heat before the garlic begins to brown. Set aside to cool.

Zest and juice the lemon.

When the walnuts, garlic, and oil have all cooled to room temperature combine the three ingredients in a food processor and blend to combine. Add the lemon juice (reserve the zest), turmeric, cumin, and miso if using. Blend to a smooth paste.

Transfer the mixture to a bowl and stir in the yogurt. Season to taste with salt and pepper.

Mollusks (Terrestrial & Freshwater)

There are 50,000 known species of mollusks. Those that are used as a food source include: snail, mussels, oysters, clams, cockles, scallop, octopus and squid. While mollusks can look dramatically different from one another, structurally and composition-wise they have many similarities. Like most animals, the specific flavor of each is affected by the environment. This environmental influence is most pronounced in filter feeders—oysters, mussels, scallops and clams— those who siphon water through their bodies to filter out the plankton and algae they feed on.

Mollusks contain less of the nitrogen and sulphur compounds that create ‘fishy’ aromas in lean and fatty fish. The majority of mollusk flavor is created by earthy and Maillard aromas, making earthy, vegetal and roasted flavors the best pairings. The distinct flavors of mollusks are outlined below. When deciding which type of mollusks to use in a recipe it is important to also consider taste factors. Here, mollusks have also been mapped out according to their taste characteristics as they may be briny or sweet and have texture that range from tender to chewy.

Common varieties

(infographic: Cartesian coordinate graph to show briny-sweet; tender-chewy)

Snail

Mussels

Oysters

Cockles

Scallop

Clam

Octopus

Squid

Taste briny, mineral to slightly sweet

Main Aromas

Oysters

1-Octen-3-one- earth, metal, mushroom

(Z)-1,5-octadien-3-one- geranium, green, metal

Mussels/Clams/Cockles

2-acetyl-2-thiazoline- caramel, popcorn, roast beef

2, 3-butanedione- butter, caramel

(Z)-4-heptanal- cooked potato

Octopus/Squid

N,N-dimethylformamide- fish, pungent

3-hydroxy-butanone- butter, cream, green pepper, rancid, sweat

Scallop

Dimethyl Sulfide- cabbage, gasoline, organic, sulphur, wet earth

Best Pairings

mushroom, grains, nuts, breadcrumbs, cream/butter, beets, asparagus

Substitute

Any mollusks may be substituted for one another. As well as other marine animals like shrimp and crab.

Pairing	Common Compounds	Aroma	Also Shared With
Clam + Cauliflower	dimethylamine 2-butanone	fish ether, fragrant, fruit, pleasant, sweet	barley, celery, radish, wine beans ,chile, garlic, thyme
Mussels + Vanilla	benzene methyl dodecanoate	solvent coconut, fat	basil, ginger, parsley, pineapple coconut, elderberry, mustard, wine
Squid + Zucchini	1-Penten-3-ol 2-methylbutanal	burnt, fish, green, meat, wet earth almond, cocoa, fermented, hazelnut, malt	cherry, ginger, lime, okra, tomato mint, pistachio, pomegranate, vinegar
Scallop + Asparagus	acetaldehyde 2-nonanone	ether, floral, green apple, pungent, sweet fragrant, fruit, green, hot milk, soap	egg, lentil, peas, sake bacon, caviar, lemongrass, rice

Barley and Cauliflower ‘Risotto’

Serves 4

The gluten-free breadcrumbs of the future! The grated, cooked and (eventually) crunchy cauliflower-barley mixture is redolent with heady flavors of anchovy, cumin and garlic making one spoonful of this the ultimate topping for opened mollusks before baking or broiling. The ‘risotto’ is also a delicious base for sautéed, roasted or grilled mollusks.

2 cups barley

8 cups water

1 tablespoon kosher salt

1 tablespoon butter

1 tablespoon extra virgin olive oil

2 garlic cloves, minced

2 anchovy filets, chopped

½ teaspoon ground caraway seed or cumin

½ cup dry white wine

1 head cauliflower, stems and florets grated

kosher salt

freshly ground black pepper

2 tablespoons minced parsley

zest of ½ lemon

Combine the barley, water and salt in a small sauce pot. Place on the stove over medium high and bring to a boil. Reduce the heat to a simmer, cover the pot and place cook until the barley is tender, 30-40 minutes. Once tender, set aside to drain in a colander. Reserve to cooking liquid.

In a separate saucepot, melt the butter and olive oil over medium heat. Add the garlic, anchovy and caraway (or cumin) and cook until sizzling and fragrant.

Stir in the grated cauliflower and turn the to high. Cook, stirring occasionally until the pot begins to sizzle again. Deglaze with the wine and continue cooking until completely dry. The cauliflower should be tender at this point. If not, continue cook adding the barley cooking water as needed.

When the cauliflower is tender, stir in the barely and season to taste with salt and pepper. Add the barley cooking water as needed to keep the mixture moist. Remove from the heat and season to taste with salt and pepper.

Mushrooms

Mushrooms with a culinary purpose can be classified into two categories: cultivated and wild. Cultivated mushrooms may be easily grown and readily available year-round. Their flavor is consistent, though it tends to be more bland and less dynamic than that of wild mushrooms. Wild mushrooms are hand harvested from the wild or they are foraged. Scarcity (relative to cultivated mushrooms) and demand for their bold flavors make wild mushrooms more sought after and expensive.

The flavor of cultivated mushrooms comes from 3-octanol. It has pronounced aromas of moss and mushroom with notes of citrus and nut. 3-octanol is prominent across wild mushrooms as well. Though not typically apparent to the nose, mushroom also contain a number of fruity esters which make for very interesting pairings with mushroom. Mushrooms have very little taste but are one of the highest natural sources of umami.

Growing regions

Italy, Poland US, France, Netherlands are top producers.

Common varieties

Cultivated

oyster, king oyster, cremini/portobello, shiitake, white (button), hon shimenji (Beech),

Wild

chanterelle, porcini, morel, enoki, porcini, hen of the woods (Maitake), black and yellow trumpet, matsutake, truffle (see page XX)

Taste umami, slightly bitter to sweet

Main Aromas

3-octanol- citrus, moss, mushroom, nut, oil

Benzaldehyde- bitter almond, burnt sugar, cherry, malt, roasted pepper

1,8-cineole- camphor, cool, eucalyptol, mint, sweet

Best Pairing

allium, parsley, origanum, thyme, lavender, fruits- dried and fresh

Substitute

Most mushrooms may be substituted for each other. Also: eggplant, zucchini/summer squash, tofu, miso, fish sauce or soy sauce.

Pairing	Common Compounds	Aroma	Also Shared With
Mushroom + Strawberry	4-hydroxydodecanoic acid lactone	apricot, flower, fruit, peach, sweet	cheese, chervil, chicken, pork, rum
	ethyl hexanoate	apple peel, brandy, fruit gum, overripe fruit, pineapple	apple, clove, corn, tomato
Mushroom + Hazelnut	1-octen-3-one	earth, metal, mushroom	allium, basil, mustard, sesame seed
	4-pentanolide	herb, sweet	barley, date, soy sauce, wild rice
Mushroom + Coconut	phenylacetaldehyde	berry, geranium, honey, nut, pungent	beetroot, celery, chicken, tamarind,
	trimethylpyrazine	cocoa, earth, must, potato, roast	almond, clam, coriander, toasted bread

Porcini-Hazelnut Chocolate Tart

Serves 6-8

Brownies with umami. Mushrooms and hazelnuts have a very complimentary pairing and, here, the two ingredients venture to a highly unexpected place: dessert. Porcini powder, while adding umami to the brownies, also help to add depth and earthiness to the chocolate.

1 cup chocolate hazelnut spread

3 eggs, beaten

pinch salt

$\frac{3}{4}$ cup almonds flour

2 teaspoons porcini powder

Powdered sugar for dusting

Combine the chocolate spread, eggs, salt, almond flour and porcini powder in a bowl and whisk until smooth.

Grease an 8-inch cake tin, line with parchment paper and pour in the cake batter.

Bake in a 350° F oven for 30 to 35 minutes.

When done baking, invert the cake on a serving dish. When cool, dust with powdered sugar.

Nuts

The culinary definition of a nut differs from the botanical classification. Botanically, almond, cashew, pecan, peanut, pine nut, pistachio, sesame and walnut are all seeds. Remember this as you proceed through the chapter

The conventional wisdom of toasting nuts to bring out their flavor holds true. The majority of a nut's flavor is contained in its natural oils and, as those oils heat, they become volatile. Once the oils are released, the nut is more aromatic and flavorful. Toasting also creates Maillard reactions which play a large part in their flavor. When improperly stored, oils in nuts can decompose and create rancid flavors. Nuts should be stored in cool, dry, dark places to minimize oxidation.

Growing regions-

Almond- US, Australia, Morocco, Spain, Iran

Peanut- China, India, Nigeria, US, Myanmar

Pecan- US is #1 by far (Argentina, Australia, Brazil, China, Israel, Mexico follow)

Hazelnut- Turkey, Italy, US, Azerbaijan, Georgia

Pistachio- Iran, US, Turkey, China, Syria

Walnut- China, Iran, US, Turkey, Ukraine

Common varietal

Almond- Marcona

Peanut- Spanish, Runner, Tennessee, Virginia, Valencia,

Pecan- Cape Fear, Desirable, Moreland, Stuart, Natives

Hazelnut- Filbert, Constantinopla, Negret

Pistachio- Noble or Sicily, Tunis, Levante

Walnut- Persian (or English), black

Taste both sweet & bitter depending on varietal

Main Aromas

Almond

Benzaldehyde- bitter almond, burnt sugar, cherry, malt, roasted pepper

Furaneol- burnt, caramel, cotton candy, honey, sweet

(E,Z)-2,4-decadienal- fat, fish, fried, green, paraffin oil

Walnut

Hexanal- fresh, fruit, grass, green, oil

Pentanal- almond, bitter, malt, oil, pungent

Peanut

2-isopropyl-3-methoxypyrazine- bell pepper, earth, green, hazelnut, pea

2-acetyl-1-pyrroline- popcorn

Pecan

2-propionyl-1-pyrroline- popcorn, roast

3-methylbutanal- almond, cocoa, cooked vegetable, malt, spice

Hazelnut

Linalool- bergamot, coriander, lavender, lemon, rose

3-methyl-4-heptanone- fruit, hazelnut

Pistachio

α -pinene- cedarwood, pine, resin, sharp, turpentine

α -terpinolene- pine, plastic, sweet

Sesame Seed

2,5-dimethylpyrazine- burnt plastic, cocoa, medicine, roast beef, roasted nut

2-ethyl-4-methyl-1H-Pyrrole- nut, sweet

Best Pairings Tropical Fruits, Honey, Chocolate, Roasted Meat, Caramel, Popcorn, Vinegar

Substitute Any nut or seed may be substitute for another.

Pairing	Common Compounds	Aroma	Also Shared With
Peanut + Lime	(E,E)-2,4-decadienal	coriander, deep fried, fat, oil, oxidized	apricot, camomile, chicken, chiles, tea, tomato
	furfural	almond, baked potatoes, bread, burnt, spice	almond, cinnamon, strawberry, licorice
Walnut + Shrimp	acetophenone	almond, flower, meat, must, plastic	clam, lentils, soy, tamarind,
	2-heptanone	bell pepper, blue cheese, green, nut, spice	coconut, ginger, pear
Almond + Asparagus	2,6-dimethylpyrazine	cocoa, coffee, green, roast beef, roasted nut	barley, beef, cheese, malt, soybean
	2-formylpyrrole	musty, beefy, coffee	allium, beer, lettuce, popcorn
Pecan + Mango	4-vinylphenol	almond shell	corn, vanilla, popcorn, raspberry
	syringaldehyde	mild plastic, woody, tonka, sweet	lime, pork, rum,

Peanut and Lime Macaron with Dulce de Leche

Makes 20 assembled cookies

Lime and peanut are fairly common combination in the cuisines of Southeast Asia, but these find these two hanging out in a cookie with dulce de leche is real treat.

If you are ready to make your own peanut macaron, then it's pretty safe to assume you're serious about this your kitchen game. So break out your scale and make sure you measure these ingredients perfectly. Don't be discouraged if they come out perfect on the first try, making great macaron is something that takes practice.

315 g confectioners' sugar

165 g finely coarsely ground peanuts

1 pinch salt

1 g ground black pepper

1 g lime zest, finely grated

115 g egg whites

Dulce de Leche, as needed

Mix half of the sugar with the ground peanuts, salt, pepper, and lime zest in the bowl of a food processor and blend for 5 seconds. Transfer the mixture to a bowl.

Beat the egg whites with a little bit of the remaining sugar on low speed to soft peaks, then add the rest of the sugar and beat on high speed for 1 minute.

In a few additions, sift the dry mixture into the meringue, and fold with a spatula.

Transfer to a pastry bag, then pipe into about 40 discs of 2" diameter on silpat mats.

Smooth the top with an offset spatula.

Bake in a 300°F oven for about 15 minutes or until the cookies are risen and feel light and hollow. Cool on the baking trays.

Assemble the cookies spreading a layer of dulce de leche on one side then topping with a second cookie.

Olives

Olive trees belong to the botanical family that includes jasmine and lilac. The fruit of the olive tree has a harsh, bitter taste and must go through a process of extensive rinsing, curing and fermentation to mellow phenolic compounds that give the raw fruit a harsh, bitter taste. Phenols are a broad category of organic compounds found in all forms of plants and have an astringent, bitter taste. Phenolic compounds are responsible for the bitterness in raw walnuts, cranberries and kale. Phenols also make up the tannins in wine and tea.

There are hundreds of olive varieties that grow in a range of shapes, sizes and colors, but an olive's final appearance depends on much more than its variety. Generally, green olives are picked when mature, size-wise, but before they have fully ripened. Most olives turn black when fully ripened, though a few varieties remain green when ripe. Some green (under-ripe) olives may turn black from oxidation during the curing process. The type of curing and fermentation the olives undergo will greatly influence their final flavor. Brine cured olives have a more mellow, well rounded flavor, while oil and dry cured olives have a more intense, concentrated flavor.

Olives are most often thought of for their salty, fatty taste, but actual olive flavor comes from strong floral and fruit aromas. Olives share 68% of their compounds with lavender, and are a 72% match for nectarines. Olive and fruit pairings have the added benefit of the salty-sweet combination.

Growing Regions

Throughout the Mediterranean, Coastal Americas, Southern Africa and Australia. 70% of the world's olives are grown in Italy, Spain, Greece, Turkey and Morocco.

Common Varietals

Black- Kalamata (Greece), Nicoise (France), Gaeta (Italy), Mission (California), Green-Manzanilla (Spain), Picholine (France), Sicilian (Italy), Castelvetrano (Italy)

Taste salty, fat, can range from slightly sweet to bitter

Main Aromas

β-damascenone- boiled apple, floral, fruit, honey, tea

nonanal- citrus, fat, floral, green, paint

(E)-dec-2-enal- fat, fish, hay, paint, tallow

Best Pairings

citrus, peach, plum, almond, honey, roasted meats, yogurt, cream

Substitute

black garlic, fermented black beans, preserved lemon, capers

Pairing	Common Compounds	Aroma	Also Shared With
Olive + Watermelon	benzaldehyde	bitter almond, burnt sugar, cherry, green almond, roasted pepper	basil, cinnamon, coco lemon balm, sage, p
	2-phenylethanol	floral, fruit, honey, lilac, rose, wine	calvados, beer, cider, coconut, elderberry, g mustard, sherry
Olive + White Chocolate	octane	alkane, fat, oil, sweet	annato, honey, wal
	ethyl alcohol	alcohol, floral, ripe apple, sweet	apple, carrot, cheeses, citrus, fig, corn, tom vinegar

**Lemon
Curd
with
Crunchy
Olives**
Serves 4

*This
recipe is
a great
example*

Olive + Star Anise	β -caryophyllene	fried, spice, wood	camomile, cinnamon, cilantro, mace, oregano,
	myrcene	balsamic, flower, fruit, herb, must	black pepper, celery, cilantro, cumin, fen ginger, turmeric
Olive + Jasmine	benzyl acetate	fruit, honey, jasmine	clove, tea, whisk
	indole	burnt, medicine, mothball	licorice, pork, shri

of how pairings can drive creativity. With sixty percent of their aromatic compounds in common, olives and citrus is a pairing the deserves more attention. Here we thrust olives into an expected setting— dessert. Olives get sweetened with honey then dehydrated in the oven to intensify flavor and create a crispy texture. Then this dessert doubles down on the olive flavor, finishing a traditional lemon curd with extra virgin olive oil rather than the typical butter. Try finishing a bowl of this creamy curd off with toasted almonds and crumbled shortbread cookies.

1 teaspoon honey
1 teaspoon olive brine
1 cup pitted nicoise olives, drained
4 egg yolks
zest of 2 lemons
4 fl oz lemon juice
 $\frac{3}{4}$ cup granulated sugar
2 fl oz extra virgin olive oil
1 oz butter

Optional garnish

toasted almonds
shortbread cookies

Preheat oven to 200°F.

Stir together the honey and olive brine until smooth. Add the olives and toss well to coat. Place olives on a parchment or aluminum foil lined baking sheet in a single layer. Place in the oven and bake until fried and crisp, about 50 minutes. Reserve.

In a medium saucepot, whisk together the egg yolks, lemon zest, juice and sugar. When smooth, place the pot on the stove over medium heat. Stir constantly, scraping the bottom to prevent scorching. Cook until the mixture thickens and begins to bubble around the edge. Remove from the heat and continue whisking for 1 minute to dissipate heat.

Transfer to a bowl and beat with an electric mixer until cooled to room temperature. To speed this process, you can set the bowl with the lemon curd over a bowl of ice. When the mixture is cool, slowly pour in the olive oil while continuing to beat on low speed. Transfer the curd to smaller container, press plastic wrap onto the surface to prevent a skin from forming and refrigerate until ready to serve. Keeps refrigerated 7 days.

To serve, spoon lemon curd into dishes and top with dehydrated olives. Optional: sprinkle with toasted almonds and crumbled shortbread cookies.

Peas

‘Peas’ refers to members of the *Pisum Sativum* species that includes small round garden peas as well as sugar snap and snow peas. While the fibrous shell of the garden peas is discarded—we just eat the tender seeds—sugar snap and snow peas are mostly eaten as whole pods. Pea flavor is dominated by green aromas with notes of fat, and nut.

Garden peas one of a just a few culinary products that are best when frozen. Fresh garden or green peas have a short eating season of late spring to summer as they must be consumed within days of harvest for optimal taste. After harvest, peas begin to convert stored sugars into starch- that means that within a couple days of picking, tender sweet peas become dry and starchy. This tendency is the reason most peas are frozen immediately after harvest.

Growing regions

China, India, US, France, Egypt are top producers.

Common varietal

Garden Pea, Snow Pea, Snap Pea

Taste sweet

Main Aromas

(E)-2-heptenal- almond, fat, fruit, metal, soap

(E)-2-octenal- fat, fish oil, green, nut, plastic

3-isopropyl-2-methoxypyrazine- bell pepper, earth, green, hazelnut, pea

3-sec-butyl-2-methoxypyrazine- bell pepper, carrot, earth, green

Best Pairings

toasted nuts, mint, alliums, fish, mushroom, berries, melon

Substitute

haricot verts, fava beans, edamame

Pairing	Common Compounds	Aroma	Also Shared With
Garden Pea + Strawberry	ethanal	ether, floral, green apple, pungent sweet	lemon, cucumber, papaya, tomato,
	methyl isovalerate	apple, fruit, pineapple	melon, mint, mushrooms, sage
Peas + Carrots	diacetyl	butter, caramel, fruit, pastry, sweet	beets, cauliflower, hazelnut, scallop, wine
	a-terpineol	anise, fresh, mint, oil, sweet	anise, cilantro, basil, pepper, savory
Sugar Snap + Shrimp	cumene	solvent	celery, mushroom, coriander, parsley, pork
	2-pentylfuran	butter, floral, fruit, green bean	capsicum, clam, red tomato
Pea + Coconut	2-undecanone	fresh, green, orange, rose	allium, asparagus, green, turmeric
	ethyl caprylate	apricot, brandy, fat, floral, pineapple	lime, mustard, pork, yam

Sweet Pea, Pork and Coconut Tacos

Serves 4

The inspiration for this recipe was the too-intriguing-to-ignore pairing of peas and coconut. We're also suckers for anything wrapped up in a taco! Begin with the light and bright pea puree, layer it with the flavors of the coconut sambal (a typical condiment found in Sri Lanka that is good on everything) and finish with juicy slices of garlicky pork.

Sweet Pea Puree

1 ounce butter

2 ounces onion, finely minced

½ ounce garlic, minced

4 fl oz coconut milk

12 ounces green peas, blanched

1 tablespoon mint, chopped

¼ cup queso fresco, grated

Sweat the onion and garlic in butter until tender. Do not let them brown.

Add the coconut milk and reduce by half.

Combine the onion mixture, blanched peas and mint in a food processor and puree.

Transfer to a small pot and reserve until ready to serve.

Coconut Sambal

Makes 1½ cups

1 jalapeno, seeded and chopped

1 shallot, chopped

½ cup cilantro, coarsely chopped

1 teaspoon turmeric

juice of 1 lime

1 cup grated fresh coconut or frozen (thawed and drained in a colander)

Combine the jalapeño, shallot, cilantro and turmeric in a food processor. Pulse until finely chopped. Add the lime juice and coconut and pulse until just combined.

Pork

Brine

6 fl ounces soy sauce

½ cup packed brown sugar

¼ cup kosher salt

½ head garlic, cut in half

2 scallions, roughly chopped

1 pc ginger, sliced

4 cups ice water

2 lbs Pork belly or shoulder

Combine all ingredients except the water in a saucepot and bring to a boil. Simmer 5 minutes to infuse flavor, then remove from the heat. Stir in the ice water. Place the pork belly in a zip top bag and add enough brine to cover completely. Seal bag and refrigerate overnight. Reserve excess brine for another use.

Remove the pork belly from the brine, and pat dry. Place the meat on a rack inside a roasting pan, tightly covered with foil. Cook at 400°F for 2 hours. Remove from the oven and rest 30 minutes on top of the stove before uncovering.

To cook sous vide: Seal the meat in a plastic bag. Set a water bath to 162°F (72°C) and

cook for 18 hours. Remove from the bath and chill thoroughly.

If desired, brown the meat under a broiler or on a hot grill before serving.

For the tacos

12 corn tortillas, warmed

Spread some the pea puree on the tortilla. Top with slices of pork, then sprinkle with generous amount of the coconut sambal.

Pomegranate

Pomegranate is an ancient fruit that is typically consumed for its seeds or juice.

Pomegranate molasses is pomegranate juice that has been slowly cooked and reduced until it reaches a thick, syrupy consistency.

Tannins can give pomegranate seeds or juice a bitter, astringent taste- especially if the fruit is under ripe and has not developed natural sugars. Pomegranate flavor is mostly fruity, with aromas of citrus, melon and tree fruit. Sharper aromas like anise and mint are also present.

Growing regions

Pomegranate trees are native to Iran and quickly spread to modern day Turkey, Spain, Tunisia, Morocco and landed in the Californian via Spanish settlers in the 18th century.

Common varietal

Wonderful, Granada, Mollar, Early Foothill, Spanish Sweet

Taste sweet, astringent

Main Aromas

3-octen-1-yl acetate- green, melon, pear, tropical

α-terpineol- anise, fresh, mint, oil, sweet

hexanol cooked vegetable, flower, green, herb, rotten

Best Pairings

citrus fruits, melon, basil, tea, lemongrass, chili, avocado

Substitute

cranberry, raspberry, blood orange

**Spicy,
Yogurt
and**

Pairing	Common Compounds	Aroma	Also Shared With
Pomegranate + Dark Chocolate	6-methyl-5-hepten-2-one	citrus, mushroom, pepper, rubber, strawberry	lemon balm, lemongrass, pistachio, plum
	isopentyl acetate	apple, banana, glue, pear	cider, grape, strawberry
Pomegranate + Chicken	2-methyl-3-buten-2-ol	earth, herb, oil, wood	black currant, yarrow, sage, tamarind
	2-ethyl-1-hexanol	citrus, green, oil, rose	cherry, litchi, mimosa
Pomegranate + Camomile	γ -terpinene	bitter, citrus, gasoline, resin, turpentine	cardamom, fenugreek, basil, pistachio
	ethyl benzoate	camomile, celery, fat, flower, fruit	kiwi, mango, rose, whisky

Lemongrass marinated Chicken with Pomegranate

Serves 4

Besides being gorgeous, this recipe gets interesting as the pomegranate, yogurt and herbs play up the hidden floral notes that each contain. Lactones in the chicken blend seamlessly with the fruit. If you're feeling bold, try substituting duck breast or boneless quail for the chicken in this recipe.

1 tablespoon minced garlic

Juice and zest of 1 lime

½ cup plain yogurt

2 tablespoons minced Cilantro stems, leaves reserved

1 tablespoon fresh mint, chopped

1 jalapeno, minced

½ piece lemongrass, thinly sliced or several dashes lemongrass bitters (p. XX)

kosher salt

3 pounds boneless, skinless chicken thighs

kosher salt as needed

garnish- sliced avocado, cilantro and mint leaves and pomegranate seeds

In a large bowl, combine the garlic, lime juice and zest, yogurt, cilantro stems, mint, jalapeno, and lemongrass. Mix well and season to taste with salt. Reserve ¼ cup of the yogurt mixture.

Add the chicken to the remaining mixture and mix to coat evenly. Refrigerate until ready to cook- minimum 2 hours, up to 24 hours.

Remove the chicken from the marinade, wiping off any excess. Lightly coat the chicken with oil and season with salt. Cook the chicken on a medium-hot, well-oiled grill until a thermometer inserted into the thickest part of the thigh registers 160°F. The chicken should reach 165°F upon resting. Rest the chicken on a rack 3-5 minutes before serving.

Serve the chicken garnished with cilantro leave, sliced avocado (salted) and pomegranate seeds.

Pommace Fruit: Apple, Pear, Quince

Botanically, apple, pear and quince are all members of the Rose Family. Thus it is no surprise that floral aromas play an important role in their respective flavors.

Green-Vegetal and other Fruit aromas also contribute to the flavor of apple, pear and quince.

Apples can be selectively bred for specific traits and uses. There are over 7500 different cultivars of apple that may be raised to consumed raw, cooked, or pressed into juice and cider.

Pears, tend to have a softer flesh than apples when ripe. Pears also contain stone cells which give the flesh what is often described as a ‘gritty’ texture. Stone cells result from the formation of lignins in cell within the flesh of pears and quince. Lignins, rarely found in foods, help form the firm structure of wood and bark.

Unlike apples and pears, quince are very rarely eaten raw, because most varieties have a hard, astringent flesh. Quince have a strong perfume and can be added to cooked dishes that include apple or pear to increase their flavor. Quince is also naturally high in pectin and make excellent jellies or jam, which are a component of cheese plates.

Growing Regions

Apples and Pears are produced worldwide with fruit maturing in early Fall to Winter (Northern Hemisphere). The apple tree is native to Central Asia. Pears originated in the Mid-East, from western Europe to Northern Africa.

Common Varietals

Apple:

McIntosh*, Braeburn*, Fuji*, Red Delicious*,

Golden Delicious**, Empire**, Cortland**, Gala**, Granny Smith**,

Northern Spy, Winesap, Macoun

*Will become mushy in baking

**Hold up best in baking

Pear:

Bartlett, Bosc, D'anjou, Seckel, Nashi (Asian Pear)

Taste Sweet to tart

Main aromas

Apple

1-hexanol- cooked vegetable, flower, green, herb, rotten

(E)-2-hexenal- fat, floral, fruit, green grass, pungent

butyl acetate- apple, banana, glue, pungent, sweet

Pear

ethyl (E,Z)-2,4-decadienoate- metal, pear

2-methylpropyl acetate- apple, banana, floral, herb, plastic

(E)- β -damascenone- boiled apple, floral, fruit, honey

Quince

α -farnesene- boiled vegetable, floral, sweet, wood

ethyl octanoate- apricot, brandy, fat, floral, pineapple

ethyl hexanoate- apple peel, brandy, fruit gum, overripe fruit, pineapple

Best Pairings

citrus, seafood, wine, brandy/bourbon, vinegar, olive/olive oil, spice, dairy, nuts

Substitute

Apples and Pears may be substituted for one another. Also: green tomato, persimmon, quince

Pairing	Common Compounds	Aroma	Also Shared With
Pommace Fruit + Basil	anethole	sweet, anise, licorice, medicinal	hyssop, coriander, fennel, lemon balm, rhubarb
	geranyl acetate	lavender, rose, sweet	celery, citrus, curry, eucalyptus, lovage
Apple + Crab	1-butanol	fruit, medicine, solvent	cream, dill, endive, mezcal
	palmitic acid	rancid, wax	coconut, corn, plum, thyme, wasabi
Quince + Hazelnut	2-pentanone	burnt plastic, ether, fruit, kerosene, pungent	banana, cheese, coffee, rum, strawberry
	methylbenzene	glue, paint, solvent	annatto, citrus, ginger, pumpkin, tea
Pear + Sage	hexyl acetate	apple, banana, grass, herb, pear	chamomile, poultry, pomegranate, rye bread
	a-terpineol	anise, fresh, mint, oil, sweet	cumin, curry, lemongrass, nutmeg, parsnip

Green Apple, Basil and Strawberry Salsa

Makes approximately 3 cups

You've probably enjoyed the combination of apples and strawberries a number of times in a fruit salad. Less common would be to add basil and tomatoes to the mix and serve it along side roasted fish. Apples and basil make for a particularly good pairing on their own but on top of fish, the flavors transform into something truly unique and delicious.

1 Green apple, cored and diced
1 cup cherry tomatoes, cut in quarters
1 cup strawberry, hulled and quartered
1 tablespoon fresh basil, chopped
kosher salt
freshly ground black pepper
1 tablespoon acid (lemon juice *or* apple cider vinegar *or* tarragon vinegar *or* wine vinegar)
3 tablespoons extra virgin olive oil

Combine the apple, tomato, strawberry in a bowl. Season to taste with salt and pepper, toss gently to combine. Add the acid, olive oil, basil and mix again.

Serve with grilled or roasted poultry or fish.

Or as a bruschetta on toasted bread with goat cheese, parmesan or pecorino.

Pork

Pork, the meat from domesticated pigs is the most commonly consumed meat in the world. It is consumed both fresh and in a variety of preserved forms known as charcuterie, including: bacon, ham, sausages, terrines and salt pork.

Pork flavor is largely determined by breed and the conditions a pig is raised in. For many years—beginning in the mid-50s's—pork was intentionally bred and raised to be extremely lean. The health conscious public readily bought the leaner cuts of pork, not noticing that the meat was also very lean on flavor. In recent years consumers have begun to demand a return to more flavorful meat.

Heritage breeds of pork, like those listed below, grow to maturity more slowly than commercially raised breeds. These pigs require more feed and space to roam and therefore more expensive. Though higher priced, heritage breeds of pork yield meat much more flavorful meat, with a higher ration of fat to lean, a richer coloring (more red than white). Like grass-fed beef pasture raised pigs develop high quantities of omega 3-fatty acids and linoleic acid.

Maillard flavors make the best pairings for pork, especially darker aromas like caramel or roasted. Pork also goes very well with fruity aromas, whether they are found in tropical fruits, chilies, or tomatoes.

Growing regions

Pigs were first domesticated in China as early as 5000 BC. Today China, EU, US, Russia and Brazil are top producers of pork.

Common varietal

Yorkshire, Duroc, Berkshire, Tamworth, Red Wattle, Hampshire, Landrace, Mangalitsa, Senese

Taste umami, sweet

Main Aromas

2-pentylfuran- butter, floral, fruit, green bean

hexanal- fresh, fruit, grass, green, oil

1-octen-3-ol- cucumber, earth, fat, floral, mushroom

Best Pairing

bourbon, brown butter, jerusalem artichoke, sweet potato

Substitute

poultry or game meats

Pairing	Common Compounds	Aroma	Also Shared With
Pork + Apple	2-acetylfuran	balsamic, cocoa, coffee, smoke, tobacco	allium, almond, coconut, potato, oyster
	1-heptanol	chemical, green, putrid, wood	citrus, mustard, olive
Pork + Hazelnut	pyrrole	nut, sweet	coffee, egg, mushroom, tamarind
	2-furanmethanethiol	coffee, roasted meat	beef, peanut, popovers, sesame seed
Pork + Clam	2-acetylthiazole	nut, popcorn, roast, sulphur	artichoke, beer, cabbage
	trimethylpyrazine	cocoa, earth, must, potato, roast	barley, coconut, corn, peanut, whiskey
Pork + Coffee	4,5-dimethylthiazole	green, nut, roast, roasted, smoke	malt, oats, peach, sugar
	2-acetylthiophene	sulphur	asparagus, tomato, paste, bread

Blueberry, Apple and Horseradish Jam

Makes 2 cups

This is a recipe of transitive properties. Pork pairs very well with fruit aromas, as lactones are major contributors to pork flavor. Blueberries have latent wood and pine aromas that complete their flavor. Put these three together and you've got a flavor explosion. This jam with a spicy, savory edge is delicious spread on toast with bacon for breakfast or makes an amazing sandwich spread for roast pork.

1 cup dried blueberries

1 teaspoon kosher salt
½ teaspoon fresh ground black pepper
2 cup apple juice
4 pods cardamom
1 lemon
1 orange
2 lbs frozen blue berries
¼ cup grated fresh horseradish
3 tablespoons low or no sugar pectin

Set the frozen berries in a bowl next to the stove to begin to thaw.

Combine the dried blueberries, salt, pepper and apple juice in a medium sauce pot and bring to a simmer.

Peel the lemon and orange with a vegetable peeler. Tie the cardamom, lemon and orange in a piece of cheesecloth and add to the pot. Continue to simmer until only ¼ cup of the juice remains.

Add the frozen berries and any juice that may have collected in the bowl. Stir well and leave to cook over medium-low heat until the mixture comes to a simmer.

Cook until thickened, about 15 minutes. Stir the pot occasionally to prevent scorching.

Remove from the heat and sprinkle the pectin over the surface of the pot and let stand 1 minute, then stir until completely dissolved. Stir in the horseradish. The mixture may be stored refrigerated or divided among jars and processed according to proper canning instructions.

Potato

All cultivars of modern potatoes can be traced back to a single origin in southern Peru. Until the ships of Spanish explorers returned from South America in the 16th century, the potato was unknown in Western Europe and the rest of the world.

Potatoes have a very mild, somewhat bland flavor, though the aroma of a baked potato is comprised of 228 different aromatic compounds. Much of a potato's flavor is generated in the cooking process, meaning Maillard aromas play an important role. Milk, butter, sour cream and cheese are the most classic pairing for potatoes and data shows them to be the best match as well. Vegetal aromas of earthy and fruit-like are also main contributors to potato aromas. Fruit-like vegetal aromas main come from tomatoes, eggplant, and capsicum—nightshades, the botanical family that potato also belongs to.

Growing regions

China, India, Russia, Ukraine and US are top potato producers.

Common varietal

Russet, Yukon Gold, Red, White, Yellow, Purple, Fingerling,

Taste bland, starchy

Main Aromas

(Z)-4-heptenal- biscuit, cream, fat, fish, rotten

2,3-diethyl-5-methylpyrazine- earth, meat, potato, roast

3,5-diethyl-2-methylpyrazine- baked, cocoa, roast, rum, sweet

Best Pairings

butter, cheese, sour cream, tomato, peppers, corn, toasted nuts, breadcrumbs

Substitute

zucchini/chayote, cauliflower, Jerusalem artichoke, parsnip

Pairing	Common Compounds	Aroma	Also Shared With
Potato + Caviar	2-propenal	burnt sweet, pungent	beer, carrot, olive, wine
	2,4-heptadienal	cucumber, fat, green, nut	corn, pea, soybean, walnut
Potato + Egg	2-methylbutanal	almond, cocoa, fermented, hazelnut, malt	allium, apple, laurel, sake,
	2-decanone	fat, fruit	beef, cheese, mushroom, shrimp
Potato + Avocado	2,3-butanedione	butter, caramel, fruit, pastry, sweet, yogurt	beetroot, blue cheese, pumpkin, scallop
	2-pentylfuran	butter, floral, fruit, green bean	broccoli, chayote, pistachio, chicken

Beet and Blue Cheese Potato Rösti

Serves 4

Earthy and fruit-like aromas are the best pairing for potatoes. Beets are loaded with both. In this flavor and color-packed take on the classic rösti, blue cheese punctuates the earthy flavors while avocado lends creamy richness.

2 lbs Yukon gold potato

kosher salt, as needed

2 tablespoons butter

1 tablespoon olive oil

4 scallions thinly sliced
1 cup red or golden beet, grated
¼ cup blue cheese, crumbled
1 recipe whipped avocado (optional), recipe follows

Pierce the potatoes with a fork and place them in a medium sauce pot. Add cold water to cover by 3 inches and season to taste with salt. Bring the pot to a boil, then reduce the heat to simmer and cover the pot. Cook until the potatoes are completely tender, about 20 minutes. Remove the cooked potatoes and allow to cool completely (ideally, 2-3 hours refrigerated). This may be done up to 3 days in advance.

Remove the skin from the cooked potatoes with a pairing knife and grate the potatoes on a box grater.

Melt the butter in a medium cast iron or non-stick pan over medium heat. Add the scallions and saute until tender, about 2 minutes. Stir in the grated potato and mix well. When the potato begins to sizzle, stir in the olive oil, grated beets and blue cheese. Press the mixture into a firm layer in the pan and leave to cook, until golden brown and crisp, about 5 minutes. With a heat proof spatula, carefully slip the cake onto a plate, then invert back into the pan to brown the other side, add more oil if needed. When both sides are golden brown, slide the cake onto a cooling rack to rest before slicing.

Serve topped with sour cream or whipped avocado.

Whipped Avocado

1 avocado
1 tablespoon lime juice
¼ cup yogurt or sour cream
¼ teaspoon cumin

kosher salt

freshly ground black pepper

Cut the avocado in half, remove the pit and scoop the flesh from the skin. Combine the avocado, lime, yogurt and cumin in a food processor and blend until smooth. Season to taste with salt and pepper.

Alternatively, mash the avocado in a bowl, then add the lime, yogurt and cumin. Whip with a hand blender until smooth and season to taste with salt and pepper.

Poultry

Poultry is the class of domesticated birds that are raised for eggs or meat. Breeds of poultry of culinary importance include chicken, turkey, guinea hen, pheasant, duck, quail, pigeon (squab) and geese. These breeds can be divided into two unofficial categories: those with light breast meat and others with dark breast meat. In all birds, breast meat is commonly referred to as light meat and legs (thigh and drumstick) are referred to as dark meat. Most domesticated birds are flightless, meaning the muscles that make up the breast meat do very little work, allowing them to remain very tender and not development much flavor. The muscles that make the leg meat, however, do significantly more work leading them to be more flavorful, while also developing more connective tissue.

Birds with light breast meat include chicken, turkey and guinea hen. These animals have a delicately flavored breast meat with leg meat that has a richer, fuller flavor. Birds with dark breast meat—duck, pheasant, quail, pigeon (squab), and geese—carry aromas of fat, nuts and faint barnyard in the meat of both the breast and leg.

Growing regions

Chickens were first domesticated in China, approximately 3500 years ago. Today China remains a top producer of poultry along with US, Indonesia, Brazil, and India.

Common varietal Chicken, Turkey, Duck, Quail, Pheasant, Guinea Hen

Taste umami, slightly sweet

Main Aromas

Chicken, Turkey, Guinea Hen

2-methyl-3-furanthiol- fried, meat, nut, potato, roasted meat

2-furfurylthiol- coffee, roasted meat

(E,Z)-2,4-decadienal- fat, fish, fried, green, paraffin oil

1-octen-3-one- earth, metal, mushroom

Duck, Pheasant, Quail, Pigeon (Squab), Geese

Nonanal- citrus, fat, floral, green, paint

2-(E)-Decenal- fat, fish, hay, paint, tallow

2-(E)-heptenal- almond, fat, fruit, metal, soap

Best Pairing

butter, cheese, root vegetable, grains, dill, chervil

Substitute

All varieties of poultry may be substituted for one another.

Pairing	Common Compounds	Aroma	Also Shared With
Turkey + Apricot	1-isopropyl-4-methylbenzene	citrus, fresh, gasoline, solvent	anise, black currant, basil, pistachio, turmeric
	benzyl alcohol	boiled cherries, moss, roasted bread, rose, sweet	citrus, cinnamon, mustard, sage
Duck + Dill	piperonal	anise, sweet	melon, blueberry, pepper, sherry, vanilla
	ethylbenzene	gasoline	cheese, egg, olive, tomato, walnut
Chicken + Banana	undecane	alkane	chervil, cream, endive, garlic, walnut
	(E)-2-hexenal (=leaf aldehyde)	fat, floral, fruit, green grass, pungent	arugula, leek, olive, radish
Poultry + Crayfish	(E,E)-2,4-decadienal	coriander, deep fried, fat, oil, oxidized	chile, corn, mushroom, pecan, sweet potato
	octanal	citrus, fat, floral, green, pungent	artichoke, caraway, coriander, ginger, oyster

Apricot, Sage and Pistachio Chutney

Makes approximately 3 cups

This chutney is extremely versatile and can be used as either a condiment or a stuffing for just about any roasted poultry- from turkey, to chicken, to duck. As a stuffing, simply roll it up in boneless cuts of breast or leg before cooking. While the flavors here are particularly well-suited for poultry, it also pairs beautifully with grilled or roasted beef and lamb.

2 cups dried apricot, diced
2 cups dry white wine
¼ cup white wine vinegar
3 branches fresh sage
1 teaspoon fennel seeds, crushed
2 tablespoons olive oil
juice and finely grated zest of ½ orange
¼ cup toasted pistachio or pecan (chopped)
1 tablespoon minced jalapeno
kosher salt
freshly ground black pepper

Combine the diced apricot, wine, vinegar, sage and fennel seeds in a small pot. Bring the mixture to a boil, reduce the heat to a simmer and cover the pot. Cook 5 minutes, then remove from the heat and let stand 15 minutes.

With a slotted spoon, transfer the apricots to a bowl. Reserve the remaining wine in the pot, discard the sage.

Mix the olive oil, orange juice and zest, pistachio and jalapeno in with the apricots and season to taste with salt and pepper. If the mixture is too dry add some of the reserved wine.

Radish

Radish is a member of the brassica family and is commonly grown for its crunchy, spicy root. Radish tops are also edible, though they are less commonly consumed. Radish leaves have a peppery bite similar to arugula. Like other members of brassica, sulphur compounds play a major role in radish flavor though the most important compounds are isothiocyanates. These nitrogen-containing sulphur compounds are responsible for the perceived 'spiciness' of raw radishes. Isothiocyanates are also found in mustard, horseradish and wasabi.

Radishes are typically classified by their growing season with the majority reaching maturity in Spring and larger varieties like Daikon, Korean and Black radishes being harvested in winter months.

Growing region

China, India, Russia, Japan, South Korea are top producers.

Common varieties

Spring radishes (also known as European radishes)- Bunntail, Cherry Bell, Champion, Red King, Sicily Giant, Snow Belle, White Icicle, French Breakfast, Plum Purple, Gala, Easter Egg, Watermelon

Winter varieties- Daikon, Korean, Black Radish

Taste bitter, spicy

Main Aromas

Phytol- floral, balsamic

Hexadecanoic acid- rancid, wax

4-(methylthio)butyl isothiocyanate- cabbage, radish

Best Pairings

chervil, basil, lettuce, peas, yogurt/sour cream, apple, citrus

Substitute

jicama, turnip, water chestnut

Pairing	Common Compounds	Aroma	Also Shared With
Radish + Apple	1,1-diethoxyethane	cream, licorice, pleasant, sweet, tropical fruit	grapefruit, plum, pork, whisky
	(E)-2-hexenal	fat, floral, fruit, green grass, pungent	apricot, cherry, cucumber, hazelnut, peach
Radish + Basil	3-carene	bell pepper, lemon, resin, rubber	caraway, fennel, ginger, honey, lime
	benzyl alcohol	boiled cherries, moss, roasted bread, rose, sweet	chicken, cinnamon, mustard, sage
Radish + Passion Fruit	3-hydroxy-2-butanone	butter, cream, green pepper, rancid, sweat, wet	carrot, mussel, tomato, yogurt
	(Z)-3-hexen-1-ol	bell pepper, grass, green leaf, herb, unripe banana	curry, dill, olive, thyme, wine

Radish and Apple with Basil Vinegar

Serves 4

Apples and mint make for two very strong pairings with radish. Here, a versatile flavor-infusing technique is employed to make a deliciously crunchy and spicy salad.

Infused vinegars make for deeply aromatic and complex dressing. Basil vinegar hits just the right notes for this recipe, but be sure to try out some of the variations listed below.

2 tablespoons basil vinegar (recipe follows)

2 tablespoons apple cider or apple juice

1 teaspoon Dijon mustard

2 tablespoon extra virgin olive oil

1 granny smith apple, thinly sliced

1 honeycrisp apple, thinly sliced

1 cup radishes, thinly sliced

1 cucumber, thinly sliced

fresh basil leaves for garnish

kosher salt

freshly ground black pepper

Combine the basil vinegar, apple cider, and Dijon mustard in a large bowl. Whisk to combine. Continue whisking while adding the olive oil.

Slice all of the vegetable on a mandolin and add to the bowl with the dressing. Toss well to combine. Season to taste with salt and pepper

Basil Vinegar

Basil vinegar makes for some of the most flavorful radishes- whether you use it to deglaze a pan of sautéed radishes or use it as a dressing for a simple salad of sliced radishes and fresh lettuces. It also makes the perfect dressing for a tomato salad.

1 teaspoon caraway seeds

6 stems fresh basil, cut into 4-5 pieces each

zest of 1 lime

16 fl oz white wine vinegar

Combine the caraway seeds, basil, and lime zest in a glass jar. Pour in the vinegar and tighten the lid on the jar. Shake vigorously promote infusion.

Store in a cool, dark place 1 week, taking the jar out occasionally to shake. The vinegar is ready to use after 1 week, however the basil and spices may remain in the jar up to 1 month. After that the vinegar should be strained, but may be kept in the jar indefinitely.

Other Vinegars to try:

Mint Vinegar: White wine vinegar, mint, lemon peel, turmeric

Fennel Vinegar: Red Wine Vinegar, Fennel tops, star anise, orange zest

Chile Vinegar: Sherry or White Wine Vinegar, Jalapeno, Cilantro

Herb Vinegar: Sherry Vinegar, Oregano, Rosemary, Thyme, Coriander Seed

Raspberry Vinegar: Red Wine Vinegar, Raspberries, Thyme, Black Peppercorn

Ginger Vinegar: Rice Vinegar, Sliced Ginger Root, Lemon, Lime and/or Orange zest

Rice

Rice is one of the oldest agricultural products in the world, with evidence of domestication dating back over 10,000 years. Evidence also suggests that all 40,000 varieties of rice can be traced back to a single cultivar.

Grains of rice are actually the seeds of a particular species of grass. After harvesting, the tough, outer seed covering—known as bran—is scrubbed off by tumbling the seeds with an abrasive. The resulting product is white rice. Brown rice has the bran layer left intact. The bran layer gives brown rice a firmer texture and nutty flavor. White rice flavor comes from a combination of floral, herbaceous and nutty aromas. Jasmine, Basmati, Thai Red, and Black Japonica are particularly aromatic varieties of rice with more pronounced floral and nutty aromas.

Growing region

Rice is native Eastern China, today it is grown all over the world. Though the 5 top rice-growing countries in the world are all in Asia: China, India, Indonesia, Bangladesh and Vietnam.

Common varieties

There are over 40,000 different varieties of rice. All varieties are typically divided into 3 divisions based on grain length. As a general rule, a grain length shortens, starch content increases.

- *Long grain* best for steamed rice, rice pilaf
- *Medium Grain* best for sushi rice or paella
- *Short Grain* best for risotto

Taste bland

Main Aromas

2-acetyl-l-pyrroline- popcorn
2-nonenal- paper
(E,E)-2,4-decadienal- coriander, deep fried, fat, oil, oxidized

Best Pairings

butter, cheese, citrus, berries, mushrooms,

Substitute

faro, barley, wheat berry, quinoa

Pairing	Common Compounds	Aroma	Also Share
Rice + Raspberry	6-methyl-5-hepten-2-one	citrus, mushroom, pepper, rubber, strawberry	annatto, lemon, mint, orange,
	benzaldehyde	bitter almond, burnt sugar, cherry, malt, roasted pepper	almond, cinnamon, co
Rice + Peas	isophorone	cedarwood, spice, tobacco	beef, blue, mushroom, ol
	2-octenal	almond, flowdandelion, fat, fruit, grass, green, soap, spiceer, meat, must, plastic	asparagus, ca, peanut, t
Rice + Crayfish	2-methylnaphthalene	moss, mothball, pungent, root	bell pepper, watercress,
	ethylpyrazine	green, iron scorch, must, peanut butter, roasted	coconut, coff, seed, p

Raspberry Rice Pudding

Serves 4

Rice pairs particularly well with fruit flavors. Among those, Berry and Citrus show the strongest affinity. In this recipe all three come together to transform a homey dessert into an unexpected and sophisticated treat.

1/2 cup Arborio rice

3 cups milk or almond milk

1/4 cup sugar

1 cinnamon stick

1/2 vanilla bean, split or 1 teaspoon pure extract

1 cup raspberry puree or 2 cups fresh raspberry sprinkled with sugar and mashed through a fine mesh strainer

Garnish

Finely grated orange zest

toasted pistachio

In a large saucepan, place all the ingredients. Bring it to a gentle boil and then turn it down to a gentle simmer, stirring occasionally to keep it from sticking to the bottom, for about 30 minutes. Stir in the raspberry and continue cooking until thickened, about 10 minutes more.

Pour into dessert bowls. You can serve it immediately or let it chill in the fridge. Before serving top with chopped pistachio and orange zest.

Root Vegetable (celery root, parsnip, salsify)

Root Vegetable refers to the root of any plant used as a food source. In this chapter, we look at a specific group of root vegetables with shared flavor characteristics. Celery root, parsnip and salsify are all white flesh roots with a mild earthy and herbaceous flavor. Like all root vegetables, they may be stored or 'cellared' for 6 months or more after harvest.

Celery root, or celeriac, is the knobby, bulbous root of a specific variety of celery (not common celery). Celery root should be firm and heavy.

Parsnip are related to carrots and parsley, but should not be confused with white colored carrots. Like carrots, parsnips should be firm and crisp.

Salsify is the root of a wildflower native to Europe. It is also known as oyster root and is often described as having a mild oyster flavor when cooked.

Growing regions

Belgium, France, Italy, US and Poland are top producers.

Common varieties

Celery Root (celeriac), Parsnip, Salsify

Taste bitter to slightly sweet

Main Aromas

Celery Root

Limonene- balsamic, citrus, fragrant, fruit, herb

β -pinene- pine, polish, resin, turpentine, wood

(E)- β -ocimene- citrus, herb, mold, sweet, warm

Parsnip

2-isobutyl-3-methoxypyrazine- bell pepper, earth, floral, green pepper, peas

myristicin- balsamic, carrot, nutmeg, spice, warm

2-(sec-butyl)-3-methoxypyrazine- bell pepper, carrot, earth, green

Salsify

valencene- citrus, green, oil, wood

Phenylacetaldehyde- pungent, green floral, sweet

1-hexanol cooked vegetable, flower, green, herb, rotten *also in apple

Best Pairings celery, cauliflower, bourbon, coffee, lemon, basil

Substitute Celery root, parsnip or salsify may be substituted for one another. Also- potato, Jerusalem artichoke, carrot

Pairing	Common Compounds	Aroma	Also Shared With
Salsify + Black Currant	cis-carveol	caraway, cool	caraway, cardamom, pepper, rosemary
	β -selinene	herb	basil, celery, ginger, o star anise
Parsnip + Cilantro	terpinolene	pine, plastic, sweet	anise, lime, ginger, bl pepper, almond
	myristicin	balsamic, carrot, nutmeg, spice, warm	caraway, lovage, fenn star anise
Celleriac + Vanilla	propylbenzene	mothball	beef, shellfish, peca watercress
	methyl laurate	coconut, fat	chile, cocoa, cocon mustard, rum

Roasted Root Vegetable with Ginger Salsa Verde

Serves 4

Citrus and Herbaceous are prominent aromas in parsnips, celery root and salsify. Too bad most cooks relegate these root vegetables to simple preparations and seasonings. Here, I take the opposite tact and play up their flavors with my ginger salsa verde, adding pleasant brightness (and some much needed love!).

2 lbs Parsnips, celery root, or salsify (about 4 cups when chopped)

¼ teaspoon coriander seeds, crushed

¼ teaspoon ground ginger

1 clove garlic, chopped

1 tablespoon extra virgin olive oil

Kosher salt

Freshly ground black pepper

2 lbs Parsnips (about 4 cups chopped)

1 recipe ginger salsa verde, recipe follows

Peel the parsnips and cut into approximately 4 inch sections. Cut the different sections in half or quarters depending on their size so that all the pieces are approximately the same size.

Whisk together the coriander, ginger, garlic, olive oil salt and pepper in a bowl. Add the parsnips and toss to coat.

Spread them into a single layer and roast in a 375°F oven until easily pierced with the tip of a knife, 12-15 minutes. Set aside to cool slightly.

Arrange the roasted parsnips on a platter and drizzle with the salsa verde.

Ginger Salsa Verde

1 bunch cilantro, leaves picked from the stems, about 2 cups

1 tablespoon capers

2 teaspoons fresh thyme leaves

1 tablespoon dried marjoram

1 clove garlic, grated

1 tablespoon grated ginger

½ jalapeno, seeds removed and chopped

¼ cup extra-virgin olive oil

2 teaspoons red wine vinegar

Combine the cilantro, capers, thyme, marjoram, garlic, ginger, jalapeno, olive oil, and vinegar in a food processor and blend in short bursts. The mixture should be finely chopped but not a smooth puree. Stir in the lemon zest and season to taste with salt and pepper.

Squash, Summer

Summer Squash is the common name for a variety of squashes of the species *Cucurbita pepo*, which also includes winter squash and pumpkin. Summer squash may be known as yellow squash, zucchini, straightneck or crookneck, patty pan or chayote. They have a generally very mild to bland flavor with mostly aromas of vegetable with faint hints of floral, nut and fat. When selecting summer squash, vegetables of smaller size are preferred. These tend to have a better ratio of firm flesh to soft seed pockets, which tends to become mushy when cooked.

Squash flowers, or blossoms, are a favorite product of early summer. They are the delicately flavored blossom that grows from the end of all varieties of zucchini and squash. There are both male and female flowers, the female will become a squash while the male flowers grow from long stems near the base of the plant. Male squash blossoms tend to be the best for stuffing.

Growing region Spain, New Zealand, Mexico, Morocco, Netherlands

Common varietal green zucchini, yellow squash, chayote, patty pan

Taste bland to slightly bitter raw, lightly sweet when cooked

Main Aromas

Phenylacetaldehyde- berry, geranium, honey, nut, pungent

3-methylbutanal- almond, cocoa, cooked vegetable, malt, spice

Leaf Aldehyde- fat, floral, fruit, green grass, pungent

Best Pairings Earthy: asparagus, grain, mushroom; lemon, cheese, garlic, fish sauce

Substitute Varieties of squash may be substituted for one another. Also cucumber.

Pairing	Common Compounds	Aroma	Also Shared With
Zucchini + Peanut	2-acetylfuran	balsamic, cocoa, coffee, smoke, tobacco	garlic, barley, jalapeno, radish
	diacetyl	butter, caramel, fruit, sweet, yogurt	avocado, cauliflower, okra, olive, scallop
Zucchini + Parsnip	2-(sec-butyl)-3-methoxypyrazine	bell pepper, carrot, earth, green	beans, beetroot, swiss cheese, wine
	3-(methylthio)propional	cooked potato, soy	curry, egg, sesame, tomato
Zucchini + Ginger	2-heptanone	bell pepper, blue cheese, green, nut, spice	annatto, beef, coconut, walnut
	methylbenzene	glue, paint, solvent	artichoke, crab, dill, peas, vinegar

Thai-Style Spicy Zucchini Salad

Serves 4

Southeast Asian ingredients like lime, peanut, ginger and fish sauce make for a great pairing with zucchini. It seemed only natural to let zucchini stand in for green papaya in this take on a traditional salad.

4 tablespoons fresh lime juice

2 tablespoons palm sugar or light brown sugar

1½ tablespoons fish sauce

1 tablespoon finely grated ginger
4 garlic cloves, minced
2 medium zucchini (about 4 cups shredded)
½ teaspoon kosher salt
1 cup cherry tomatoes, cut in half
4 scallions, thinly sliced
2 tablespoons cilantro leaves
1 bird eye chili, thinly sliced
¼ cup toasted peanuts, coarsely chopped

In a small bowl, whisk together the lime juice, brown sugar, fish sauce, ginger and garlic. Set aside to combine.

Cut the zucchini into a fine julienne on a mandolin, or grate on a box grater. Place the cut zucchini in a large bowl and add the salt. Toss well to combine. Let stand 10 minutes, then transfer to a colander and squeeze dry. Break apart the squeezed zucchini and return to the bowl.

Add the tomatoes, scallions, chili and reserved dressing. Toss well to combine. Adjust seasoning to taste with salt or fish sauce and lime juice. Transfer to plates and garnish with peanuts.

Squash, Winter

**Note: Though not technically a winter squash, sweet potato is included here because it has very similar flavor profile and culinary uses to those of winter squash.*

There are several species of winter squash within the genus of Cucurbita. Squash and pumpkin make up the edible species of this genus. Unlike zucchini, also a Cucurbita, squash and pumpkin are rarely eaten raw and the skin is almost always removed before consuming.

Squash and Pumpkins may grow in a wide variety of colors, shapes and sizes. Their interiors are most commonly orange and remain bitter and starchy when raw and become tender-sweet when cooked. Despite its orange color, the flavor of winter squash is made up of strong green-vegetal aromas. The strongest pairings are with citrus and maillard flavors.

Growing regions

China, India, Russian Federation, US, Egypt are top producers.

Common varietal

Acorn, Butternut, Calabaza, Delicata, Kabocha, Hubbard, Long Island Cheese Pumpkin, Field Pumpkin, Fairytale Pumpkin

Taste

bitter-starchy when raw; sweet when cooked

Main Aromas

(Z)-3-Hexenol- bell pepper, grass, green leaf, herb, unripe banana

n-Hexanol- cooked vegetable, flower, green, herb, rotten

Dimethyl Disulfide- cabbage, garlic, onion, putrid

Limonene- balsamic, citrus, fragrant, fruit, herb

Best Pairings

citrus, butter, cream, cheese, caramel, toasted nuts, cocoa, mushroom

Substitute

Any variety of squash or pumpkin may be substituted for another. Also: sweet potato, zucchini, chayote, summer squash

Pairing	Common Compounds	Aroma	Also Shared
Pumpkin + Melon	3-(methylthio)propanal	cooked potato, soy	apple, citrus, pineapple scallop, sesame
	2-phenylethanol	fruit, honey, lilac, rose, wine	blue cheese, citrus elderberry, mushroom tequila
Sweet Potato + Clam	2,3-pentanedione	bitter, butter, caramel, fruit, sweet	beer, cream, ham tamarind
	ethanol	almond, caramel, cooked, roasted garlic, spice	coconut, chile, peach
Pumpkin + Okra	2-acetylfuran	balsamic, cocoa, coffee, smoke, tobacco	almond, chocolate mushroom, wine peanut
	2,5-dimethylpyrazine	burnt plastic, cocoa, medicine, roast beef, roasted nut	chestnut, soy squid

Winter Squash Pickles and Melon Salad

Winter squash and melons are closely related, giving them similar flavor profiles and a good pairing score. Unfortunately, one grows in spring, the other in winter, so they rarely have a chance to meet in the same dish. But Brooke and I came up with a solution- make winter squash pickles and then add them to a July fresh melon salad. We were blown away by the results!

Winter Squash Pickles

Makes 1 cup

6 fl oz white wine or rice wine vinegar

2 fl oz water

1 tablespoon yellow mustard seeds

1 teaspoon kosher salt

2 teaspoons sugar

½ teaspoon curry powder

1 cup thinly sliced winter squash

Combine all ingredients except the squash in a saucepot and bring to a boil, reduce the heat to simmer and cook 2-3 minutes. Add the squash and simmer until just tender.

Transfer to a clean container to cool. Store in tightly covered in a cool dark place up to 6 months.

Melon Salad

½ cup greek yogurt

1 teaspoon curry powder

1 teaspoon minced mint leaves

zest and juice of ½ lime

1 kirby cucumber, thinly sliced (about 1 cup)

2 cups melon (honeydew, cantaloupe, watermelon or a combination)

1 cup cherry tomatoes

1 cup winter squash pickles

2 tablespoons winter squash pickling liquid

1 red cow horn pepper (medium heat long red chile)

2 tablespoons olive oil

kosher salt, to taste

2 tablespoons picked basil leaves

In a small bowl combine the yogurt, curry powder, mint and lime. Whisk until smooth and set aside for flavors to develop.

In a separate bowl, combine the cucumber, melon, tomatoes, pickles, pickling liquid, chile, and olive oil. Gently toss to combine and season to taste with salt.

To serve, spoon some of the yogurt on the base of a plate and spread with the back of a spoon. Mound the melon on top of the yogurt and garnish with basil leaves.

Stone Fruits

Stone fruits, or 'drupes,' are a class of fruits that have thick flesh surrounding a pit which contains a kernel. This chapter looks specifically at the drupes of the genus *Prunus* which are peach, plum, apricot, nectarine and cherry. All of these fruits derive their distinctive flavors from lactones, a class of intensely fruity aromas that are often used in the production of fragrances and flavorings. While there are similarities in flavor across the category, stone fruits can be placed into subgenus with further correlations.

Peaches and Nectarines- yellow-fleshed fruits that are at peak ripeness from June to early September. Their flavor comes from a combination of citrus, floral and almond aromas.

Plums and Apricots- gold to orange fleshed fruits that are at peak ripeness in August. Their flavor comes from fruity lactones with aromas of apricot, banana and orange.

Cherry- red fleshed fruits that are at peak ripeness in July. Their flavor comes from a combination of almond, fruit and vegetal aromas.

Growing region (top 5 producers, globally)

Peach- China, Italy, Spain, US, Greece

Plum- China, India, Serbia, Romania, Chile

Apricot- Turkey, Iran, Uzbekistan, Algeria, Italy

Nectarine- China, Italy, Spain, US, Greece

Cherry- Turkey, US, Iran, Italy, Spain

Common varieties (subspecies)

Prunus Amygdalus

Peach- Clingstone: Santa Rosa, Red Beauty; Freestone: O'Henry, Elegant Lady, Red Top

Nectarine- White Fleshed: Zephyr, Jade, Emeraude and Arctic varieties; Yellow Fleshed:

Prunus Prunus

Plum- Damson, Greengage, Mirabelle, Satsuma Plum, Victoria, Yellowgage

Apricot- Gold Kist, Autumn Royal, Blenheim, Early Golden

Redgold, Sunglo, Honey Royale

Prunus Cerasus

Cherry- Bright Red Sour Cherries, Bing, Dark Hudson, Ranier, Yellow

Taste sweet to sour

Main Aromas Fruity- *Tree fruit, Citrus*

Peach, Nectarine

Linalool- bergamot, coriander, lavender, lemon, rose

Benzaldehyde- bitter almond, burnt sugar, cherry, malt, roasted pepper

decano-1,5-lactone- coconut, fruity, peach, creamy

Plum, Apricot

γ -decalactone- apricot, fat, peach, pleasant, sweet

methyl octanoate- fruit, orange, sweet, wax, wine

hexyl acetate- apple, banana, confectionery, grass, herb

Cherry

Benzaldehyde- bitter almond, burnt sugar, cherry, malt, roasted pepper

Geranylacetone- green, hay, magnolia, fruity

benzyl alcohol- almond, boiled cherries, moss, roasted bread, rose

Best Pairings

cream, yogurt, basil, citrus, lemongrass, fruit brandies, corn, peppers, wine

Substitute

All stone fruits may be substituted for one another.

Pairing	Common Compounds	Aroma	Also Shared With
Cherry + Beer	isopentyl benzoate	balsamic, sweet	cocoa, lychee, quince, vinegar
	2-hydroxybenzoic acid	sour, sparkling, sweet	cheddar cheese, honey, peanut, vanilla
Apricot + Sage	α -phellandrene	citrus, mint, pepper, turpentine, wood	cumin, fennel, lime, pepper, pistachio
	linalool	bergamot, coriander, floral, lavender, lemon, rose	chile, lemongrass, prickly pear, star anise
Plum + Elderberry	butyl hexanoate	fruit, grass, green	capers, melon, passion fruit, apple
	methyl 2-hydroxybenzoate	almond, caramel, medicine, peppermint, sharp	basil, coffee, olive, strawberry
Peach + Soy Sauce	4-hydroxypentanoic acid lactone	herb, sweet	chicken, hazelnut, swiss cheese, toasted bread
	ethyl benzoate	camomile, celery, fat, flower, fruit	mushroom, sherry, tamarind, red wine

Pan Roasted Pork Tenderloin with Coffee, Soy and Peaches

Serves 4

Pork also has latent fruit aromas, which are drawn out by the peaches. Soy sauce and bourbon flip the switch on peaches giving them a savory edge. This is a dead simple

recipe to pull out when ever you are looking to impress in hurry. It packs in unexpected flavors at every step and can be on the table in about 20 minutes.

2 teaspoons kosher salt
1 teaspoon ground coffee bean
1 teaspoon ground coriander
½ teaspoon freshly ground black pepper
¼ teaspoon ground cinnamon
1 pound pork tenderloin, trimmed and cut crosswise into 12 medallions
2 tablespoons butter
2 tablespoons extra virgin olive oil
2 peaches, pit removed and sliced (about ¼ inch thick)
2 shallots, thinly sliced
2 branches fresh sage
2 fl oz bourbon
1 fl oz soy sauce
8 fl oz chicken stock
1 teaspoon fresh thyme

Combine the salt, coffee, coriander, pepper and cinnamon in a small bowl and mix well.

Spread the sliced pork medallion on a plate or baking pan and sprinkle evenly on both sides with the spice mixture.

Heat a large cast-iron skillet over medium-high heat. Add 1 tablespoon each of the butter and olive oil.

Cook the pork medallions in batches, browning them well on both sides before removing to a rack to rest. Halfway through the cooking, discard the fat in the pan, wipe the pan out and add fresh butter and oil.

When all the pork has been cooked, add the shallots and sage to the pan and cook 1 minute before adding the peach slices. Saute the peaches until tender, 1-2 minutes more.

Deglaze the pan with the bourbon and reduce until nearly dry. Return the pork to the pan along with the chicken stock. Simmer 1 minute. Sprinkle with fresh thyme leaves and serve immediately.

Tomato

Tomatoes are the edible fruit of the tomato plant, a member of the nightshade family, which originated in Central and South America. Today, tomatoes are an essential part of cuisines around the world despite having not arrived on the European, Asian or African continents until the mid-1500s. Approximately 7,500 different varieties of tomato are grown worldwide, with production totaling over 170 million tons annually.

Tomatoes are highly aromatic, with nearly 500 different aromatic compounds coming together to create their flavor. Vegetal and fruit aromas are the most prominent in tomato flavor, though no single compound stands out as largely responsible for tomato flavor.

The taste of tomatoes is a result of the balance of natural sugars and acid present in the fruit and can range from tart to sweet. When seasoning tomatoes, taste can be adjusted by adding vinegar or sugar to fine tune the acid-sugar balance. Tomatoes are also a natural source of umami.

Growing region

China, India, US, Turkey, Egypt are top producers. Tomatoes thrive in climates with warm dry days and cool nights. In the Western Hemisphere, tomatoes reach their peak in summer months, June through September.

Common varieties

Beefsteak, Plum, Cherry, Grape, Campari, Brandywine, Cherokee Purple, Green Zebra

Taste sweet to sour, umami

Main Aromas

3-methylbutanal- almond, cocoa, cooked vegetable, malt, spice

(Z)-3-hexenal- apple, bell pepper, cut grass, green, lettuce

3-methyl-1-butanol- banana, cocoa, floral, gasoline, malt

6-methyl-5-hepten-2-one- citrus, mushroom, pepper, rubber, strawberry

Best Pairings

eggplant, corn, avocado, peppers, peach, sesame, tamarind, lettuce

Substitute

strawberry, tomatillo, persimmon, peppers

Pairing	Common Compounds	Aroma	Also Shared With
Tomato + Coconut	2-phenylethanol	fruit, honey, lilac, rose, wine	cheese, lemon, tamarind, truffle, watercress
	hexanal	fresh, fruit, grass, green, oil	avocado, cardamom, cucumber, mussels, sesame
Tomato + Tea	methyl 2-hydroxybenzoate	almond, caramel, medicine, peppermint, sharp	chile, honey, sage, vanilla
	β -cyclocitral	mint	apricot, date, ginger, olive
Tomato + Passion Fruit	geranylacetone	green, hay, magnolia	clam, lemon balm, pork, rice
	trans-linalool oxide	citrus, flower, fresh, lemon	cilantro, citrus, cocoa, strawberry

Tomato + Celery	dimethyl sulfide	cabbage, gasoline, organic, sulphur, wet earth	crab, corn, okra, shrimp paste
	4-methylacetopheno ne	bitter almond, floral, fruit, spice, sweet	mint, parsley, peach, pepper

Black Tea Tomato Sauce

Makes 3 cups

The flavors of tea and tomato are an excellent match. Black tea in particular gives a deep savory edge with herbaceous and phenolic aromas, the ideal match for tomatoes. Tea is extremely powerful though, so it must be used judiciously. Just a pinch, really, lends the perfect seasoning to a pan of tomato sauce.

2 tablespoons olive oil

1 cup minced yellow onion

4 garlic cloves, minced

28 ounces canned tomato

½ teaspoon loose black tea

1 cup water

Pinch of sugar (optional)

Heat the olive oil in a medium saucepot. Add the onions and garlic and season with salt and a few grinds of black pepper. Cook over a medium flame until tender but not browned, about 5 minutes.

Add the canned tomatoes, tea and water. Bring the mixture to a boil, and season again to taste with salt and pepper. Reduce the heat to a simmer, cover the pot and cook 15 minutes.

Taste the sauce a final time, adjust the seasoning with salt and another pinch of tea if desired. The sauce may pureed if desired or used as is. Keep sauce refrigerated for up to 2 weeks or freeze to keep up to 6 months.

Tropical Fruit (Mango, pineapple, banana, coconut, passion fruit, papaya)

Tropical fruit refers to a group of aromatic fruits that grow in tropical regions throughout the world. In this chapter, we have chosen to look at them as a large group because of the commonalities they share in flavor profiles, while also noting their unique differences below under the heading 'Main Aromas.'

Tropical fruit flavors are mainly created by lactones. The tropical scent of many sunscreens and lotions also owes the distinctive aromas to lactones. *Methyl Hexanoate* is the common thread among all tropical fruits. It plays a significant role in the flavor of: passion fruit, papaya, kiwi, oysters, pineapple, white wine, cider, cranberry, melon, olive, raspberry and strawberry. It also contributes to the aroma of blue cheese and is one of the main factors that contribute to the surprising pairing of pineapple and blue cheese.

Growing regions of Common Varieties

Mango- India, China, Thailand, Indonesia, Pakistan

Pineapple- Costa Rica, Philippines, Brazil, Thailand, India

Banana- India, Chad, Uganda, Philippines, Ecuador

Coconut- Indonesia, Philippines, India, Brazil, Sri Lanka

Passion Fruit- Brazil, Ecuador, Columbia, Peru

Papaya- India, Brazil, Indonesia, Nigeria, Mexico

Taste sweet to sour

Main Aromas

Mango

delta-3-carene- citrus, fir, needle

ethyl 3-methylbutanoate- apple, citrus, pineapple, sour, sweet

(Z)-3-hexenyl butanoate- apple, wine, green

Pineapple

Furaneol- burnt, caramel, cotton candy, honey, sweet (mainly responsible for toasted)

methyl 3-methylthiopropionate- vegetable, radish, horseradish

3-(methylthio)propanoic acid ethyl ester- sulphur, pineapple, onion, garlic, tomato

δ-octalactone- coconut, sweet, cream

Banana

Isoamyl acetate- apple, banana, glue, pear

isoamyl alcohol- banana, cocoa, floral, gasoline, malt

2-methyl-1-propanol- apple, bitter, cocoa, fusel, solvent

Passion Fruit

3-hydroxy-2-butanone- butter, cream, green pepper, rancid, sweat

2-heptanol- citrus, coconut, earth, fried, mushroom, oil

ethyl hexanoate apple peel, brandy, fruit gum, overripe fruit, pineapple

Coconut

δ-octalactone- coconut, peach

δ-nonolactone- nut, peach

ethyl octanoate- apricot, brandy, fat, floral, pineapple

Papaya

Linalool- bergamot, coriander, lavender, lemon, rose

isoamyl butanoate- green fruit

decanal- floral, fried, orange peel, penetrating, tallow

Best Pairings

peach, vanilla, thyme, honey, cream

Substitute Any tropical fruits may be substituted for one another

Pairing	Common Compounds	Aroma	Also Shared
Banana + Jalapeno	isopentyl butanoate	green fruit	camomile, pomegranate
	isopentyl 3-methylbutanoate	fruit	beer, lettuce, plum
Coconut + Lamb	5-hydroxyoctanoic acid lactone	coconut, peach	beef, chicken, tomato
	2-undecanone	fresh, green, orange, rose	corn, lemon, scallion, turnip
Mango + Dill	(Z)- β -ocimene	citrus, flower, herb, mold, warm	basil, cardamom, fennel
	cis-3-Hexenyl acetate	banana, candy, floral, green	lemon, olive, wine
Pineapple + Blue Cheese	butanoic acid	butter, cheese, rancid, sour, sweat	mushroom, red bread
	methyl pentyl ketone	bell pepper, blue cheese, green, nut, spice	annatto, asparagus, yogurt
Passion Fruit + Mustard	methyl benzoate	herb, lettuce, prune, sweet, violet	black currant, vanilla, wine
	citronellol	citrus, green, rose	cinnamon, ginger, rosemary
Papaya + Tequila	2-methylthiophene	sulphur	crab, crayfish, soybean
	carvacrol	caraway, spice, thyme	coriander, oregano, basil

Crab, Mango, Dill and Poblano Salad

Serves 4

Mango and dill is a surprising and delicious combination. It mainly comes from the woody/pine aromas found in each. Combined with crab and avocado, they create a remarkable dish that hits all of the right notes.

1 lb lump crabmeat

1 cup diced cucumber

½ Lemon, zested and juiced

2 teaspoons fresh dill, chopped

3 tablespoons mayonnaise

1 poblano, roasted and chopped

kosher salt

freshly ground black pepper

1 mango

¼ cup celery leaves

Combine the crab, cucumber, lemon, dill, mayonnaise, and poblano in a bowl. Stir gently to combine. Season to taste with salt and pepper.

Peel the mango and cut the flesh from the pit. Slice the mango flesh into long pieces, 1/8 – 1/16th inch thick. Curl each slice and place on top of the salad. Garnish with Celery Leaves.

Truffles

Truffles are the most highly prized fungus in the culinary world. They grow just below the surface of the ground near the roots of certain trees. The most desirable truffles, *Tuber melanosporum* (Black Truffles) and *Tuber magnatum* (White Truffle) grow naturally in specific locations throughout southern France, Italy and Croatia. Truffles only grow under certain conditions at specific times of the year and must be located by specially trained dogs who sniff out their hiding places just below the earth's surface. It is the truffle's scarcity and difficulty in obtaining that causes fresh truffles to demand incredible prices. During their short seasons black truffles can sell for \$700+ and the rarer white truffle \$1,200+ per pound.

The complex flavor of fresh truffles is largely created by sulphur compounds and sweet fruit aromas. Fresh white truffles lose their aroma quickly after slicing and are best shaved directly onto food immediately before eating. White truffles should not be cooked. Black truffles release more of their aroma when heated and should be added to dishes in the last moments of cooking.

If purchasing truffle products like canned truffle, truffle butter, or truffle oil, always check the labels to insure the ingredient list contains *tuber melanosporum* or *tuber magnatum* rather than less flavorful species or imitation flavors.

Growing region

White Truffle- Italy, Croatia, Istria

Black Truffle- France, Spain, Italy, Croatia, Slovenia

Common varieties

Black truffle (Black Perigord truffle), Black Summer Truffle (or Black Burgundy Truffle)

White Truffle, "White-ish Truffle," Garlic Truffle, Pecan Truffle

Taste bland

Main Aromas

2,3-butanedione- butter, caramel, fruit, sweet, yogurt

dimethyl disulphide- cabbage, garlic, onion, putrid

ethyl butyrate- apple, butter, cheese, sweet, pineapple, red fruit, strawberry

Best Pairing

earthy: grain, rice, mushroom; asparagus, egg, tomato, butter, cheese, toasted nuts

Substitute

No direct replacement for truffle aroma. Substitute mushrooms- esp porcini, and hen of the woods.

Pairing	Common Compounds	Aroma	Also Shared With
Truffle + Beetroot	ethanol	alcohol, floral, ripe apple, sweet	blue cheese, cocoa, pistachio, walnut
	dimethyl sulfide	cabbage, gasoline, organic, sulphur, wet earth	asparagus, corn, pea, shellfish
Truffle + Tomato	geranylacetone	green, hay, magnolia	brandy, hyssop, lemon, sage
	2-propenal	burnt sweet, pungent	beef, beer, potato, wine
Truffle + Strawberry	isobutanol	apple, bitter, cocoa, fusel, plastic, solvent	camomile, chicken, gin, radish,
	2-octanone	fat, fragrant, gasoline, mold, soap	chile, fig, cream, peanut, vanilla

Truffle-Roasted Beet Salad

Yield: Serves 6

Truffle flavor is a combination of sulphur and fruit aromas. In this recipe, the earthy and fruity aromas of beets blend perfectly with truffle. Blue cheese accentuates the pungency of truffles, but may be substituted for more mild goat cheese or robiola.

2 pounds medium-sized red beets

1 fluid ounce olive oil

4 branches fresh thyme

2 bay leaves

3 tablespoons truffle peelings, divided

½ cup red wine vinegar

garnish- crumbled blue cheese, toasted walnuts or pistachio, pea shoots or tendrils

Preheat oven to 350° F

Trim and rinse the beets; toss with the olive oil and salt and pepper. Place the red beets in a deep, oven-proof dish. Add the thyme, bay leaves, vinegar, and 2 tablespoons of the truffle peelings to the dish. Cover with aluminum foil and roast in the oven until tender, 45 to 60 minutes.

Set the cooked beets aside to cool slightly. When cool enough to handle—but still hot—peel the beets cut into wedges or slices.

Strain the remain juices in the bottom of the roasting pan. Add the remaining truffle peelings and pour over the beets. Toss well to coat. Season to taste with salt and pepper. Transfer salad to plates and garnish.

Vanilla

Vanilla is a flavoring derived from a specific genus of orchid (genus- Vanilla) that is native to Mexico. Vanilla was introduced to Europe by early Spanish conquistadors. However, because the vanilla plant is pollinated by a bee that is indigenous to Mexico, the vanilla orchid was not successfully cultivated elsewhere. It wasn't until the discovery of a technique for hand pollinating—in the mid 1800s—that vanilla production was able to spread outside of Mexico.

Vanilla pods must be hand harvested because pods on the same plant will ripen at different times. When vanilla beans are ready to harvest, they look like plump green beans. After harvest the fruit of the vanilla plant must go through a process of curing to develop its flavor.

Curing begins with killing the fruit. This is typically done by heating or freezing to begin enzymatic reactions that create flavor. Sweating is the next step in the curing process. The beans are stored in high temperature (115°F to 150°F) and high humidity conditions to further develop their aroma. After sweating, the pods are dried to lock the aroma into the pod, dried vanilla pods retain 25-30% moisture content. Finally pods go through conditioning, in which they are stored for up to six months as fragrance develops. The cured vanilla pods are then graded. Grading systems vary from country to country but are based on length, appearance and moisture content of the pods.

The aromatic compound *4-hydroxy-3-methoxybenzaldehyde* also known as *Vanillin* is mostly responsible for the characteristic aroma of vanilla. Vanillin is used to create artificial vanilla flavoring or vanilla 'scented' products. However, real vanilla aroma is the result of a combination of over 250 different compounds.

Growing region

Indonesia, Madagascar, Mexico, Papua New Guinea, China

Common varieties

Bourbon vanilla beans, Tahitian vanilla bean, Mexican vanilla bean,

Taste slightly bitter

Main Aromas

vanillin (4-hydroxy-3-methoxybenzaldehyde)- sweet, vanilla

p-hydroxybenzaldehyde- slightly nutty, almond, balsamic, vanilla

anisaldehyde- almond, anise, caramel, mint, sweet

Best Pairings

milk/cream, peach, nectarine, citrus, lemongrass, whisky, mint

Substitute

Tonka beans may be substituted for fresh vanilla beans. Substitute almond extract or bourbon whisky for vanilla extract.

Pairing	Common Compounds	Aroma	Also Shared With
Vanilla + Corn	vanillin	sweet, vanilla	citrus, dill, fish, sherry
	4-hydroxybenzaldehyde	roast	beer, pineapple, shrimp paste, tomato

Vanilla + Tomato	anisaldehyde	almond, anise, caramel, mint, popcorn, sweet	anise, basil, coffee, hazelnut
	2,3-butanediol	cream, floral, fruit, herb, onion	cider, melon, pecan, vinegar
Vanilla + Asparagus	3-hydroxy-2-butanone	butter, cream, green pepper, rancid, sweat	barley, blue cheese, clam, honey
	acetovanillone	clove, flower, vanilla	beans, pork, soy sauce, wine

Vanilla Butter

Makes 1 cup

Vanilla is too often relegated to just the final course of a meal. With it's incredible aroma vanilla can add an enticing flavor to many food. We like to a savory vanilla scented butter and use it for finishing vegetables, roasted meats and seafood—think brushing over shrimp as they come off the grill. Vanilla will deepen those smoky, roasted flavors of the grill.

8 ounces butter, softened to room temperature

½ vanilla bean, split and scraped

2 teaspoons kosher salt

½ teaspoon freshly ground black pepper

½ teaspoon freshly ground coriander seeds

Combine the butter, scraped vanilla seeds, salt, pepper and coriander seeds in a bowl and blend until smooth and well mixed.

Transfer to an airtight container and store in the refrigerator up to 2 weeks. Transfer to the freezer if keeping longer.

Part III

Inspiration

[TK section: Additional resources]

Suggestions for further reading:

Books:

Chemistry in Your Kitchen by Matthew Hartings (Royal Society of Chemistry, 2017)

Ingredient: Unveiling the Essential Elements of Food by Ali Bouzari (HarperCollins, 2016)

Cognitive Cooking with Chef Watson: Recipes for Innovation from IBM & the Institute of Culinary Education (Sourcebooks, 2015)

Websites:

www.Foodb.ca (food component database)

www.Vcf-online.nl (Volatile compounds in food database)

[TK section: Acknowledgments]

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[Note to indexer: Please include categories of recipes in index, as below]

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Tahini Vinaigrette (Artichoke)

Beer-Pickled Mustard Seeds (Asparagus)

Umami Vinaigrette (Brassica Oleracea)

Banana and Pepper Hot Sauce (Capsicum)

Gin and Brown Butter Emulsion (Jerusalem Artichoke)

Green Apple, Basil and Strawberry Salsa (Apple)

Apricot, Sage and Pistachio Chutney (Poultry)

Blueberry and Horseradish Jam (Pork)

Black Tea Tomato Sauce (Tomato)

Vanilla Butter (Vanilla)

Bacon Tapenade and Melon Bruschetta (Melon)

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Benne Seed and Avocado Salad with Fig Vinegar (Avocado)

Beet and Carrot Salad with Orange Hazelnut Dressing (Beetroot)

Cucumber, Berry and Hazelnut Salad with Tamarind Vinaigrette (Cucumber)

Tamarind Glazed Lettuce with Creamy Dressing (Lettuce)

Radish and Apple with Basil Vinegar (Radish)

Thai-Style Spicy Zucchini Salad (Summer Squash)
Winter Squash Pickles and Melon Salad (Winter Squash)
Crab, Mango, Dill and Poblano Salad (Tropical Fruit)

Vegetables

Broccoli, Peanut and Pumpkin Stir Fry (Cauliflower, Broccoli)
Fried Eggplant with Roasted Red Pepper, Walnut and Pomegranate Molasses Sauce
Spice Roasted Grapes (Grape)
Crunchy Green Bean “Granola” (Green Beans)
Sungold Kiwi and Cranberry Relish (Kiwifruit)
Beet and Blue Cheese Potato Rösti (Potato)
Roasted Root Vegetable with Ginger Salsa Verde (Root Vegetable)

Seafood

Molasses-Stained Swordfish Kabobs (Cane Syrup/Molasses)
Grilled Oysters with Watercress Butter (Cress)
Shrimp and Lamb Gumbo (Crustacean)
Coffee Cured Salmon (Fish)
Creamy Coconut Oats with Shrimp and Jalapeno (Grain)

Meats

Cocoa and Chile Rubbed Beef (Beef)
Chicken and Mushroom Burger with Strawberry ‘Ketchup’ (Berries)
Turnip, Chicken, Apple, Hazelnut, Radish, Thyme (Brassica Rapa)
Game Meat Seasoning (Game Meats)
Lamb Seasoning Rub (Lamb)
Sweet Pea, Pork and Coconut Tacos (Peas)

Spicy, Yogurt and Lemongrass marinated Chicken with Pomegranate (Pomegranate)
Pan Roasted Pork Tenderloin with Coffee, Soy and Peaches (Stone Fruit)

Sweets

Chocolate Mousse with Beet Meringues (Cocoa Bean)
Corn and Coconut Crème Brulee (Corn)
Bourbon Roasted Fennel and Plums with Coffee Ice Cream (Fennel)
Fig, Olive and Walnut Relish (Fig)
Porcini Chocolate Tart (Mushrooms)
Peanut and Lime Macaron (Nuts)
Lemon Curd with Crunchy Olives (Olive)
Raspberry Rice Pudding (Rice)

Cocktails

Ginger Mule with Pistachio Vodka (Ginger)
Spicy Honey Syrup (Honey)
Lemongrass Bitters (Lemongrass)